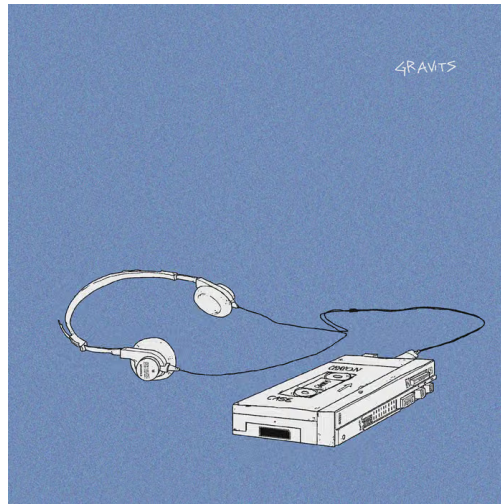


Tugas 1 IF4020 Kriptografi

Kriptanalisis pada Algoritma Kriptografi Klasik



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2026

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Pernyataan Orisinalitas

Tugas dan laporan ini disusun sepenuhnya tanpa bantuan kecerdasan buatan (*Generative AI*), melainkan hasil pemikiran dan analisis mandiri.



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Bab 1

Pendahuluan

1.1 Latar Belakang

Marisa sedang berusaha menyusun sebuah arsip rahasia yang memuat pemetaan wilayah, entitas penguasa, dan kronologi insiden penting di Gensokyo. Berkas ini merupakan gabungan dari catatan riset pribadinya serta dokumen-dokumen sensitif yang berhasil ia kumpulkan dari berbagai pihak. Demi menjaga keamanan informasi, seluruh naskah di dalam arsip tersebut dienkripsi dengan berbagai jenis *cipher* klasik. Sebagai rekan riset yang dipercaya, tugas kami adalah membantu Marisa mendekripsi serangkaian dokumen dalam arsip ini agar seluruh pengetahuan rahasia tersebut dapat terungkap.

1.2 Ringkasan Soal

Laporan ini terdiri atas lima soal wajib dan satu soal bonus. Setiap soal menuntut kriptanalisis terhadap sebuah *cipher* klasik yang berbeda.

No.	Judul	Algoritma <i>Cipher</i>
2.1	Beast Metropolis	Substitusi abjad-tunggal (<i>monoalphabetic</i>)
2.2	Horizon Chartreuse	Vigenère <i>Cipher</i>
2.3	Domain of Hopes and Dreams	Playfair <i>Cipher</i>
2.4	Trinitarian Rhapsody	Affine <i>Cipher</i>
2.5	Silent Goddess	Hill <i>Cipher</i>
3.1	Ex Machina (Bonus)	Enigma Machine M3 (termodifikasi)

Tabel 1.1: Daftar soal dan algoritma *cipher* yang digunakan.

1.3 Ketentuan Pengerjaan

Sesuai spesifikasi tugas, seluruh proses enkripsi pada *plaintext* asli dilakukan setelah teks diubah menjadi huruf kapital dan hanya pada karakter alfabetik [A-Z]; karakter lain (angka, spasi, tanda baca) dibuang dan tidak dienkripsi. Untuk soal wajib, kriptanalisis dilakukan hanya dengan *approach* yang telah diajarkan di kelas. Bantuan skrip program, lembar kerja, atau alat bantu komputasi lain diperbolehkan untuk menghitung frekuensi karakter dan pola.

Bab 2

Problem Set (Wajib)

2.1 Beast Metropolis

Deskripsi Soal

Di antara megahnya pencakar langit Alam Satwa yang dipenuhi persaingan liar, Marisa menemukan sebuah lembaran catatan yang membahas fasilitas misterius dengan struktur yang menyerupai lubang kunci. Namun, catatan mengenai benteng perlindungan tersebut kini tidak dapat dibaca karena seluruh teksnya telah dienkripsi menggunakan *cipher substitusi abjad-tunggal* (*monoalphabetic cipher*). Untuk dapat memahami struktur dari kawasan terisolasi itu, dokumen riset ini harus dikembalikan ke dalam *plaintext* seperti semula.

Bantulah Marisa mendekripsi *ciphertext* tersebut meskipun kunci pemetaannya tidak diketahui. Gunakan teknik **Analisis Frekuensi** serta analisis pola kata umum dalam **bahasa Inggris** untuk memecahkan sandi monoalfabetik ini.

Berkas lampiran:

- *Attachment:* https://drive.google.com/file/d/14DWEicqv6_2K00cA9mbXf_ZwhPrEiak/view?usp=sharing
- md5sum: dd67736bf3c6ef6db2d5f014d55d870a
- sha256sum: a6478218a83091712dc99972553f89e92da8436b2038c5b93f782268742b4972

Berkas *Ciphertext*

```
UMRQARPQNCKICIRQQIRPVQIRZABXUASQTVUXQYQIODFDQIMZIHQSUIRDUOQUACIWQTRQAIGUOUIQJUNOXQTMFUITUCAQSCDICIVXZKQ
FDMZIDMRDFZVPCCITUFZAUCIUUAUOAMQVRZAQRPFDMZIDMRTZCDRTZYBUNUCIFCYZSUWUFBDROAQMQRZAQRPFDMZIDMZGCNURQYZB
UNUCIDFUBUNUDFUBUNUOAMQVRZAQUIKRPQFDMZIDMCTPCHZRUCCIFUIKUMZFZDFUOAMQVRZAQURRPQTUNQRCNQRPAQWQAQUXTDYQIOD
FDQIMZICIVPSDFZAQSCDIUIKFBZTPZMDAQJUNOXQRPQUIVCQIRRDNHTDMNDRTPCCIUWUCIRPQVCRBDMUTUPCFBDRDOAMQVRZAQUIKRP
QUIVCQIRRDNHTDMRDNCFUSUCIRPQVCRBDMDFUBUNUUNDISDRPQATRPQXUASQTRDIQCTRPQRDFZRDNUUIVCQIRRDNHRPQRDRUXXQISRPMR
PQNDZIKCTUHDZRRWDPZIKAKQCSPRBNQRQATWPCVPCTNDAQRPUIRPAQRCNQTXUASQARPUIRPQXUASQTRNDZIKRDNHDZRRQCSPRBNQR
QATCIRPQXURQAQACDKMRQBUBDCOQACDKUIKRTUAQUUIKVUUVCRBUAQMUSAQURQARPUIRPQXURRQAUXRPDZSPRPQNDZIKRDNHTWCR
PUMADIRUIKHUVFVCAVZXUATPUOQWQAQUXAKUBTVURRQAQKRPADZSPDZRGUOUIKZACISRPQBUBDCOQACDKRPQQAUXBMDANTDMRPQYQIODF
DQIMZIWQAQNDZRRBEQARCVUXVUEQTRBXQTRDIQVPUNHQATWCRPHUNHDDTPUOQKWDDKQIVDMMCTWPCVPWQAQKCMMAQIRMADNRPNQNDZIKR
DNHTKZACISRPQBUBDCOQACDKNDAQDEQAQXABYQIODFDQIMZIPUKVDNNDIMQURZAQTTZVPUTUXDWQAUIKWCKQAMADIROUARDVNOUAQKRDR
PQHUVFVCAVZXUAOUARUIKRPQVDNHCIURCDIUIKODTCRCDIDMSAUEQSDDKTWAQRPQTUNQCITCKQRPQVDMMIWQAQGUQKUIKNCAADATWPCX
QDZRTCKQRPQVDMMIWQAQXUASQIZNHQADMRAUISZXUAQSKQCECIHQUTRNCADATCADIWQUODITUSACVZXZRAUXUIKMCTPCISRDDXTU
IKTDDIAQSUAKCISRPCTRUCCVPCADTPCAUCTPCHQXCQEQRPURURRPURRCNQEUAACDZTMDAVQTCIFCIDXQKHBRRPBUNURUCFCISKDNUIKRPQ
TQRDCIXUIKTQUMDANQKUIUXCUIVQUIKDHURUCIQKACADIUIKEUACDZTVZXZRAUXAQXCVTMADNFDAQUHBKMQMURCISMDAVQTTZVPUTIZFDFZ
```

UIKRPQCRDFCISKDNTWPCVPVDIRADXXQKRPQSQIFUCTQUDIRPQDRPQAPUIKBDTPCADFDIKDUXTDNQIRCDIQKRPURNDRDMPRQYQIODFDQIM
 ZIHZCXRDI MXRUXUIKWAQWQXDA SUICYKUIKRPQNDZIKTWQAQXTDH ZCXR XUBQA HXBUBQAUMRQAVDITDXCKURCDIVDITCKQACISRPQT XD
 OQUIKUAUISQNIQIRDMRPQNDZIKTVCAVZXUANDZIKTWQAQVDTTRAZVRQKCIUKKCRCDIHADIYQNC AADATMADNNUCIXUIKVP CUIUIKRPQWUGC
 IKQIMADNRPTUNQOQACDKCIKCVURQRPURPQAQWUTUIQJVPUI SQHQRWQKIRPQWVDZIRACQTURRPURRCNQRPQAQMDAQCRCCTOQVZXURQK
 RPURRPQQA XBUOQQAUIVQDMRPQYQIODFDQIMZINUBPUEQHQQICIMXZQIVQKHBVCECXQISCIQQACISRPQVIDXDSBMADNNUCIXUIKVP CUIU
 IKRPQODXCR CVUXUIKCKQXDCSVUXCIMXZQIVQCTAQMXQVRQK CIRPQGD CIRPVNITRAZVRCDIDMRPQYQIODFDQIMZIHBEUACDZTUXXCUIVQNN
 NHQATUTOADMDMRPQUXXCUIVQURRPQTUNQR CNQRPQAQAQXUTDRDNHTMCEQMDZAU IKRPAQQDMRPQFUNQINZICIRPQMA DIRM RPQDPZBZUI
 RDNH C I QUTRQAIGUOUIKZACISRPQTUNQOQACDKUTCTPCHZRUCFDMZIUNDISRPQNRDNHMC EQCTR PQQUAXCQTRFDMZICIQUTRQAIGUOUIPTCT
 RDABCIKCVURCISUODTTCHXQVDIIQVRCDIHQRWQKIRPQFUYZTUOADECIVUIKRPQBUNURUCFCISKDNURRPURRCNQ CIRPQMDZARPVQIRZABP
 DFZACFZAQSCDIFUIRDAQSCDIRDUFUCAQSCDIUIKRPDPFZAQSCDIWQAQNUCIXBVDNODTQKDMADIRUIKAQUARDNHTWPCVPSAUKZUXXBH QVUN
 QYQIODFDQIMZIRWDRDRAQQSQIAURCDITXURQAASUA K CISRPCTBUNURDRUFQAZODCIRKQDZRRPURUXRPDZSPRPQDFGFCUIKICPDITPD
 FCNQIRCDIQKRPURRPQMDZASQIAUXTU IKGUOUIQTQRUFQYZIUIKDRPQATPUNQASQKQUTRGUOUICIRDRPQRQAACRDA BDMRPQGUOUIQTQND
 IUA V PBRPADZSPNZXRCOXQQJQKCRCDITPQHQCQEQRUPURCWRUTUVVRZUXBFZIZIDFZICXDVRQK CIRPQIDHCOXUCIRPURMDANQKUIUXXC
 UIVQCIQUTRGUOUI NUCIXBVDITCTRCISDMAQUASAEQTCIRPQOUTRUIKRPURUMRQAR PQKQURPDMPCNCFDRPQBXDTRRPQHURRXQWCRPRPQBU
 NURUCFCISKDNDAIQSDRCURQKQOVVQZIKAR PQXQKQATPCODMRPQBUNURDFCISTPCOCIRPQQIKRPQUXXCUIVQDMRPQK DSTXUEQFCISKDN
 IKRPQUXXCUIVQDMRPQECXPDATQFCISKDNQASQK CIRDDIQMDANCISRPQGUOUIQTQNDIUA V PBRPQIRPQXQKQATDMQUTRUIKWTRGUOUIW
 QAQKCTRCISZCTPQKHBRPQYQIODFDQIMZUIKRPQMA DIRUIKAQUARDNHTMADNR PQKIDMRPQMDZARPVQIRZABDRPQCMRPVQIRZABXUASQ
 YQIODFDQIMZIAQTQNHXCISDFMZICINDZIRRUCKUHQSUIRDUOQUACIOAQTQIRKUBSZINUOAQMQRVZAQXUASQYQIODFDQIMZIPUEQUXTDHQ
 QIHZCXR CIOXUVQTTZVPUBUNUIUTPCOAQMQRVZAQCHUAUF COAQMQRVZAQIUIKVPCHUOAQMQRVZAQUMRQARPNCKMDZARPVQIRZABRPQKUWU
 RDNHTDACSUIUXBVIQIRQAQKUADZIKRPQBUNURDFDMZICIRPQTDZRPQUTRDMRPQIUUAHU TC IUIKRPQABZHQIFDMZISAUKZUXXBTPCMRQKRD
 RPQKUCTQIFDMZICIRPQIDARPRPQXUASQYQIODFDQIMZIDMKUWUWUTRPQMZAZCVP CFDMZIVXZTRQAUIKNDY ZRDNHTRPQVQIRQADMSAUECRB
 DMRPQSAQURFCISTRDNHTPCMRQKUSUCICIRPQCMRPVQIRZABDRPQUTVCQIRVCRBRDNH SADZOUIKRPQPZIKAKRDISZQKHAKRDNH SADZO
 CIRPQDTUFUOXUCIAQSUAKCISRPQAQUTDITMDARPQNZXRCOXQRAUITM QATMRPQTQXUASQYQIODFDQIMZIHUTQKDIAQTQUAVPDIRPQFZTPC
 GCUIKICPDITPZGCRPQGUOUIQTQVUUKNCVVDNNZICRBPUTOADODTQKECQWTTZVPUTRPQRPDABDMKBIUTRBUXRAIURCDIUIKRPQRPDAB
 DMQLZQTRAUIRACHQTV DILZQACISKBIUTRCQTRDQJOXUCIDIRPQDRPQAPUIKPCADTQFUYZDOADODTQKRPURRPQTTRAZVRZAQTMRPQTUTUFC
 FDGCTPCYZFCUIKNUFCIDUIVCQIRRDNH SADZOTWQAQTCNCXUAUIKADNR PQXURQMDZARPVQIRZABDRPQTQVDIKPUXMDMRPQCMRPVQIRZA
 BRPQPQKUTDMRPQMDZAUICVQIRRDNH SADZOTGDCIRXBOXUBQKUA DXQCIRPQBUNURDFCISTPCOUMRAKQURPRPQBKQCMCQKUTRPQSUAUCUI
 KQCRBDMRPQBUNURDADBUXODWQAUIKUXTD OADODTQKRPURRPQVDITRAZVRCDIDMRPQXUASQYQIODFDQIMZIPUKUKRQAAQIRQMMQVRDIEUA
 CDZTAQSCDITURRPQTUNQR CNQ CIRPQTQVDIKPUXMDMRPQCMRPVQIRZABIDXUASQTVUXQYQIODFDQIMZIWQAQVDITRAZVRQKCIUAQUTT ZVP
 UTFCHCOADECIVQZOOQANUBQUIKPBZSUOADECIVUIKRPQT VUXQDMRPQYQIODFDQIMZICISDFCTPCVPCKDSAU KZUXXBKQVAQTQKUVVDAK
 CISRDRPQCITVACORCDITDIGCGCRPQCADITWDAKZIQUAR PQKMA DNR PQCIUWUNDZIRUCIFDMZUIKRPQXUASQTDWDAKZIQUAR PQKMA DNR PQQR
 UMZIU BUNUFDMZIQNOQADABZABUFZURRPURRCNQAQSUAKQKPCNTQXMTRPQFCISDMRPQWDA XKUIKPCTUZRPD ACBWTATAQMXQVRQK CIRPQMU
 VRRPURDIXBPQVDIRCIZQRDHCXKXUASQYQIODFDQIMZICIRPQGUOUIQTQUAVPCOQXUSD CIRPQTCJRPVQIRZABRPQPDACYDIRUXVUEQTRB
 XQTRDIQVPUNHQADMRPQFCIUCTBTQRNHQSUIRDHQDOOZUACYQKCIYQIODFDQIMZIUURRPQTUNQR CNQRPQBHSUIRDUOQUACIUAQUTT ZVP
 TTQRTZOADECIVQDWUACOADECIVQZGCOADECIVUIKQEQICIFUIRDAQSCDIWPCVPWQAQVDITCKQAQKUTHUTQTMDARPQQTURHXTPNQIRDM
 RPQNOQADAFQCRUCUNDISRPQNZQIDPUKUOUARCVZXUAXBXUASQIZNHQADMYQIODFDQIMZIVDNOUAQKRD DIXBRPCARBIQICIFDYZFQOADE
 CIVQKZACISRPQTUNQOQACDKZQIDUXDIQPUKICIRBTEQIWPXCQIFUIRDRPQAQWQAQRWDPZIKAKT CJRQJUTHDRPFCIDUIKDCYPUISKQ
 VAQUTQKDIXBRPQFUIRDAQSCDIDIRCIZQRDVIDITRAZVRXUASQYQIODFDQIMZIRPADZSPDZRRPQTCJRPVQIRZABCIKCVURCISRPURRPQBU
 NURDNDIUA V PBRURRPURRCNQAQXCQKQKUECXBDIRPQUTRQAIFCISKDN CIHDPQVDIDNCVUIKNCXCRUABUTOQVRTMADNR PQKIDMRPQTCJRP
 VQIRZABDRPQHQSICICISDMRPQTQE QIRPVQIRZABXUASQTLZUAQIUKVCAVZXUARDNHTAQOXUVQKRPQMA DIRUIKHQVUNQRPNUCITRAQUNC
 IRPQNCCKXQDMRPQTQE QIRPVQIRZABRPQBQEDXEQK CIRDDVRUSDIXRDNHTAQSUAKCISRPQAQUTDIMDARPQVQTTURCIDMVDITRAZVRCDID
 MYQIODFDQIMZIRUCCVPCADTPCAUCTPCHQCQEQTRPURCRCTAQXURQKDRPQAQMDANTDMVADWIOACIVQTPDRDFZUIKTD SUIDZNUF DUTWQXX
 UTRPQQTRUHXCTPNQIRDMRPQIURCDIUXNUIZMUVRZACISTBTRQNWPCXQRPQMCIXUYQIODFDQIMZIU AQHQCQEQRD HQUTUNUNDZIRUCIFDM
 ZUIKADISNUVPCVXCNTRPURRPQFDMZIWQAQHZCXRCIRPQMCATRPXMDMRPQTQE QIRPVQIRZAB

Langkah Penyelesaian

Monalphabetic cipher rentan terhadap analisis frekuensi. Jika kriptanalisis mendapatkan frekuensi yang mendekati distribusi huruf pada kata-kata di Bahasa Inggris, sebuah *ciphertext* dapat diterka dengan mencocokkan monogram, bigram, bahkan hingga trigram.

Langkah-langkah yang dilakukan:

1. Lakukan penghitungan frekuensi monogram pada *ciphertext*.

```

uv run 21/solver.py
rank cipher cipher% -> plain plain%
1 Q 12.31 -> E 12.02
2 R 9.27 -> T 9.10
3 U 8.83 -> A 8.12
4 I 8.60 -> O 7.68
5 D 8.06 -> I 7.31
6 C 6.64 -> N 6.95
7 A 6.10 -> S 6.28
8 P 5.61 -> R 6.02
9 T 4.30 -> H 5.92
10 K 3.56 -> D 4.32
11 Z 3.56 -> L 3.98
12 M 3.07 -> U 2.88
13 V 2.89 -> C 2.71
14 X 2.81 -> M 2.61
15 N 2.73 -> F 2.30
16 S 2.01 -> Y 2.11
17 O 1.96 -> W 2.09
18 B 1.78 -> G 2.03
19 F 1.77 -> P 1.82
20 H 1.34 -> B 1.49
21 W 0.96 -> V 1.11
22 E 0.64 -> K 0.69
23 Y 0.59 -> X 0.17
24 G 0.39 -> Q 0.11
25 J 0.15 -> J 0.10
26 L 0.05 -> Z 0.07

```

Kita dapatkan pemetaan (kasar) berikut:

cipher	ABCDEFGHIJKLMNOPQRSTUVWXYZ
plain	SGNIKPQBOJZUFWRETYHACVMXL

2. Untuk memperkuat dasar pencocokan, hitung juga frekuensi bigram dan trigram pada *ciphertext* (`freq_ngram(ciphertext, n)` dengan $n = 1, 2, 3$), lalu bandingkan sepuluh besarnya dengan referensi Bahasa Inggris.

Monogram				
Peringkat	Cipher	Freq (%)	Referensi	Freq (%)
1	Q	12.31	E	12.02
2	R	9.27	T	9.10
3	U	8.83	A	8.12
4	I	8.60	O	7.68
5	D	8.06	I	7.31
6	C	6.64	N	6.95
7	A	6.10	S	6.28
8	P	5.61	R	6.02
9	T	4.30	H	5.92
10	K	3.56	D	4.32

Bigram				
Peringkat	Cipher	Freq (%)	Referensi	Freq (%)
1	RP	3.88	TH	3.56
2	PQ	2.99	HE	3.07
3	CI	2.21	IN	2.43
4	UI	1.83	ER	2.05
5	QI	1.80	AN	1.99
6	AQ	1.70	RE	1.85
7	IR	1.64	ON	1.76
8	QA	1.49	AT	1.49
9	DI	1.42	EN	1.45
10	IK	1.39	ND	1.35

Trigram				
Peringkat	Cipher	Freq (%)	Referensi	Freq (%)
1	RPQ	2.91	THE	1.81
2	UIK	1.03	AND	0.73
3	QIR	0.62	ING	0.72
4	MZI	0.62	ENT	0.42
5	DMR	0.61	ION	0.42
6	MRP	0.61	HER	0.36
7	IRP	0.59	FOR	0.34
8	CIS	0.56	THA	0.33
9	QAQ	0.52	NTH	0.33
10	KRP	0.51	INT	0.32

Bentuk distribusinya memang tidak identik urutan-per-urutan, tetapi tiga huruf/pasangan/tripel tersering pada *ciphertext* (Q, R, U dan RP, PQ dan RPQ) menempati rentang persentase yang sebanding dengan huruf/pasangan/tripel tersering Bahasa Inggris (E, T, A dan TH, HE dan THE). Hal ini konsisten dengan hipotesis awal bahwa *cipher*-nya adalah substitusi abjad-tunggal murni terhadap teks Bahasa Inggris, sehingga pencocokan berbasis frekuensi dapat dilanjutkan.

- Lakukan substitusi berdasarkan pemetaan (kasar) yang telah dibuat.

AUTESTREFNDONOEETOTRCEOTLSGMASYEHKAMEXEWIPIEOLUBOYAOTIAWWEASNOVEHTESQAWAOEJAFWMEHIUPAOHANSEYNIONO
CMLDEPIULOIUTIPLCRNNOHAPLSANOASAWSEUECTLSETREPIULOIUTHNLTHLXLGAFANOPNXYAVAPGITIWSEUECTLSETREPIULOIUI
LQNFATEXLGAFANOIPAGAFAPAGAFWSEUECTLSEAODTREPIULOIUNHRLTANNOPAOAULPLIPAWSEUECTLSEATTREHAFETNFETRE
SEVESEAMHIXEOWIPIEOLUNOCRLYIPLSEYNIOAODPGLHRLUISEJAFWMETREAOACNEOTTIFBHIUFIITHRNOAVANOTRECNTGUAHARN
PGITIWSEUECTLSEAODTREAOACNEOTTIFBHIUTIFNPAYANOTRECNTGIUPAGAFAAFIOTITRESHTREMASYEHTIOENHTRETITPLTIFAAOC
NEOTTIFBTRETITAMMEOYTRIUTREFILODNHABILTTVIRLODSEDENYRTGFETESHVRNCRNHFISETRAOTRSEETNFEHMASYESTRAOTREMA
SYEHTFILODTIFBIUABILTENYRTGFETESHNOTREMATESWESNIDIUTREGAGINWESNIDAODNTHASEAAODCAWACNTGASEUASYSEATESTR
AOTREMATTESAMTRILYRTREFILODTIFBHVNTRAUSIOTAODBACPCNSCLMASHRAWVESEAMSEADGHGATTESEDTRSILYRILTQAWAODLSN
OYTREGAGINWESNIDTREEASMGUISFHIUTREXEWIPIEOLUOVESEFIHTMGKESTNCAMCAKEHTGMEHTIOECRAFVBESHVNRBAFBIHRAW
DVIIDEOCIUUNOHVRNCRVESEDNUVESEOTUSIFTREFILODTIFBHDLSNOYTTREGAGINWESNIDFISEIKESEASMGXEWIPIEOLURADCCIFF
IOUEATLSEHHL CRAHAMIVESAODVNDUSIOTWASTCWFASEDTITREBACPCNSCLMASWASTAODTRECIFBNOATNIOADWIHNTNIOIUYSA
KEYIIDHVESETREHAFENOHNDRECIUUNOVESEQADEAODFNSSISHVRNMEILTHNDETRECIUUNOVESEAMASYEOLFBESIUTSNAOYLMASE
DYEDKNNOEBEAHTFNSSISHNSIOVEAWIOHAYSNCMLTMSAMAODUNHRNOYTIIMHAODHIIIOSEYASDNOYTRNHTANNCRNSIHRNSANHRNBEMN
EKEDTRATATTRATTNFEKASNILHUISCEHNOPNOIMEDBGTREGAFATANPNODYIFAODTREHETINOMADHEAUISEFEDAOAMNNOCEAODIBTA
NOEDNSIOAODKASNILHCLMTLSAMSEMNCUSIFPISEABGDEUEATNOYUISCEHHL CRAHOLPIPLAODTRENTIPNOYDIFHVRNCRCIOTSIMME
DTREYEOPANHEATOTREITRESAODGIHRNSIPIODIAMHIFEOTNIOEDTRATFIHTIUTREXEWIPIEOLUBLNMTIOMATMAODVESEVEMMI
SYAONXEDAODTREFILODHVESEAMHIBLNMTMAGESBGMAGESAUTESCIOHIMNDATNIOCIOHNDENSOYTREHMIWEAODASSAOYEFEOIUTRE
FILODHNCNSCLMASFILODHVESECIOHTSLCTEDNOADDNTNIOBSIOXEFNSSISHUSIFFANOMAODCRNOAAODTREVAQNODEOUSIFTREHAFEW
ESNIDNODNCATETRATTRESEVAHAOEJCRAOYEBETVEEOTRETVICILOTSNEHATTRATTNFETRESEUISENTNHHWECLMATEDTRATTREEASM
GAWWEASAOCEIUTREXEWIPIEOLUOFAGRAKEBEEONUOMLEOCEDBGCNKNMEOYNOEESNOYTECROIMIYGUSIFFANOMAODCRNOAAODTREW
IMNTNCAMAODNDEIMIYNCAMNOUMLEOCENHSEUMECTEDNOTREQINOTCIOHTSLCTNIOIUTREXEWIPIEOLUBGKASNILHAMNNOCEFEF
BESHAWHSIIUIUTREAMNNOCEATTREHAFETNFETRESEASEAMHITIFBHNKEUILSAODTRSEEIUTREPAFEOLFONOTREUSIOTIUTRERIL

GLAOTIFBNOEAHTESQAWAODLSNOYTREHAFEWESNIDAHNRNBLTANPIULOAFIOYTREFTIFBUNKENHTREEASMNEHTPIULONOEAHTESO
 QAWAOHRNHTISGNODNCATNOYAWIHHNBMECIOECTNIOBETVEOTREPAXLHAWSIKNOCEAODTREGAFATANPNODYIFATTRATTNFENOTRE
 UILSTRCEOTLSGRIPLSNPLSEYNIOPATISEYNIOTIPANSEYNI OAODTIRIPLSEYNI OVESEFANOMGCIFWIHEDIUUSIOTAODSEASTIFBH
 VRNCRYSADLAMMBECAFEXEOWIPIEOULOTVITITRSEYEYESATNIOHMATESSEYASDNOYTRNHGAFATITAPESLWINOTEDILTTRATAMTR
 ILYRTREPIQNPNAODONRIOHRIPNFEOTNIOEDTRATTREUILSYEYESAMH AODQAWAOEHETAPEXLOAODITRESHRADFESYEDEAHTQAWAONO
 TITRETESNTISGIUTREQAWAOEHEFIOASCRGTRSILYRFLMTNWMEJWEDNTNIOHREBEMNEKEDTRATNTVAHACTLAMMGLOLOIPLONMIC
 ATEDNOTREOIBNWMANOTRATUISFEDAOAMMNAOCENOEAHTQAWAOFANOMGCIOHNHTNOYIUSEASYSAKEHNOTREWAHTAODTRATAUTESTRE
 DEATRIURNFPITREGMIHTTREBATTMEVNTRTREGAFATANPNODYIFISOEYITNATEDWEACELODESTREMEADESHRNWIUTREGAFATIPNOY
 HRNWNOTREODTREAMMNAOCEIUTREDIYHMAKEPNOYDIFAODTREAMMNAOCEIUTREKNMRISHEPNOYDIFESYEDNOTIOEUISFNOYTRE
 QAWAOEHEFIOASCRGTREOTREMEADESHIUEAHTAODVEHTQAWAOVESEDNHTNOYLNHREDBGTREXEOWIPIEOULO AODTREUSIOTAODSEAST
 IFBHUSIFTREEODIUTREUILSTRCEOTLSGTITREUNTRCEOTLSGMASYEXEOWIPIEOULOSEHEFBMNOYPIULONOFILOTTANDABEYAOITIA
 WWEASNOWSEHEOTDAGYLOFAWSEUECTLSEMASYEXEOWIPIEOULORAKEAMHIBEEOBLNMTNOWMACEHLCRAHGAF AOAHRNWSEUECTLSENB
 ASAPNWSEUECTLSEAODCRNBWSEUECTLSEAUTESTREFN DUILSTRCEOTLSGTREDAVATIFBHSYNNOAMMGCEOTESEDASILODTREGAFAT
 IPULONOTREHILTREAHITUTREOASABAHNOAODTRESGLBEOPULYOYADLAMMGHRNUTEDTITREDANHEOPIULONOTREOISTRTREMASYE
 XEOWIPIEOULOIUDAVAVAHTREULSLNCRNPULOCMLHTESAODFXLTIFBHTRECEOTESIUYSAKNTGIUTREYSEATPNODYITFBHRNUTEDA
 YANONOTREUNTRCEOTLSGTITREAOCEOTCNTGTIFBYSILWAODTRERLODSEDTIOYLEDNBSDTIFBYSILWNOTREIHAPAWMANOSEYASDN
 OYTRESEAHIOHUISTREFLMTNWMETSAOHUESHIUTREHEMASYEXEOWIPIEOULOB AHEDIOSEHEASCRITOTREPLHRNQNAODONRIOHRLQNTR
 EQAWAOEHEACADEFNCCIFFLONTGRAHWSIWIHEDKNEVHHL CRAHTRETREISGIUDGOAHTGAMTESOATNIOAODTRETREISGIEZLEHTSAOT
 SNBEHCIOZLESNOYDGOAHTNEHTIEJWMANOIOTREITRESRAODRNSIHEPAXLWSIWIHEDTRATTREHTSLCTLSEHIUTREHAHAPNPIQHNRN
 XLPAODFAPNOIAOCNEOTTIFBYSILWHVESEHNFNMAS AODUSIFTREMAUEUILSTRCEOTLSGTITREHECIODRAMUIUTREUNTRCEOTLSGT
 REREAHIUTREUILSAOCNEOTTIFBYSILWHQINOTMGWMAGEDASIMENOTREGAFATIPNOYHRNWAUTESDEATRREGDENUNEDAHTREYLASD
 NAODENTGIUTREGAFATISIGAMWIVESAODAMHIWSIWIHEDTRATTRECIOTSLCTNIOIUTREMASYEXEOWIPIEOULORADADETESSEOTEUU
 ECTIOKASNILHSEYNIOHATTREHAFETNFENOTREHECIODRAMUIUTREUNTRCEOTLSGOIMASYEH CAMEXEOWIPIEOULOVESECIOHTSLCT
 EDNOASEAHHL CRAHPNBNWSIKNOCELWESFAIGE AODRGLYAWSIKNOCEAODTREHCAMEIUTREXEOWIPIEOULON OYIPNHRNCRNDIYSADLA
 MMGDECSEAHEDACCISDNOYTTIRENOHCSNWTNIOHQNTRENSIOHVISDLOEASTREDUSIFTRENOAVAFILOTANOPIULO AODTREMASYE
 HVISDLOEASTREDUSIFTREETAULOAGAFAPIULOEFWESISGLSGAPLATTRATTNFSEYASDEDRNFHEMUAHTREPNOYIUTREVISMADOARNH
 ALTRISNTGVAHSEUMECTEDNOTREUACTTRATIOMGRECIOTNOLEDTIBLNDMASYEXEOWIPIEOULONOTREQAWAOEHEASCRNWEMAYINOTR
 EHNJTRCEOTLSGTRERISNXIOTAMCAKEHTGMEHTIOECRAFBSIUTREPNOANHGHTEFBEYAOITBEWIWLMASNXEDNOXEOWIPIEOULOATTR
 EHAFETNFETREGBEYAOITIAWWEASNOASEAHHL CRAHHETTHLSIKNOCEIVASNWSIKNOCELQNWSIKNOCEAODEKEONOPATISEYNIORVNC
 RVESECIOHNDESEDAHBAHEHUISTREEHTABMNRHFEOTIUTREEFWESISPENTANAFIOYTREFLEOIRADAWASTNCLMASMGMASYEO LFBESI U
 XEOWIPIEOULOCIFWASEDTIIOGMTRNSTGONENOPIXLPEWSIKNOCEDLSNOYTREHAFEWESNIDLEOIAMIOERADONOTGHEKEOVRNMENO
 PAOTITRESEVESETVIRLODSEDH NJTEOAHBITRPNIOAODINXRAOYDECSEAHED IOMGTREPAOTISEYNIOCIOTNOLEDTICIOHTSLCTMAS
 YEXEOWIPIEOULOTRSILYRILTTRHNJTRCEOTLSGNODNCATNOYTRATTREGAFATIFIOASCRGATTRATTNFSEMNEDREAKNMGIOTREEAH
 TESOPNOYDIFNOBITRECIOIFNCAODFMNTASGAHWEC THUSIFTREEODIUTREHNJTRCEOTLSGTITREBEYNOON OYIUTREHEKEOTRCEOTL
 SGMASYEHZLASEAODCNSCLMASTIFBHSEWMACEDTREUSIOTAODBECAFETREFANOHTSEAFNOTREFNDMEIUTREHEKEOTRCEOTLSGTREG
 EKIMKEDNOTIICTAYIOAMTIFBHSEYASDNOYTRESEAHIOUISTRECEHATNIOIUCIOHTSLCTNIOIUXEOWIPIEOULOTANNCRNSIHRNSAN
 HRNBEMNEKEHTRATNTNHSEMATEDTITRESEUISFHIUCSIVOWSNOCEHRITIPLAODHIYA OILFAPIAHVEMMAHTREEHTABMNRHFEOTIUTRE
 OATNIOAMFAOLUACTLSNOYHGHTFVRNMETREUNOAMXEOWIPIEOULOASEBEMNEKEDTIBEAHAFAFILOTANOPIULO AODSIOYFACRNCMAN
 FHTRATTREPIULOVESEBLNMTNOTREUNSHTRAMUIUTREHEKEOTRCEOTLSG

4. Cocokkan melalui informasi kecocokan bigram dan trigram, lalu lanjutkan dengan iterasi yang sama: setiap kali sebuah bigram/trigram/kata umum Bahasa Inggris mulai terlihat pada hasil dekripsi semantara, huruf *cipher* yang bersangkutan diganti agar sesuai. Tiga huruf – R, Q, U – sudah benar sejak tebakan awal (langkah 1) dan tidak pernah diubah lagi. Untuk sisanya, dapat dilihat pada tabel di bawah yang disusun berdasarkan *log* dari *src/21/log/plain/* secara kronologis.

Ganti	Konteks (sebelum → sesudah)	Petunjuk
P→H	AUTESTREFNDONOETEEOTR → AUTESTHEFNDONOETEEOTH	delapan karakter pertama UMRQAR PQ → AUTESTHE, trigram THE
I→N	UTESTHEFNDONOETEEOTHCEOTL UTESTHEFNDNNTEENTHCENTL →	mengarah ke NINETEENTH
C→I	AUTESTHEFNDNNTEENTHCEN AUTESTHEFIDNINETEENTHCEN →	kata NINETEENTH terbentuk
A→R	AUTESTHEFIDNINETEEN → AUTERTHEFIDNINETEEN	bigram ER
Z→U	TEENTHCENTLSGMASYEHCAMEXE TEENTHCENTURYMARYEHCAMEXE →	mengarah ke CENTURY
B→Y	ENTHCENTLSGMASYEHCAMEXENW ENTHCENTURYMARYEHCAMEXENW →	kata CENTURY lengkap
M→F	AUTERTHEFIDNINET → AFTERTHEFIDNINET	kata umum AFTER THE
N→M	AFTERTHEFIDNINETEENTHCE AFTERTHEMIDNINETEENTHCE →	kata MID (AFTER THE MID-...)
D→O	EHCAMEXENPIPIENFUNBEYANTI EHCAMEXENPOPOENFUNBEYANTO →	mengarah ke ZENPOKOENFUN
T→S	NTURYMARSEHCAMEXENPOPOENF NTURYMARYESCAEXENPOPOENF →	bigram ES
F→K	SCAPEXENPOPOENFUNBEYANTOA SCAPEXENPOKOENFUNBEYANTOA →	mengarah ke ZENPOKOEN
S→G	CENTURYPARYESCAPEXENPOKOE CENTURYPARGESCAPEXENPOKOE →	kata LARGE
O→P	GESCAPEXENLOKOENFUNBEGANT GESCAPEXENPOKOENFUNBEGANT →	kata ZENPOKOEN
X→L	NTHCENTURYPARGESCAPEXENPO NTHCENTURYLARGESCALEXENPO →	kata LARGE SCALE
W→W	TOAPPEARINGESTERNQAPANEJA TOAPPEARINWESTERNJAPANEXA →	kata IN WESTERN
G→J	RINGESTERNQAPANEJAMPLESOF RINWESTERNJAPANEXAMPLESOF →	kata JAPAN
J→X	TERNQAPANEJAMPLESOFKANSAI TERNJAPANEXAMPLESOFKANSAI →	kata EXAMPLES
Y→Z	LARGESCALEXENPOKOENFUNBEG LARGESCALEZENPOKOENFUNBEG →	istilah ZENPOKOENFUN terbentuk
E→V	WEREMOSTLYKERTICALCAKESTY WEREMOSTLYVERTICALCAVESTY →	kata VERTICAL CAVE
L→Q	LARGESZUAREANDCIR → LARGESQUAREANDCIR	kata SQUARE

Artefak

Skrip *solver* yang digunakan untuk menghitung frekuensi monogram/bigram/trigram, menyusun tebakan awal, menerapkan pemetaan hasil analisis, serta mencatat setiap hasil dekripsi ke log/plain/ dan metadatanya (pemetaan & frekuensi n-gram) ke log/metadata/. Modul ini bergantung pada `src/frequencies.py` yang berisi tabel referensi frekuensi Bahasa Inggris (dipakai bersama untuk seluruh soal substitusi/Vigenère pada laporan ini, sehingga isinya tidak diduplikasi di sini).

```

1 import json
2 import os
3 import sys
4 from datetime import datetime
5 from typing import Dict, List
6
7 sys.path.insert(0, os.path.dirname(os.path.dirname(os.path.abspath(__file__))))
8 from frequencies import MONOGRAM, MONOGRAM_ORDER
9
10 CORRECTIONS = [
11     ("R", "T"), ("P", "H"), ("Q", "E"), ("I", "N"), ("C", "I"),
12     ("U", "A"), ("Z", "U"), ("A", "R"), ("B", "Y"), ("M", "F"),
13     ("O", "P"), ("W", "W"), ("N", "M"), ("D", "O"), ("T", "S"),
14     ("X", "L"), ("G", "J"), ("F", "K"), ("S", "G"), ("J", "X"),
15     ("Y", "Z"), ("E", "V"), ("L", "Q"),
16 ]
17

```

```

18 class Solver:
19     def count(self, string: str) -> Dict:
20         data = {}
21         for c in string:
22             data[c] = data.get(c, 0) + 1
23         return dict(sorted(data.items(), key=lambda item: item[1], reverse=True))
24
25     def sort(self, string: str) -> List:
26         return list(self.count(string).keys())
27
28     def freq_ngram(self, string: str, n: int = 1) -> Dict:
29         freq_data = {}
30         total_ngrams = len(string) - n + 1
31         for i in range(total_ngrams):
32             ngram = string[i:i+n]
33             freq_data[ngram] = (freq_data.get(ngram, 0) + 1)
34         for ngram in freq_data:
35             freq_data[ngram] = (freq_data[ngram] / total_ngrams * 100)
36         return dict(sorted(freq_data.items(), key=lambda item: item[1], reverse=
True))
37
38     def solve(self, string: str) -> Dict:
39         cipher_order = [c for c in self.sort(string) if c.isalpha()]
40         mapping = {}
41         for cipher_letter, plain_letter in zip(cipher_order, MONOGRAM_ORDER):
42             mapping[cipher_letter] = plain_letter
43         return mapping
44
45     def replace_char(self, mapping: Dict, char0: str, char1: str) -> Dict:
46         mapping[char0] = char1
47         return mapping
48
49     def decrypt(self, string: str, mapping: Dict) -> str:
50         return "".join(mapping.get(c, c.lower()) for c in string)
51
52     def table(self, string: str) -> str:
53         cipher_freq = self.freq_ngram(string)
54         rows = ["rank cipher cipher% -> plain plain%"]
55         for i, cipher_letter in enumerate(cipher_freq):
56             if not cipher_letter.isalpha():
57                 continue
58             plain_letter = MONOGRAM_ORDER[i] if i < len(MONOGRAM_ORDER) else "-"
59             plain_pct = MONOGRAM.get(plain_letter, 0.0)
60             rows.append(
61                 f"{i+1:>4} {cipher_letter:^6} {cipher_freq[cipher_letter]:>6.2f}
-> "
62                 f"{plain_letter:^5} {plain_pct:>5.2f}"
63             )
64         return "\n".join(rows)
65
66     def load_ciphertext(base_dir: str) -> str:
67         with open(os.path.join(base_dir, "s.txt")) as f:
68             return f.read().strip()
69
70     def write_plain(base_dir: str, timestamp: str, decrypted: str) -> str:
71         path = os.path.join(base_dir, "log", "plain", timestamp + ".txt")
72         os.makedirs(os.path.dirname(path), exist_ok=True)
73         with open(path, "w") as f:
74             f.write(decrypted)

```

```

75     return path
76
77 def write_metadata(base_dir: str, timestamp: str, mapping: Dict, freq: Dict) -> str
78 :
79     path = os.path.join(base_dir, "log", "metadata", timestamp + ".json")
80     os.makedirs(os.path.dirname(path), exist_ok=True)
81     payload = {
82         "timestamp": datetime.now().isoformat(timespec="seconds"),
83         "mapping": mapping,
84         "freq_ngram": {str(n): freq[n] for n in freq},
85     }
86     with open(path, "w") as f:
87         json.dump(payload, f, indent=2)
88     return path
89
90 if __name__ == "__main__":
91     base_dir = os.path.dirname(os.path.abspath(__file__))
92     content = load_ciphertext(base_dir)
93
94     solver = Solver()
95     mapping = solver.solve(content)
96     for cipher_letter, plain_letter in CORRECTIONS:
97         mapping = solver.replace_char(mapping, cipher_letter, plain_letter)
98
99     decrypted = solver.decrypt(content, mapping)
100     freq = {n: solver.freq_ngram(content, n) for n in (1, 2, 3)}
101
102     timestamp = datetime.now().strftime("%Y-%m-%d_%H%M%S")
103     plain_path = write_plain(base_dir, timestamp, decrypted)
104     metadata_path = write_metadata(base_dir, timestamp, mapping, freq)
105
106     print("mapping (cipher -> plain):")
107     print("".join(sorted(mapping)))
108     print("".join(mapping[c] for c in sorted(mapping)))
109     print()
110     print("hasil dekripsi:")
111     print(decrypted)
112     print()
113     print(f"[tersimpan ke {os.path.relpath(plain_path)} dan {os.path.relpath(
114         metadata_path)}]")

```

src/21/solver.py

Plaintext Hasil Dekripsi

AFTER THE MIDNINETEENTH CENTURY LARGE SCALE ZENPO KOEN FUN BEGAN TO APPEAR IN WESTERN JAPAN. EXAMPLES OF KANSAI REGION INCLUDE KOFUN OF TOKUCHI IN SAKURAI NARA PREFECTURE, THE KOFUN OF TSUJITSU, YAMAIKIZUGAWA KYOTO PREFECTURE, THE KOFUN OF FUJIMATEZUYAMA IN OKAYAMA OKAYAMA PREFECTURE AND THE KOFUN OF ISHIBUTAI IN KANFUKU OKAPREFECTURE AT THE SAME TIME THERE WERE ALSO ZENPO KOEN FUN IN CHUGOKU REGION AND KYUSHU. FOR EXAMPLE THE ANCIENT TOMBS OF MOTOSHI IN NAWA IN THE CITY OF ASAHIKYOTO PREFECTURE AND THE ANCIENT TOMBS OF TOMIKAGA IN THE CITY OF OKAYAMA AMONG OTHERS. THE LARGEST ONE IS THE TOKUTOMA ANCIENT TOMB. THE TOTAL LENGTH OF THE MOUND IS ABOUT TWO HUNDRED EIGHTY METERS, WHICH IS MORE THAN THREE TIMES LARGER THAN THE LARGEST MOUND TOMB OF ABOUT EIGHTY METERS IN THE LATER PERIOD OF THE YAYOI PERIOD AND ITS AREA AND CAPACITY ARE FAR GREATER THAN THE LATTER. ALTHOUGH THE MOUND TOMBS WITH A FRONT AND BACK CIRCULAR SHAPE WERE ALREADY SCATTERED THROUGHOUT JAPAN DURING THE YAYOI PERIOD, THE EARLY FORMS OF THE ZENPO KOEN FUN WERE MOSTLY VERTICAL CAVE STYLE STONE CHAMBERS WITH BAMBOO SHAPED WOODEN COFFINS WHICH WERE DIFFERENT FROM THE MOUND TOMBS DURING THE YAYOI PERIOD. MORE OVER EARLY ZENPO KOEN FUN HAD COMMON FEATURES SUCH AS A LOWER AND WIDER FRONT PART COMPARED TO THE BACK CIRCULAR PART AND THE COMBINATION AND POSITION OF GRAVE GOODS WERE THE SAME INSIDE THE COFFIN. THERE WERE JADE AND MIRRORS WHILE

EOUTSIDETHECOFFINWEREALARGENUMBEROFTRIANGULAREGDEDIVINEBEASTMIRRORSIRONWEAPONSAGRICULTURALANDFISHINGTOOLSANDSOONREGARDINGTHISAIICHIROSHIRAIISHIBELIEVEDTHATATTHATTIMEVARIOUSFORCESINKINOLEDBYTHEYAMATAIKINGDOMANDTHESETOINLANDSEAFORMEDANALLIANCEANDOBTAINEDIRONANDVARIOUSCULTURALRELICSFROMKOREABYDEFEATINGFORCESSUCHASNOKUKUANDTHEITOKINGDOMSWHICHCONTROLLEDTHEGENKAISEAONTHEOTHERHANDYOSHIROKONDOALSOMENTIONEDTHATMOSTOFTHEZENPOKOENFUNBUILT ON FLATLAND WERE WELL ORGANIZED AND THE MOUNDS WERE ALSO BUILT LAYER BY LAYER AFTER CONSOLIDATION CONSIDERING THE SLOPE AND ARRANGEMENT OF THE MOUNDS CIRCULAR MOUNDS WERE CONSTRUCTED IN ADDITION BRONZE MIRRORS FROM MAINLAND CHINA AND THE WAJI N D E N FROM THE SAME PERIOD INDICATE THAT THERE WAS AN EXCHANGE BETWEEN THE TWO COUNTRIES AT THAT TIME THEREFORE IT IS SPECULATED THAT THE EARLY APPEARANCE OF THE ZENPOKOEN FUN MAY HAVE BEEN INFLUENCED BY CIVIL ENGINEERING TECHNOLOGY FROM MAINLAND CHINA AND THE POLITICAL AND IDEOLOGICAL INFLUENCE IS REFLECTED IN THE JOINT CONSTRUCTION OF THE ZENPOKOEN FUN BY VARIOUS ALLIANCE MEMBERS AS PROOF OF THE ALLIANCE AT THE SAME TIME THERE ARE ALSO TOMBS FIVE FOUR AND THREE OF THE KAMEN MUN IN THE FRONT OF THE HOUYUAN TOMB IN EASTERN JAPAN DURING THE SAME PERIOD AS ISHIBUTA IKO FUN AMONG THEM TOMB FIVE IS THE EARLIEST KO FUN IN EASTERN JAPAN HIS TORY INDICATING A POSSIBLE CONNECTION BETWEEN THE KAZUS A PROVINCE AND THE YAMATAI KINGDOM AT THAT TIME IN THE FOURTH CENTURY HOKURIKU REGION KANTO REGION TOKAI REGION AND TOHOKU REGION WERE MAINLY COMPOSED OF FRONT AND REAR TOMBS WHICH GRADUALLY BECAME ZENPOKOEN FUN TWO TO THREE GENERATIONS LATER REGARDING THIS YAMATO TAKERU POINTED OUT THAT ALTHOUGH THE KOJIKI AND NIHON SHOKI MENTIONED THAT THE FOUR GENERALS AND JAPANESE TAKEZUN AND OTHERS HAD MERGED EAST JAPAN INTO THE TERRITORY OF THE JAPANESE MONARCHY THROUGH MULTIPLE EXPEDITIONS SHE BELIEVED THAT IT WAS ACTUALLY KUNUNOKUNI LOCATED IN THE NOBI PLAIN THAT FORMED AN ALLIANCE IN EAST JAPAN MAINLY CONSISTING OF REAR GRAVES IN THE PAST AND THAT AFTER THE DEATH OF HIMIKO THEY LOST THE BATTLE WITH THE YAMATAI KINGDOM OR NEGOTIATED PEACE UNDER THE LEADERSHIP OF THE YAMATO KINGSHIP IN THE END THE ALLIANCE OF THE DOGSLAVE KINGDOM AND THE ALLIANCE OF THE EVIL HORSE KINGDOM MERGED INTO ONE FORMING THE JAPANESE MONARCHY THEN THE LEADERS OF EAST AND WEST JAPAN WERE DISTINGUISHED BY THE ZENPOKOEN FUN AND THE FRONT AND REAR TOMBS FROM THE END OF THE FOURTH CENTURY TO THE FIFTH CENTURY LARGE ZENPOKOEN FUN RESEMBLING KO FUN IN MOUNT TAIDABEGAN TO APPEAR IN PRESENT DAY GUNMA PREFECTURE LARGE ZENPOKOEN FUN HAVE ALSO BEEN BUILT IN PLACES SUCH AS YAMANASHI PREFECTURE IBARAKI PREFECTURE AND CHIBI PREFECTURE AFTER THE MID FOURTH CENTURY THE DAWA TOMBS ORIGINALLY CENTERED AROUND THE YAMATO KO FUN IN THE SOUTHEAST OF THE NARABAS IN AND THE YUBEN KO FUN GRADUALLY SHIFTED TO THE DAI SEN KO FUN IN THE NORTH THE LARGE ZENPOKOEN FUN OF DAWA WAS THE FURUICHI KO FUN CLUSTER AND MOZUTO MB THE CENTER OF GRAVITY OF THE GREAT KING TOMBS SHIFTED AGAIN IN THE FIFTH CENTURY TO THE ANCIENT CITY TOMB GROUP AND THE HUNDRED TONGUE DBIRD TOMB GROUP IN THE OSAKA PLAIN REGARDING THE REASONS FOR THE MULTIPLE TRANSFERS OF THESE LARGE ZENPOKOEN FUN BASED ON RESEARCH ON THE KUSHIJI AND NIHON SHUJI THE JAPANESE ACADEMIC COMMUNITY HAS PROPOSED VIEWSSUCH AS THE THEORY OF DYNASTY ALTERNATION AND THE THEORY OF EQUESTRIAN TRIBES CONQUERING DYNASTIES TO EXPLAIN ON THE OTHER HAND HIROSE KAZUO PROPOSED THAT THE STRUCTURES OF THE SAKI KOJISHI ZUKI AND MAKINO ANCIENT TOMB GROUPS WERE SIMILAR AND FROM THE LATE FOURTH CENTURY TO THE SECOND HALF OF THE FIFTH CENTURY THE HEADS OF THE FOUR ANCIENT TOMB GROUPS JOINTLY PLAYED A ROLE IN THE YAMATO KINGSHIP AFTER DEATH THEY DEIFIED AS THE GUARDIAN DEITY OF THE YAMATO ROYAL POWER AND ALSO PROPOSED THAT THE CONSTRUCTION OF THE LARGE ZENPOKOEN FUN HAD A DETERRENT EFFECT ON VARIOUS REGIONS AT THE SAME TIME IN THE SECOND HALF OF THE FIFTH CENTURY NO LARGE SCALE ZENPOKOEN FUN WERE CONSTRUCTED IN AREAS SUCH AS KIBI PROVINCE UPPER MAOE AND HYUGA PROVINCE AND THE SCALE OF THE ZENPOKOEN FUN IN GOKISHI CHIDO GRADUALLY DECREASED ACCORDING TO THE INSCRIPTIONS ON JIJITHE IRON SWORD UNEARTHED FROM THE INAWA MOUNTAIN KO FUN AND THE LARGE SWORD UNEARTHED FROM THE ETAFUNAYAMA KO FUN EMPEROR YURYAKU AT THAT TIME REGARDED HIMSELF AS THE KING OF THE WORLD AND HIS AUTHORITY WAS REFLECTED IN THE FACT THAT ONLY HE CONTINUED TO BUILD LARGE ZENPOKOEN FUN IN THE JAPANESE ARCHIPELAGO IN THE SIXTH CENTURY THE HORIZONTAL CAVE STYLE STONE CHAMBER OF THE KINAI SYSTEM BEGAN TO BE POPULARIZED IN ZENPOKOEN FUN AT THE SAME TIME THEY BEGAN TO APPEAR IN AREAS SUCH AS SETSU PROVINCE OWARI PROVINCE UJIPROVINCE AND EVEN IN KANTO REGION WHICH WERE CONSIDERED AS BASES FOR THE ESTABLISHMENT OF THE EMPEROR KEITAI AMONG THEM UENO HAD A PARTICULARLY LARGE NUMBER OF ZENPOKOEN FUN COMPARED TO ONLY THIRTY NINE IN KOZUKE PROVINCE DURING THE SAME PERIOD UENO ALONE HAD NINETY SEVEN WHILE IN KANTO THERE WERE TWO HUNDRED SIXTEEN AS BOTH KINO AND OIZHANG DECREASED ONLY THE KANTO REGION CONTINUED TO CONSTRUCT LARGE ZENPOKOEN FUN THROUGHOUT THE SIXTH CENTURY INDICATING THAT THE YAMATO MONARCHY AT THAT TIME RELIED HEAVILY ON THE EASTERN KINGDOM IN BOTH ECONOMIC AND MILITARY ASPECTS FROM THE END OF THE SIXTH CENTURY TO THE BEGINNING OF THE SEVENTH CENTURY LARGE SQUARE AND CIRCULAR TOMBS REPLACED THE FRONT AND BECAME THE MAINSTREAM IN THE MIDDLE OF THE SEVENTH CENTURY THEY EVOLVED INTO OCTAGONAL TOMBS REGARDING THE REASON FOR THE CESSATION OF CONSTRUCTION OF ZENPOKOEN FUN AIICHIROSHIRAIISHIBELIEVES THAT IT IS RELATED TO THE REFORMS OF CROWN PRINCESHOTO KU AND SOGANO UMAKO AS WELL AS THE ESTABLISHMENT OF THE NATIONAL MANUFACTURING SYSTEM WHILE THE FINAL ZENPOKOEN FUN ARE BELIEVED TO BE AS A MOUNTAIN KO FUN AND RONGMACHI CLAIM THAT THE KO FUN WERE BUILT IN THE FIRST HALF OF THE SEVENTH CENTURY

Plaintext dengan Format Asli

After the mid-19th century, large-scale Zenpokoefun began to appear in Western Japan. Examples of Kansai region include Kofun of Tokuchi in Sakurai, Nara Prefecture, the Kofun of Tsui Otsuzuyama in Kizugawa, Kyoto Prefecture, the Kofun of Ujima Tezuyama in Okayama, Okayama Prefecture, and the Kofun of Ishibutai in Kanda, Fukuoka Prefecture. At the same time, there were also Zenpokoefun in Chūgoku region and Kyushu. For example, the ancient tombs of Motoshi Inawa in the city of Asahi, Kyoto Prefecture, and the ancient tombs of Tomikaga in the city of Okayama, among others, the largest one is the Tokutoma ancient tomb. The total length of the mound is about 280 meters, which is more than three times larger than the largest mound tomb of about 80 meters in the later period of the Yayoi period, and its area and capacity are far greater than the latter. Although the mound tombs with a front and back circular shape were already scattered throughout Japan during the Yayoi period, the early forms of the Zenpokoefun were mostly vertical cave-style stone chambers with bamboo-shaped wooden coffins, which were different from the mound tombs during the Yayoi period. Moreover, early Zenpokoefun had common features, such as a lower and wider front part compared to the back circular part, and the combination and position of grave goods were the same. Inside the coffin were jade and mirrors, while outside the coffin were a large number of triangular edge divine beast mirrors, iron weapons, agricultural and fishing tools, and so on. Regarding this, Taiichiro Shiraishi believed that at that time, various forces in Kinki, led by the Yamatai Kingdom, and the Seto Inland Sea formed an alliance, and obtained iron and various cultural relics from Korea by defeating forces such as Nukoku and the Ito Kingdoms, which controlled the Genkai Sea. On the other hand, Yoshiro Kondo also mentioned that most of the Zenpokoefun built on flat land were well organized, and the mounds were also built layer by layer after consolidation. Considering the slope and arrangement of the mounds, circular mounds were constructed. In addition, bronze mirrors from mainland China and the "Wajinden" from the same period indicate that there was an exchange between the two countries at that time. Therefore, it is speculated that the early appearance of the Zenpokoefun may have been influenced by civil engineering technology from mainland China, and the political and ideological influence is reflected in the joint construction of the Zenpokoefun by various alliance members as proof of the alliance.

At the same time, there are also tombs 5, 4, and 3 of the Kamen Mun in the front of the Houyuan Tomb in Eastern Japan during the same period as Ishibutai Kofun. Among them, tomb 5 is the earliest Kofun in Eastern Japan's history, indicating a possible connection between the Kazusa Province and the Yamatai Kingdom at that time. In the 4th century, Hokuriku region, Kantō region, Tōkai region and Tōhoku region were mainly composed of front and rear tombs, which gradually became Zenpokoefun two to three generations later. Regarding this, Yamato Takeru pointed out that although the "Kojiki" and "Nihon Shoki" mentioned that the Four Generals and Japanese Takezumi and others had merged East Japan into the territory of the Japanese monarchy through multiple expeditions, he believed that it was actually Kununokuni located in the Nōbi Plain that formed an alliance in East Japan mainly consisting of rear graves in the past and that after the death of Himiko, they lost the battle with the Yamatai Kingdom or negotiated peace under the leadership of the Yamato Kingship. In the end, the alliance of the Dog Slave Kingdom and the alliance of the Evil Horse Kingdom merged into one, forming the Japanese monarchy. Then, the leaders of East and West Japan were distinguished by the Zenpokoefun and the front and rear tombs. From the end of the 4th century to the 5th century, large Zenpokoefun resembling Kofun in Mount Taida began to appear in present-day Gunma Prefecture. Large Zenpokoefun have also been built in places such as Yamanashi Prefecture, Ibaraki Prefecture, and Chiba Prefecture.

After the mid-4th century, the Dawa Tombs, originally centered around the Yamato Kofun in the southeast of the Nara Basin and the Ryuben Kofun, gradually shifted to the Daisen Kofun in the north. The large Zenpokoefun of Dawa was the Furuichi Kofun Cluster and Mozu Tombs. The center of gravity of the Great King's Tomb shifted again in the 5th century to the ancient city tomb group and the hundred-tongued bird tomb group in the Osaka Plain. Regarding the reasons for the multiple transfers of these large Zenpokoefun, based on research on the "Kushiji" and "Nihonshu Ji", the Japanese academic community has proposed views such as the theory of dynasty alternation and the theory of equestrian tribes conquering dynasties to

explain. On the other hand, Hirose Kazuo proposed that the structures of the Sasaki, Koji, Shizuki, and Makino ancient tomb groups were similar, and from the late 4th century to the second half of the 5th century, the heads of the four ancient tomb groups jointly played a role in the Yamato Kingship. After death, they deified as the guardian deity of the Yamato royal power and also proposed that the construction of the large Zenpokoefun had a deterrent effect on various regions. At the same time, in the second half of the 5th century, no large-scale Zenpokoefun were constructed in areas such as Kibi Province, Upper Maoye, and Hyūga Province, and the scale of the Zenpokoefun in Gokishichidō gradually decreased. According to the inscriptions on Ji Ji, the iron sword unearthed from the Inawa Mountain Kofun, and the large sword unearthed from the Eta Funayama Kofun, Emperor Yūryaku at that time regarded himself as the king of the world, and his authority was reflected in the fact that only he continued to build large Zenpokoefun in the Japanese archipelago.

In the 6th century, the horizontal cave-style stone chamber of the Kinai system began to be popularized in Zenpokoefun. At the same time, they began to appear in areas such as Settsu Province, Owari Province, Uji Province, and even in Kantō region, which were considered as bases for the establishment of the Emperor Keitai. Among them, Ueno had a particularly large number of Zenpokoefun, compared to only 39 in Kōzuke Province during the same period. Ueno alone had 97, while in Kanto there were 216. As both Kino and Oizhang decreased, Only the Kanto region continued to construct large Zenpokoefun throughout the 6th century, indicating that the Yamato monarchy at that time relied heavily on the Eastern Kingdom in both economic and military aspects. From the end of the 6th century to the beginning of the 7th century, large square and circular tombs replaced the front and became the mainstream. In the middle of the 7th century, they evolved into octagonal tombs. Regarding the reason for the cessation of construction of Zenpokoefun, Taiichiro Shiraishi believes that it is related to the reforms of Crown Prince Shōtoku and Soga no Umako, as well as the establishment of the national manufacturing system, while the final Zenpokoefun are believed to be Asama Mountain Kofun, and Rongmachi claims that tKofun were built in the first half of the 7th century.

Sumber:

- <https://en.wikipedia.org/wiki/Zenp%C5%8Dk%C5%8Denfun>

Kunci

cipher	ABCDEFGHIJKLMNOPQRSTUVWXYZ
plain	RYIOVKJBXNDQFMPHETGSACWLZU

Referensi

Tabel referensi frekuensi monogram, bigram, dan trigram Bahasa Inggris yang digunakan pada langkah-langkah di atas mengacu pada [5, 10]. Konsep *cipher* substitusi abjad-tunggal mengacu pada bahan kuliah [7].

2.2 Horizon Chartreuse

Deskripsi Soal

Di lembar catatan riset berikutnya, Marisa menemukan laporan mengenai fenomena aneh yang menimpa langit malam. Seseorang telah mencuri bulan purnama yang asli dan menggantinya dengan purnama palsu yang redup. Laporan yang memuat kronologi hilangnya bagian dari langit tersebut kini tidak dapat dibaca karena seluruh teksnya dienkripsi menggunakan **Vigenère Cipher**.

Manfaatkan metode **Kasiski Examination** untuk mengidentifikasi perulangan pola karakter pada *ciphertext* dalam memperkirakan panjang kuncinya. Setelah panjang kunci diperoleh, terapkan analisis frekuensi bahasa Inggris pada tiap sub-teks untuk merekonstruksi kunci dan memulihkan seluruh isi naskah.

Berkas lampiran:

- Attachment: https://drive.google.com/file/d/1LyCID57V3ckzlvQ9Wnj-mGg_YMHoP50e/view?usp=sharing
- md5sum: 6b933809bbdb3c5c0f1fd9e2b1076cd
- sha256sum: d6ecf3775c56593f833db5c62e1447f2668b41a310562e7a9ccd6a1486d5f641

Berkas *Ciphertext*

MHPQVEQYKKUJLRXGWGJEIFHWFVIOBLCLRXPZODJYGVAYZTZYHDYBYIISWXINTMLRXPQIHWFJIEFWQFHMURXEONALUEHGFBOUZXBZRDN
 SOUZCGDGFZNUYMUOIKVUKMLUXQRAAGUWSBXORVIRPDYGLQVFXLIYNFIWHTGZNMHDPWEWSHAOZDRGRZYXWGLZVYMGZMQVVIKBCNVNYV
 BEJUWALMYEQEACQMOMPVJRVGBKPFJRMJUIHSHVJLKGPKYCFDRFYLSHGMVRRCBKGAAXNWXYRMJNXISJIEFWQFHCVEGCMGYTHPHVRQGG
 AXMGKIWRMPGABIMKFPZMIEBEEQIQPGXBIEKZWYWAYEHXYNBXLCYPWMMVAOHLFLXVZAZOTNYRSBNAZOHTKXYRGSZBGEIWEWSAAYBXX
 VEEUYEXUDZVFWSLAIJISLALZOXCEZEPVZVMBVVLFAKZMLNKKVZIOBHRVFFBSYAAUGPEBRVUZXFHQRITFPGYFRRVFCIOHVVHKGKNVOYH
 HORBKUIZDTMBQGVBKSPVVMFAVUDLEYIIGPKYHFRJJCQXPMONOWKBZGNXMCVLZGAIGHFVYINRTRPGMBPIKUQTNNBNWYJRLZVWCSPRL
 PATTHFKEGRAYOXFDCBPKYLBVNMNXUOAAFLWKUMNVFYBICLXIXPRUXXFGPKFHOXDMFSJUBGXNUMVMBLKMVGQVOMIVFCAJENNZKVYUF
 OMXUBGIGIEPECVBEPGARQWFXGUZAYUDWEGJLKGUOIKBARLXJHPCRDKUWOELRXGPKKTSKGIVAMSFPUFBKIGWHLLYRQFLGANLPUFFIIGIO
 UMBNWMKZCYAAUIHXRMTZGHRRCVWCLKZHOXFCRSMBNWSANLWIRVKTMTZVNNBIXVAPUDXPRUEEQZDBFYEIMRZEIHNUHVJBKMHUHV
 KBBGRXUADGKVTHEIAHLVLQIHGLBSIYZOUTHQOIKUMXOTHQOIKUYWKIOIDFWYOXMRWSLNUYMBRVLIVVKGIGIKIOCOSWCAJFPPIZLW
 UGWLVRMLHHTHRJBSEVMBRUIJNPATTHGKEKFKRGIJVVYRZKPMNFIEIRNLXJRUFWAENVTFJDCJUIVWXHFWSSRJUYXFRWYRTRNEUOCLA
 LKYMUXHXYRBGZDISUIJBTPGGAJKEKRDKMYLBXFCRBNLYVVMKFKSEBPMBCYPMCFVEZQBNHNMUHFJEMAXUGVLLZITZAOZDRJHAAHEFL
 PEZABGPGUFDJVQYATHPHJIBUOABNFDXDBAVOLRFSEFBGMFLVZZACPMBNQYERIXAAFLHZZYBNLYIEHWKBNSHZCPDTCNKDAYEHECYBN
 LXPVOMETMTZHELRRRGCXHEFLJEKUMXZPCGKMEGPGAYIEHWKUFGESEBHEESQMBKYFDVVTIZOXLRGMEGWGFUYOFLVTJPGAUXQRANONLRVW
 DNTRLKNUDRYHUGULNUHVRVHVWBOBOSIRLVBIRJWIVEIRPVYDZVNBXVBXVVVNLLOUZVBROJNUOKLNNPSLABGPGISGSCYANVPWNQLLZITZI
 IFVMSYGTVMVRDAREMUMMBVLLTMJPLUFWIIQCTUHCSSKSQUQTNBMQRRVJMCVGNHESYMYXHWRCUIZZAYVVRKGPKEARJRRJGRZYVVGURKVG
 AUBMISWIEOCZUHAHICFMCHLCAWIRAZLWCAWLVCZUIEYZVSJUMYLMHWFVVBLLNVJEKRNUYAYEVICSWXYTNUHVJUMZYBYQWSZQWTAECX
 HXYVAZOXSEHYJRLZVMBVVOZALUMMBVQKJBBNLRMURYCQLUPMSRDLIRISRMUHOERETVMBVQKROWAAMBRHRVZGGUWBNGRFVLKHAJJSXRB
 ATLGHHZABNLXHQVLVQMIPIWYQWSMVAOATBHPEEJPUHEMBOMMRLOUMBRISIRAZAHFRWLVEBGRXWNUIFSBNLBMFXIJUMHYHOTKXSMCIHIXVE
 CBVMNKCZMRIRAHVHEFLRGITFAOZDRNBCKRYCAGIOGZKTXFLUEIRITKWCISIMTHTZAHARWEYBTJVNNUXGRZYVVGQBPXUAMBKVRDFCRBUYX
 ZHVIJHKNHGISIIFIQBRUJIKIRIXLRIHLVEMOZTYHUSGRITTHFLSEYIWHYMQIOGBUEUXHEJPISXNZDRJVWTHLBELPCIWOJXQNVITU
 WOUZUYOXYEWANACGLVRYTYHLOFEXCOMZDXYADPRXMUMZFVXVAQTNBPVWIRALGMHLRVXFSKYYUAWKIRMTZMIBGEDNVYPHBIYELQKSWCA
 JWNZRLMCGVLFHTJIXUADABJIXKLCTKXSHBYAKUAJICLQZYOYEBWFBNNPAGVMMJFIOKMBNWXZZMYAHJFIPFJQTNPCGKMEGPOZFUAVMFABN
 LLWNUPVGLKCBFZDRJVWTPMMARXRZMZHIBUUVVZQRPTMPDVCRBGCTGLVJYISHIXLRSXWXLCHVJBVGSFUVGLRIMEVNUYUIRQGVJYGKE
 KNAOHLERGSROEVNLCVUBVSFEUQBFLGQIVNFQXXRBCOTNNFXLNTFRIRHPIRABZVNSHRVKHVGAXFLVLVJIYHIJNUIEGTEBGMHGFVFBALBE
 UGFSZDZCTPVUGLRXJQZOMBKRYDNVHLYIEHLVENOUXCYOHFVBSFLYIWRXCEHLILRYUBLGPESFWYNRTDAUGHZVEGUBPUAWSSIQUBLFLVLVU
 IJUNHOIEBZJLKYQWSJGIEOHGRWLVLZIOKAUFQSTUWJXVHWFBNZLXUHVVDVZYXMFWDRCZUAXWGLSEPPGYFSBXOEBOZOTNLRYNBVZIXUO
 OIKBLUHSGSKMETILAXLFXRIVAKZAYFDMUVVNLKBRDVKBVBYZHRDGVSCRHGAXRYVVMFGJNUXZPCRHKCFKEGCMTPGAFWRWYRTRQNMJSSNKQ
 OHGRRTRBNLFCFWVVFAMLMGLVVBQNLVUGVXYBCOMOMFVJLYGUMVHUWVGPGAMIBZEJYMLANHFSSBRVTVHNUHVGYYIILBHTHRJBSEVBUAPE
 KPPZOXMVOIEPMUMMBVVTCKKOHQRYIIVBOZGNQOIRXQZOTMNSIRPMLBEUGPSJCPKXYFVNIRFTKLICAJWFHTZOXLRLEWBAONGISDRPQQYAN
 LODRTRVGANLRFVHFVZFTHQIWEYOBHTZMEQQYAAFRPVFWAYVYBIWFHVJAACFLWKUMTLMBRUAFETJAAYYDRUBNZOXRHDHUMXLBMAWECV
 DOUZBHPREENVEDAYEHXFOMLVNHQWLVTUZZMMGKIIRPUDXPRUIEWHEHCHIPPNNZLKFVIIJBTGKRSHBYBBATVMUJDVBNZOBMGKIDBAZCBPV
 GEEQTGYZYFTCNKPKGNULWHHQAPIOHKUMNHDOTBSBHZUBRIHPYBBVVDONJEIQTMLKQNVTFALKYBHTZLVGPKYMGHPGUMXTBMGUIJFIH
 VNNNPSJGZKJXHLRTVLKUMUGWLRGUUTXHGVLVSAUWBRUPRQGNLTXVQKKBEGYWBURKFBZPFCAJEYYIJFROLXOFLWATNUEHCFHAZPEFVJR
 FEQTNMBNWIYJPGAWILRYDRITIRNUDXNUIZFHOEHRFGMBLUGJVVBNBOAMBNWNUGEVNLRUEXTMJHUIHWCFFZJBFAHWFJPKHYFQXIRUKTU
 YEEIZAOXHZARGFLGQZZXYZVXYBNLKFNGCENAGDTLRRJKUMZYHOOOIRFEKSEMBBSLJMXLMUYNMETIHVNNGKIDBWTPTGPHVKNQTSRUJDVVB
 NZOTNLRYFATEZTCQWLRGITKLIFXHURVRFBNYRSBFTORXHBRVFBGRBHTDGKVWZHQUBHFABEVNBNIYRTWEVNGHHLYJPEZHDHVXBVLJPGAL
 RYIRBUVNNEHPZNRJLTHLZEPGGAOZDRNRAGDMBRXRYRZJHRQNVFVBKPYPYOMNVTRKHCGWLVAUUDTSCOIRFMJVGNOHLLPPTXUALIZFPG
 SEABWSFVZKHEFLPIRABCOTNVVEZQIHVNHHQVYVQGIQBFVWCKUWANASHBYBBAGPSBZXNZGPKNAIFWPPZQYAKYFVSWGPKZTCTBSLWQLHFCYBV
 RTOKKRIHPYNNQZKHSBXLRIHGURCQHENUMXLMITRQPYIJFHZPRYIFMGSBNBIXYRUGZPYFYSFGABYZQRAEJMRSPRQXLNTRFYCAGECRIJAA
 UGVAYRZKFHOEHAIBVMSTXLBYPHSUFHOEHPGWTNWURRINEVVPYEDRUUIBLUQDMDFWAMUXHWVIMTSHHTHVPBCYOHOYGGFAKKUMLNWIPIB
 CXHMNNFODBZKSHIXBSLEJGJDCFVSLALMKXHHHPBCSBROLXOFJIIYXYUOCNBZXPXNESLGGUBFOTRMETWAAFTBQIJBANLWYPLHVQBUATER
 DGKVWTRIBRUWVYNYPGRWLVBXVVGAYAWJCWCLKZHOIEBCMOMIPDYJRAAJAUQLWRFBKYTNRDQFSBCVPJHOHEGJKHUUQLHVNQZDTMCHETRNA
 SHLVWFPBSKKIYNFIWHTNVPIYHVRYTEVNNENLAVEMAWLYGZLRGAXLTFYBLRCKXUBHTLWKUIZDBNURYKNVEVGYGDOZAOVMCPHWLQLKUESGKI
 WHTRTHIAKEJQQYHJRDVQXVFAHQWFXGUZGCTKXVIMTVGNUMHRZTPZYUFXTGBAKKYOOYQFBVGBIBNBIZQFAOUZVJLCKZCKXHGVMKSZUTU
 YVQKWHTRPMMARAFALKYMBNWLZITZTLRQXRJIXLHZGKIJVBAHMCBQXYBCMOBNFRRLBNLMCALIJGJOAHZQLJWRZKUYVSRVEVBNBFAVLFIJ

```

MBLKUZRSEGPGBMAWJLYTJVXMAWJLAKZPHHNWECYQZZTHRVTVPQGESSEPUMKVKSILBEPVZNUYMBBVINUWJPLFVNIJHVRPZBGDLLZITHGXL
RYBNQGAUXZRJKJWBLGNHUIUBCZPGNBJIEFWQFHMZLHEVONAAIHUWKBZKAKCRYIKUMSPLMVQKWEIMTXHGRJKUMSVHHNQHIRAZVKYGIWHR
THIAWLVEOSEMGRTKUMTPZBGLJERMJIXYIHRZSQZIXWBPIJNVOTIYELWYNJRLGCTKXRFAATFYEGVRJAZVTWYRWVWZTNWUWMDRZKTTCVAF
VSWXLMBRKEIIMYAFIBQXYRBKHFISKYDNVUGUWSBXORVAZVINUHRZTPZHFBZMETITJBYAWZZEOASTNRRFARKZHICYHSWEWAUWIOMITGAOUM
BRVMXUBGZFUYYOKVZIOBHVBQKFEJGKMKUTLPVFWASTHQWLVIYNXMGVTYRZKOHQVWVYRLUPGAGKIJRLGFLMUHAFALKYXXNVYRAZHKYQDXKU
MYLHVWHGKFBOTXXBHWEBBLSHQVQXYVAVSTWRKMJGWXFKYCHEKFQZZXFSVLVGWUDTMVQKVAAURRI

```

Langkah Penyelesaian

Vigenère Cipher rentan terhadap analisis **Kasiski Examination** jika kunci yang digunakan relatif pendek. Metode ini mengasumsikan bahwa pengulangan rangkaian karakter (*substring*) pada *ciphertext* terjadi ketika *substring plaintext* yang sama dienkripsi menggunakan bagian kunci yang sama.

Penyelesaian soal ini dibagi menjadi dua tahap: **identifikasi panjang kunci** dengan *Kasiski Examination*, lalu **pencarian kunci** dengan *Frequency Analysis* bahasa Inggris.

Tahap 1: Kasiski Examination

1. Cari semua rangkaian karakter yang berulang pada *ciphertext* dengan panjang minimal 3 karakter (`VigenereSolver.find_repeated_sequences` pada `src/22/solver.py`).
2. Hitung jarak antarkemunculan dari masing-masing *substring* berulang tersebut.
3. Cari faktor dari setiap jarak tersebut dan hitung frekuensi kemunculan setiap faktor (`VigenereSolver.get_kasiski_factors`). Faktor yang paling sering muncul berpeluang besar merupakan panjang kunci atau kelipatannya.

Hasil pencarian jarak dan pemfaktoran menunjukkan beberapa faktor yang dominan:

Faktor	Frekuensi Muncul
2	2144
5	1898
10	1866
4	1014
20	888

Tiga faktor teratas yang paling sering muncul berturut-turut adalah 2, 5, dan 10. Mengambil nilai maksimum dari sekumpulan faktor dengan frekuensi tertinggi ini memberikan tebakan panjang kunci yang sangat meyakinkan, yaitu **10**.

Tahap 2: Frequency Analysis

Setelah panjang kunci $L = 10$ diketahui, *ciphertext* dapat dibagi menjadi L bagian (*sub-text*), di mana *sub-text* ke- i berisi huruf-huruf pada posisi $i, i + L, i + 2L, \dots$ dari *ciphertext*. Karena setiap *sub-text* dienkripsi dengan satu karakter kunci yang konstan, *cipher* ini dapat dipandang sebagai gabungan L buah sandi **Caesar Cipher**.

1. Untuk setiap *sub-text*, hitung frekuensi kemunculan hurufnya.
2. Lakukan uji coba pergeseran (*shift*) dari 0 hingga 25.
3. Untuk setiap pergeseran, hitung *dot product* antara frekuensi huruf yang dihasilkan dengan probabilitas frekuensi huruf tunggal (*monogram*) bahasa Inggris (`MONOGRAM` pada `src/frequencies.py`).
4. Pilih pergeseran dengan nilai *dot product* terbesar sebagai huruf ke- i dari kunci (`VigenereSolver.guess_key`).

Berdasarkan analisis tersebut, kunci sepanjang 10 karakter berhasil ditebak, yaitu **UNDERNIGHT**.

Artefak

Skrip *solver* yang digunakan untuk melakukan *Kasiski Examination*, menemukan kunci dengan analisis frekuensi, mendekripsi *ciphertext*, serta menyimpannya ke berkas log.

```

1 import json
2 import os
3 import sys
4 from datetime import datetime
5 from typing import Dict, List, Tuple
6 from collections import Counter
7
8 sys.path.insert(0, os.path.dirname(os.path.dirname(os.path.abspath(__file__))))
9 from frequencies import MONOGRAM
10
11 class VigenereSolver:
12     def __init__(self):
13         self.english_freq = {c: pct / 100.0 for c, pct in MONOGRAM.items()}
14
15     def find_repeated_sequences(self, text: str, min_len: int = 3, max_len: int =
16 6) -> Dict[str, List[int]]:
17         seq_positions = {}
18         for length in range(min_len, max_len + 1):
19             for i in range(len(text) - length + 1):
20                 seq = text[i:i+length]
21                 if seq not in seq_positions:
22                     seq_positions[seq] = []
23                 seq_positions[seq].append(i)
24
25         repeated_seqs = {seq: pos for seq, pos in seq_positions.items() if len(pos)
26 > 1}
27         return repeated_seqs
28
29     def get_kasiski_factors(self, text: str, max_factor: int = 20) -> List[Tuple[
30 int, int]]:
31         repeated_seqs = self.find_repeated_sequences(text)
32         distances = []
33         for seq, pos in repeated_seqs.items():
34             for i in range(len(pos) - 1):
35                 distances.append(pos[i+1] - pos[i])
36
37         factor_counts = Counter()
38         for dist in distances:
39             for factor in range(2, max_factor + 1):
40                 if dist % factor == 0:
41                     factor_counts[factor] += 1
42
43         return factor_counts.most_common(10)
44
45     def guess_key(self, text: str, key_length: int) -> str:
46         key = ""
47         for i in range(key_length):
48             subtext = text[i::key_length]
49             counts = Counter(subtext)
50             N = len(subtext)
51
52             best_shift = 0
53             max_dot_product = -1
54
55             for shift in range(26):

```

```

53         dot_product = 0
54         for c in "ABCDEFGHIJKLMNOPQRSTUVWXYZ":
55             shifted_c = chr((ord(c) - 65 + shift) % 26 + 65)
56             observed_freq = counts[shifted_c] / N if N > 0 else 0
57             expected_freq = self.english_freq.get(c, 0.0)
58
59             dot_product += observed_freq * expected_freq
60
61         if dot_product > max_dot_product:
62             max_dot_product = dot_product
63             best_shift = shift
64
65         key += chr(best_shift + 65)
66         return key
67
68     def decrypt(self, text: str, key: str) -> str:
69         decrypted = []
70         key_len = len(key)
71         key_idx = 0
72         for char in text:
73             if char.isalpha():
74                 shift = ord(key[key_idx % key_len].upper()) - 65
75                 decrypted_char = chr((ord(char.upper()) - 65 - shift) % 26 + 65)
76                 decrypted.append(decrypted_char)
77                 key_idx += 1
78             else:
79                 decrypted.append(char)
80         return "".join(decrypted)
81
82     def load_ciphertext(base_dir: str) -> str:
83         with open(os.path.join(base_dir, "v.txt")) as f:
84             return f.read().strip().replace('\n', '').replace(' ', '')
85
86     def write_plain(base_dir: str, timestamp: str, decrypted: str) -> str:
87         path = os.path.join(base_dir, "log", "plain", timestamp + ".txt")
88         os.makedirs(os.path.dirname(path), exist_ok=True)
89         with open(path, "w") as f:
90             f.write(decrypted)
91         return path
92
93     if __name__ == "__main__":
94         base_dir = os.path.dirname(os.path.abspath(__file__))
95         content = load_ciphertext(base_dir)
96
97         solver = VigenereSolver()
98
99         print("1. Kasiski Examination")
100         factors = solver.get_kasiski_factors(content)
101         print(f"Faktor yang paling sering muncul dari jarak antar substring berulang:")
102         for factor, count in factors:
103             print(f"Faktor {factor}: {count} kali")
104
105         # Ambil nilai maksimum dari 3 faktor yang paling sering muncul
106         best_key_length = max([factor for factor, count in factors[:3]])
107         print(f"\nPanjang Kunci yang Ditebak (dari Kasiski): {best_key_length}")
108
109         print("\n2. Frequency Analysis")
110         guessed_key = solver.guess_key(content, best_key_length)
111         print(f"Kunci yang Ditebak: {guessed_key}")

```

```

112     decrypted = solver.decrypt(content, guessed_key)
113
114     print("\n3. Plaintext Preview")
115     print(decrypted[:200] + "...")
116
117     timestamp = datetime.now().strftime("%Y-%m-%d_%H%M%S")
118     plain_path = write_plain(base_dir, timestamp, decrypted)
119
120
121     print(f"\n[tersimpan ke {os.path.relpath(plain_path)}]")

```

src/22/solver.py

Plaintext Hasil Dekripsi

SUMMERISDRAWINGTOACLOSESOONCHIRPINGCRICKETSWILLTAKEOVERFORBUZZINGCICADASGENSOKYOSHOTNIGHTSARECOOLINGDOWNMA KINGITACOMFORTABLETIMEFORHUMANSANDYOUKAIALIKETIMESAREPEACEFULASUSUALFROMAHUMANPERSPECTIVELEASTWISETOUHOUEI YASHOUETERNALNIGHTVIGNETTEIMPERISHABLENIGHTTHEREISANOLDMANSIONLOCATEDATTHEEDGEOFGENSOKYOITSAPEARANCERADIA TESTHEFEELINGOFITSHISTORYANDAKINDOFATMOSPHEREATHATREJECTSVISITORSITEMSTHATLOOKASTHOUGHTHEYMIGHTBEFROMTHEOUT ERWORLDHAVESOMEHOWFOUNDTHEIRWAYINTOTHISHOUSESTRANGEMACHINESBOOKSTHATMAKENOSENSEMAGAZINESONESUCHOBJECTANIRO NBOXTHATWASONCEATELEVISIONNOWSERVESTHEROLEOFSPIRITUALSTORAGESPIRITSCANLIVEWELLINSOMETHINGTHATUSEDTODISPLAY HUMANSHAPESSHEDSAYTOHERSHIKIGAMITHISISTHEHOMEOFYUKARIYAKUMOTHEYOUKAIOFBOUNDARIESEVERSINCEBECOMINGAWAREOFAS LIGHTABNORMALITYINGENSOKYOSHEHASNTBEENABLETOSLEEPWELLENVDURINGTHEDAYTHEENEMYWASYETTOBESEENBUTSHEWASCERTAI NTHATITMUSTHAVETAKENSOMEONEPOWERFULTOPULLTHATOFFSHEDOESNTGETOUTOFTENHOWEVERANDITWILLBEVERYBOTHERSOMEFORHER TOTAKEANACTIONALONEHEYICANROPEHERINANDLETHERHANDLETHISPROBLEMSOSHESETOUTFORTHESHRINEANOTHERBUILDINGLOCATED ATTHEEDGEOFGENSOKYOTHERESA HUMAN THAT SHE KNOWS THERE ITS A CAREFREE PERSON WHO ALWAYS HAPPENSTO BE BORED S SHE LLGLADLY UN DERTAKETHETASKOFRESOLVINGWHATEVERTROUBLETHEREISITSMELLSOMINOUSITISSAIDTHATTHEFORESTEATSHUMANSUSUALLY MAINTAINASAFEDISTANCEFROMITITSATMOSPHERECONSTANTLYBRIMSWITHANUNEARTHLYEVILTHEFORESTOFMAGICAPLACEWHEREALLTH EEVILINGENSOKYONATURALLYGATHERSWITHINTHATFORESTSMALLHUMANFIGURESAREGATHEREDINTOASMALLBUILDINGHUMANFIGURESS MALLERTHANHUMANSTHERAINBOWCOLOREDPUPPETEERALLICEMARGATROIDISREADINGBOOKSAMIDSTAMOUNTAINOFDOLLSHOWCANHUMANSP OSSIBLYNOTBEAWAREOFTHISHUGEDISASTERDUNNOIFNOTHINGISDONEWONTBEABLETOENJOYTHATSHEISNTTHETYPESPERSONWHOGET SOUTOFTENBUTNOONEELSEWASINTERESTEDINTHEPROBLEMSOSHESETOUTTOINVESTIGATEFORHERSELFFORRATHERSHETRIEDTOIDONTLIK ETHISTHEYREUSEDTOTHISKINDOFTHINGSOTHEYSHOULDDOITYESHEALREADYKNOWSABOUTTHEENEMYANDHADNOIDEAHOWTOGETS TARTEDINTHEENDSHEDECIDEDTOVISITAHUMANWHOALSO LIVEDINTHEFORESTTOTELHERTAKECAREOFTHEISSUESHEBROUGHTAFEWBOOKSA LONGGRIMOIRESBOOKSTHATANYHUMANWOULDFINDEXTRMELYRAREANDDIFFICULTTOGETAHOLDOFTHATPERSONWOULDNEVERBEABLETORE FUSESUCHANOFFERSAKUYAWHEREAREYOU THERE IS A EUROPEAN MANSION LOCATED NEXT TO A LAKE AS A SCARLET MANSION A SHRILL VOICE WAS ECH OING ALL THROUGH IT SHALLSASUSUAL BETWEEN A LAKE OF GLISTENING WHITE AND A FOREST OF VERDANT GREEN STOOD A MANSION OF FUNYIELDIN GSCARLETITSHOULDBEANAWKWARDSIGHTBUTSTRANGELYITSVERYSOOTHINGITISSAIDTHATTIMESTOPSFLOWINGWITHINTHISMANSSIONTH ESCARLETDEVILMANSIONITSNOTAMETAPHORREMILIASCARLETAVAMPIREWASLOOKINGFORHERPERSONALMAIDHAVEYOUALREADYDONETHA TASIASKEDIBEGYOURPARDONMYLADYBUTICOULDNTGETWHATACTUALLYYOU MEANTTOUNFORTUNATELYSHEWASAPPARENTLYUNSUCCESSFUL ATCOMMUNICATINGWITHTHEHUMANBEFOREHERFINEILLDOITMYSELFSAKUYASOYUODODAILYSTUWELLWHATEVERYOUWANTOBVIOUSLYSHEH ADNOTBEENORDEREDTOSTAYHOMETHEMAIDHASNOCHOICEBUTTOATTENDHERMISTRESSASAPROTECTIONCHARMYOUKNOWTHATYOUWONTBEAB LETODOANYTHINGAFTERSUNRISESHE SAID IN HER HEART ITSSOPEACEFULANDNOTHINGINPARTICULARISHAPPENINGSOSHELLJUSTGOBACK HOMEONCETHEMISTRESSGETSTIREDTHEMAIDTHOUGHTLIGHTLYOFCOURSETHATTOOWASLEFTUNSPOKENNOOTHERPLACEINGENSOKYOCANMA TCHTHE SILENCE OF THIS PLACE HOWEVER IT ISNT BLEAK IT HAS A PEACEFUL ATMOSPHERE LIKE A SLEEPING SOUL THERE IS NO SIGN OF ANY DISTU RBANCENATURESBOUNTYANDREFRESHINGWINDISTHE SOLE SOURCE OF SOUND THIS IS THE NETHER WORLD THE LAND OF THE DEAD THERE ISNT ALI VINGHUMANANYWHERE TO BE FOUND THE GHOSTS THERE HOWEVER ENJOYALIVELYAFTERLIFESOLADYYUYUKOSNOTAWAREOFTHISTHEMOSTVIVI DANDLARGESTPLACEINTHISQUIETWORLDTHEHAKUGYOKUROUYOUMUKONPAKUAGARDENERWASPONDERINGWHETHERTOTELHERMISTRESSAB OUTAMOSTRECENTINCIDENTATTHATMOMENTSHEFOUNDHERLADYHEADINGTOWARDHERGOODTIMINGAHLADYYUYUKOYOU MUAREYOU STILLIGN ORINGTHATEWHWHATDOYOU MEANBYTHATWHATYOUARENOTEVENAWAREOFITTHATSWHYOURERAGGEDABOUTYOURDULLNESSSHEDOESNTREMEMB ERBEINGRAGGEDBUTITSEEMSTHATHERLADYWAS AWARE OF THE TROUBLE AS WELL SO YOU WERE TALKING ABOUT THE MOON I AM CERTAINLY AWARE O FTHATYOUONLYSAIDTHATANDSOSUDDENLYITLOOKSLIKENOONESTAKINGACTIONSOWHYDONTYOUHAVEAGOOUMEHWHYSOJUSTKIDDINGY OURETOO UNRELIABLE ANYWAY THAT HUMAN WE SAW THE OTHER DAY WAS BETTER WELL I WILL DO IT THEN NOWAY PLEASE DONT BE SUCH A MEANIE ISHA LLGOTOOIREALLYMEANTWHATISAIDABOUTUNRELIABILITYTHOUGHUYUKOSAIGYOUJITHEGHOSTLYMISTRESSOFTHESAIGYOUJIFAMILYR

AGGEDYOU MUWAITDOYOU HAVE ANY IDEA WHERE TO GOMY LADY OF COURSE A LOT OF THEM AS WE SHOOT STUFF DOWN WELLS EVENTUALLY FIND A LEAD
 AT WHERE YOU'RE WRONG LADY YOU YUKO YOU'RE LYTOO MUCH ON RAW POWER AND HAVE BAD AIMS OITTAKE SEVEN LONGER YOU SHOULD CONCENTRATE
 YOU ATTACK MORE LOOK YOUR BACK ISSOUND DEFENDED YOU MU YUYUKO WAS REALLY WORRIED ABOUT YOU MUGOING OUT ALONE SO SHE DECIDED TO TAKE
 ACTION BY HERSELF SINCE THE OPPONENT IS POWERFUL ENOUGH TO CAUSE SUCH A DISASTER A TEAM OF TWO WOULDNT BE A BAD IDEA IT WAS PEACEFUL
 FOR IT LOOKED PEACEFUL HOWEVER ALL YOU KAI WERE UPSET WHATS REALLY HAPPENING IS THAT WITHOUT ANYONE TAKING NOTICE SUDDENLY THE
 FULL MOON HAS DISAPPEARED FROM GENSOKYO SINCE EVEN ON THE NIGHT OF A SUPPOSED FULL MOON A BIT OF MISSING LIGHT PREVENTS IT FROM
 BEING FULL ITS NOW WONDER THAT HUMANS AREN'T AWARE OF THE SITUATION THOUGH IT'S ONLY THE THIN EST BIT OF DIFFERENCE FOR NON HUMANS SHOW
 EVER A MOON THAT ISN'T FULL DOESN'T FUNCTION AT ALL ITS AN ESPECIALLY CRITICAL PROBLEM FOR THOSE WHO DISLIKE SUNLIGHT A HUMAN AND
 YOU KAI A TEAM OF TWO VENTURED OUT INTO GENSOKYO MIDNIGHT HOUR STORE TO RETRIEVE THE MISSING FRAGMENT OF THE MOON AND RESTORE THE FULL
 MOON THEY WILL STOP THEN IF NEEDED BEEVEN IF IT BECOMES AN IMPERISHABLE NIGHT ASSUMER DRAWSTO A CLOSE NOT MUCH TIME REMAINS
 BEFORE THE HARVEST MOON THE TEAM OF HUMAN AND YOU KAI STOP THEN NIGHT AGLOW IN ANCIENT VIRGULATE OBJECT A PILE OF FOUND OBJECTS IN THE
 HESIGHT AS SMALL GEMASHINING ORBA FRAGILE SOUL AND THE LARGEST SPHERE HOW IS SHE DOING THESE DAYS SHE WONDERED AS SHE STARED AT THE
 ESE OBJECTS TIME DOES NOT FLOW IN THIS PLACE HISTORY REPEATS ITSELF SHE TOO WAS IN GENSOKYO

Plaintext dengan Format Asli

Summer is drawing to a close. Soon, chirping crickets will take over for buzzing cicadas. Gensokyo's hot nights are cooling down, making it a comfortable time for humans and youkai alike.

Times are peaceful as usual.

From a human perspective, leastwise.

There is an old mansion located at the edge of Gensokyo. Its appearance radiates the feeling of its history and a kind of atmosphere that rejects visitors. Items that look as though they might be from the outer world have somehow found their way into this house. Strange machines, books that make no sense, magazines.

One such object, an iron box that was once a television, now serves the role of spiritual storage.

"Spirits can live well in something that used to display human shapes," she'd say to her shikigami.

This is the home of Yukari Yakumo, the youkai of boundaries.

Ever since becoming aware of a slight abnormality in Gensokyo, she hasn't been able to sleep well, even during the day.

The enemy was yet to be seen, but she was certain that it must have taken someone powerful to pull that off. She doesn't get out often, however, and it will be very bothersome for her to take an action alone.

"Hey, I can rope her in and let her handle this problem."

So she set out for the shrine, another building located at the edge of Gensokyo.

There's a human that she knows there.

It's a carefree person who always happens to be bored,

so she'll gladly undertake the task of resolving whatever trouble there is.

It smells ominous. It is said that the forest eats humans. Humans usually maintain a safe distance from it. Its atmosphere constantly brims with an unearthly evil.

The Forest of Magic, a place where all the evil in Gensokyo naturally gathers.

Within that forest, small human figures are gathered into a small building.

Human figures smaller than humans.

The rainbow-colored puppeteer, Alice Margatroid, is reading books amidst a mountain of dolls.

"How can humans possibly not be aware of this huge disaster?"

"Dunno."

If nothing is done, we won't be able to enjoy that. She isn't the type of person who gets out often, but no one else was interested in the problem, so she set out to investigate for herself.

Or rather, she tried to.

"I don't like this, they're used to this kind of thing, so they should do it."

"Yeah, really."

She knew nothing about the enemy and had no idea how to get started.

In the end, she decided to visit a human who also lived in the forest, to let her take care of the issue.

She brought a few books along.....

Grimoires. Books that any human would find extremely rare and difficult to get ahold of.

That person would never be able to refuse such an offer.

"Sakuya, where are you?"

There is a European mansion located next to a lake; a scarlet mansion. A shrill voice was echoing all through its halls, as usual.

Between a lake of glistening white and a forest of verdant green stood a mansion of unyielding scarlet. It should be an awkward sight, but strangely, it's very soothing.

It is said that time stops flowing within this mansion, the Scarlet Devil Mansion. It's not a metaphor.

Remilia Scarlet, a vampire, was looking for her personal maid.

"Have you already done that as I asked?"

"I beg your pardon, my Lady, but I couldn't get what actually you meant to..."

Unfortunately, she was apparently unsuccessful at communicating with the human before her.

"Fine! I'll do it myself, Sakuya, so you do daily stu..."

Well, whatever you want."

Obviously, she had not been ordered to stay home. The maid has no choice but to attend her mistress as a protection charm. "You know that you won't be able to do anything after sunrise," she said in her heart.

It's so peaceful and nothing in particular is happening, so she'll just go back home once the mistress gets tired, the maid thought lightly. Of course, that too was left unspoken.

No other place in Gensokyo can match the silence of this place. However, it isn't bleak. It has a peaceful atmosphere like a sleeping soul. There is no sign of any disturbance; nature's bounty and refreshing wind is the sole source of sound.

This is the Netherworld, the land of the dead.

There isn't a living human anywhere to be found. The ghosts there, however, enjoy a lively afterlife.

"So, Lady Yuyuko's not aware of this?"

The most vivid and largest place in this quiet world. The Hakugyokurou.

Youmu Konpaku, a gardener, was pondering whether to tell her mistress about a most recent incident.

At that moment, she found her lady heading toward her. Good timing.

"Ah, Lady Yuyuko..."

"Youmu, are you still ignoring that?"

"Eh? ... What do you mean by 'that'?"

"What, you're not even aware of it?"

That's why you're ragged about your dullness."

She doesn't remember being ragged, but it seems that her lady was aware of the trouble as well.

"So you were talking about the Moon? I am certainly aware of that.

You only said that, and so suddenly..."

"It looks like no one's taking action, so why don't you have a go, Youmu?"

"Ehh, why so?"

"Just kidding. You're too unreliable anyway.

That human we saw the other day was better... well, I will do it then."

"No way, please don't be such a meanie... I shall go too."

"I really meant what I said about unreliability, though."

Yuyuko Saigyouji, the ghostly mistress of the Saigyouji family, ragged Youmu.

"Wait, do you have any idea where to go, my Lady?"

"Of course, a lot of them. As we shoot stuff down, we'll eventually find a lead."

"That's where you're wrong, Lady Yuyuko.

You rely too much on raw power and have bad aim, so it takes even longer.

You should concentrate your attack more, look..."

"Your back is so undefended, Youmu."

Yuyuko was really worried about Youmu going out alone, so she decided to take action by herself.

Since the opponent is powerful enough to cause such a disaster, a team of two wouldn't be a bad idea.

It was peaceful.

Or, it looked peaceful.

However, all youkai were upset.

What's really happening is that, without anyone taking notice, suddenly...

The full moon has disappeared from Gensokyo's night.

Even on the night of a supposed full moon, a bit of missing light prevents it from being full.

It's no wonder that humans aren't aware of the situation, though; it's only the tiniest bit of difference.

For non-humans, however, a moon that isn't full doesn't function at all.

It's an especially critical problem for those who dislike sunlight.

A human and youkai, a team of two, ventured out into Gensokyo's midnight hours to retrieve the missing fragment of the moon and restore the full moon.

They will stop the night if need be, even if it becomes an imperishable night.

--- As summer draws to a close, not much time remains before the harvest moon.

The team of human and youkai stop the night.

A glowing, ancient, virgulate object.

A pile of round objects in the sight.

A small gem. A shining orb. A fragile soul. And the largest sphere.

"How is she doing these days?" she wondered as she stared at these objects.

Time does not flow in this place. History repeats itself.

She, too, was in Gensokyo.

Sumber:

- https://en.touhouwiki.net/wiki/Imperishable_Night/Story/Prologue

Kunci

Panjang kunci: 10

Kunci hasil rekonstruksi: UNDERNIGHT

Referensi

Tabel frekuensi *monogram* bahasa Inggris yang digunakan pada Tahap 2 mengacu pada [5]. Referensi mengenai konsep dasar dekripsi Vigenère *Cipher* mengacu pada [3]. Konsep Vigenère *Cipher* dan metode Kasiski mengacu pada bahan kuliah [8].

2.3 Domain of Hopes and Dreams

Deskripsi Soal

Dalam suatu catatan simposium, Marisa menemukan rekaman mengenai keberadaan sebuah dimensi buatan yang terisolasi dari dunia luar. Batas-batas menuju wilayah rahasia ini dikunci menggunakan segel jimat yang dienkripsi dengan **Playfair Cipher** agar tidak dapat diakses secara sembarangan.

Tugas kami adalah memulihkan naskah petunjuk akses tersebut. Susun kembali matriks kunci dengan menganalisis pola kemunculan pasangan huruf (*bigram* maupun *n-gram* lainnya), distribusi karakter, serta

tebakan kata-kata umum dalam **Bahasa Inggris**. Setelah matriks kunci berhasil direkonstruksi, dekripsikan seluruh dokumen rahasia tersebut.

Berkas lampiran:

- *Attachment:* <https://drive.google.com/file/d/10Dljf85XJXS41PSSqrfaqU0QywtuneuX/view?usp=sharing>
- md5sum: b6f21903cbdb62ca899d6bb2694814b1
- sha256sum: e4bb0e2a2ac8425ef0133b78dd0103308f781d2afdfcc46712b2a23efb7ee597

Berkas *Ciphertext*

```

ONSFLRQLEDQLUKBVUNIBGBUCXMMZTVMSTUKDSVGIPPQMSATNUGCDSBFIAGCYGVSPATGSZDIFNUKBFBELSDYIESQBFTUWTIYMUURTYDSON
SFNTNSIDTUFBLTIAADGNVLVTIFBPDMAUBDEMSIVZUNLBTRUBVOIACVTOSSFBFBYQSSFGBCBUCRTDSQINWUHCUYSGPMZTFFBQLFTLPUATA
BANFBESDIVAUDSYGSFPZYASIMPOSITLYYPAAQTVIDSZGLLOLPUQNTUIAICPADYGHCUYSGPSYUBDQUAVGPNTVFIAMNGIUOLSTOIFNSTOI
YBLQLYUISBGPSTBZIDSWISZTINDILYTLGLSDULCBDILYTLBUUTYGTGEBYMPPOSYULSBMDCYVTYITCBTGBVTOLMETUICVTYFTUKLFUZDU
KPDSCBTBFBGPASZSQBFTIZSUBIAEAPWSLIYFIYCNQIOMSLASZOLNWUMQAGBIZUAVUYADINHNUGCBFTITITAYFIYFTUKPQUOBFUCUUBFIIF
IZBRTFFBGBUAFSAQNEFTPAIFNQUTUDIZLSULOQNAIQPOUOIQNLISIAUQBEGBMFSTUCPAMNGIUPSADIAULYIINUMBFUAQPKISDFLUKELFI
FLICMDIEUISLICTVTYEBTEMQDUSRZSSBQMYTSQSDGBVZVTNUFUPAIFAQUOUILHOUGNVTLKNNRNTSNRUISBGBBKUTKSSTTYDSRFBGPMYAVT
ICBYMNDYICGBYAFIFNSFYSRFGBRUNWNIVGINBFYAHKLSBFBDFLTGSXUTIFLSIFUNNIAPRUBUGCRDSIDDIUACUNUHKEBLPCUECIEMNDUYT
DPIPATTVPVRUBVIFAQUOUIBSMSYIDSPLSUASPNTYAPOVYITICUMURTVCUYACKBTMGLOLAUESIAQDOLGBWSTFASANOUZMGCVTDBICUMYS
LSIFUNBWINQIMICVCYELFTGTIBBFDLSUFFBTBCYCUPLATBSIDACUAQYGHCUYSGPMDOUCREBACUAFEAQOYSLRIOTBEBWSUABTDSQAYS
YODUFNUKSMQWNTFBGBVTITTBICYSGHCUYSGPSYCVBYDSCBPAQYAEBSITRFBPEIAQLYGIOMPOSITLYNUGCBFTIMWADBFTITITULIFBFTG
TSIFPABCIIEVBVTINZLMYNYGTSFNQUAPAUQBQLNWUBUKTYBFYAHKTIBFIDYSMWIDBKIAVGRNLWIBGBUCXMMZTVMSTUKDSVGIPPQMSYSVB
QYBFTVDUMFUADSQYQDELZBQTCSSSTYDIACIDDATIBFTVSLFIUADBVBYGSXBEESYLBKQUBFTVBNTYDLABKSQELITUCPRISURINIFFLYS
EVFBLDGVPAZQIDFWCVBUODINHCBCRCVCUBFVTVBUUFAFASIRYRTVEVTPRYSMWIDIZGUINUHSAUKSDINHLDYADIVTICMSAYIVPSTBZIDSSI
NYOSYTEAQBDATBEQLDSGMDUQMNSGEBICGBRTSICRBFYTUMRQLEDQLUKBVUNIBGBUCXMMZTVMSTUKDSVGIPPQMSULIFBFZSXZMWIDGY
TVOUZLICYACRBTVDUUMYAINVTDQOUVTNIEVFBGBUCXMMZTVISTGTIVFADNWAMGIIISUFFBLQSTHKFTIYCRDSBNIZTROUNUCRBIGUINIBFW
CVDUVTBFCUBVDPYBSIBFBGATTBICGBYBSQBFTYMPPOSITLYTIBFIDOUSLHKECBGSYFTUBBUBYCUFIBGVTDSTAISNYDSIVYCNDTGLNVGIPPQ
MSSZSYBFFUAIABQVTUTUBFYAHKESYAVBTIFIZFTUKDSVGEIYTTPESPBPABSIDEWFMZOLUPUGIZYFMDPUBVDPYBSIRLNNBIMUOMTCUBV
DPYBSIRSSPBFYABFTYBSIYCSIYCEURDUUPZLMVBUIBGBAIFGBVTINUCXBSPTVTIQLNWIMDISFYQSSSTBSSGTSIFLSSXTYITPASQVGIP
PQMCSBITTBIFGBQIDEBFIDBFTVDUVTMSNGIUMWPDILYTLQWSDSOLCXUATAIFQLCRBFBYUATOSQGLOLIPZBDNOSTTNYAOIACDUITUMMRUO
BTBFBGLYIAYSFRGBRUNWNIVGYOCUTEGBTSQDMYUAYBICINZUFVIAEPQIUNUAFESAPOZSYGBFTUWISDUANUGTIATEGBMSNGIUIZDUNWAMVB
TESDNQDUEZDIADIVGSQBFTUXTNOSTTNYAPITYIFSDGBTSGBIZFIDPICCFRZICXUMTFOISLZIBNBEQLFIPDUTESQLPIEDMANIOYBTUGDIAC
LTBUUTYGBFBDIVVTICDLNICKTUOTIAGVZSIYUMLAIFLDUBLGUFEGPRMRGMQATYVGGIPQMSQLYAPDQSLAUIFZIDSSIPDUTESDNCPAPFRLA
DEASLXTFQLNWBWLYIAYOSYTPRITUKBFTYLOTYDUIIMDIACUAQUACZICXIADQSLAUIFZIDSSFLFISTHKBFLYIAPICVBTUBDQTVIFYQLU
TBBFBDYSZLZTOIQMYCRTNWUBTQIEIVTUKIPZPACVBUTYTIGAINTRUTTEGBAUZMLNTDITVDIAUMLRPATBVXQLCFUAQAKRGIEIVTNUYUAF
SZTGSZBFBGLYIAYIACTVBFTVLNDYRISPDYUCMWSAPIMAPAVGWFLYIAVIUBTIYIBKIAEVUMZMZTTFZMLNTAIYECDUIFSQUOPDUYIVTLAFT
IMSPDYUCRSSPLOPAFEGBGLOLIPTILDUBDSLSTFIDDUACYZTVEIXIYBOLSUOIOUZMZTMSIYUOBFIDBFTVTUPIADBOUATIFSIDSZUAVGIPPQ
MSELTQAICBFUAFQCFLFIGATFBEIVBQLETSMLODUUKCUYSGPSYENVZVTICQSWSDNCPAPECBYIUICIUUCBZTFQMIIEBFGUYAQNTIOSVBVFCU
YSGPMZIOTXDUEBTCVMSNGIUDSOIYGBFBYTTUWYTIQEBZINWIBGBGLOLIPUAVGIPPQMSCVTIFRSZAPGISFGBXSIVOSYTBFIADSSXTYTVYG
BFGUYAQNTFGBGLOLIPZBCTUUKVIUBTCSEGBMSNGIUIFVFCVBEGBMSNGIUGCCYZLEDQMUCNDQMBFTVDSSXGYCUDSBFTVMSNGIURSSPVICP
AUUKZIPOIADIACVINUUKYZTVBFBGLYIAOIACD SOLGCYIVTBFBGLYIAOTPADIZLGCICSDGBTSNDEIINDPACTFASAIACEIXIQAXLFBVZSUK
ONBFBFTVTGSZBEGQIDEBFIDIZLNTVQLPIEBYBVBFBTRSSPLOPAFEGBGLOLIPTIUAVGIPPQMSMTTYFTICTVTYFISDQLFIQMLRFTMTIETBFB
IDIFGBDNYTBFCUBFCYPZSZGLOLIPGBVTIZCYTFUIHBDVTIDIVTYBFUAQPRTCRBFTIZLOUVTBFDUAUOLIMUADOPZYAIKSZIEAIEBIAQDOA
SAQIUAIFTFIEASLXGBVTCUQDUKTVBTIALSINOIVTSXSDBDEBQDACGLOLIPDNAPTMTVYGBFBYIABADUXILDYACRFBFYDIFNUKIAIDIVINUM
RSCVBCTSQDACAMKXUAFUEISNIPOLNWNBUQVBGBUPYAAMPQIDTFRSSPBFIDYCSIYCEURTUFICVBYOFUEBLAQSSFLAIDBCIEAPRLETYDPI
BFUZBEMZSZVSLDYACSVTQDDSSISZVICPAUUKYADSKRQSAQIATEGBVTRLNWNIBMUOMTCYUBICMSGKLYIAIRSZTFDSAMPQIDYSTFRNLNWNAGY
TVVGECSZBTTRMSIEFLCXSTFNUGCBFIDIZCUCRBFQUSSFLAIDTYQCUYSGPSYGLOLIPIZDUAUOLIMUADOPZMZQLBFOYCUCTEMQLYIATA
KUTFOEUAFAUESGLFLQAICBFUAFQCFLFIYSYBFBFUAUVYAVTDBURTGZSPSBFTVIFQMEVWMQLETSYABFUAQKPWPWYBMWIDOTUFAUESNUGCSXAT
DYSIBURSTYYCDESQDPYIVTLLSSXYAQAGTYGVSBUAFUQAELBYGAURTVBVTQXSTBTUADIYSYBFBFUAATCBYCSRBDISGUABVTFBQLFIMAPA
QUITVBVTQLRUIKLATICVBEGBCNUBIOTGSZVGDIDMIEMWIDTIUBZUIBRQLBRTVGUIDINFNAIDMIENIEVFBGBVTIOPAQNAIQPDNYBDSVGIP
PQMSIAMVUOLRIOQLRUAIQLZTFTYBQSNWAMRLICQLBFPATYBISUQAICFUYGIQNUONDMIEBFBGLYIAYSFRGBRUNWNIVGDINHLDYAESTVYTUW

```

TYBFBEGBYBZLMRQLDIVTMSLESTUKUAFEYQCUYSGPADRFTVINIMSQTSIFSQINBMGMEOLOYZBEYQCUYSGPADUPZLSUDNYBDSVGIPPMQSDSYT
 CGUCRHIUBFUABFDYIEITTBZMWITOFWUATYQLYGCBDPYBYBVTDSIVTICRFSNOYSXTINUBCAPATTIPNTINBFAUFYNUGCOSVBSIOSFIYSMNND
 TYQSWFLYIAIFPZSQTSFGBTSQITFFBSDBGYTOQLYGTAOINPYCIZBTQYTOIZBCTUUKIFQLBRUNNBDUYTDSCTVTIBRTFMIEGLOLIPCBNUGY
 UAFUICGBPYSMIOUELFTQAICBFUAGPUZGVSDEBLPNEKGIUBFUAVGIPPQMSTUBFTVULIFCBNUGYUAFEQLBRTVTVUDNUYTAPEVFLICGBYAFI
 ATBVSZABBFUGYGFISOYCTCBUTRUUKBFIDQYTOGYCUCVTIIFTACCVBTGYTVSEFLUKVTINUMBKIAVGQACKBVBDOBEYQCUYSGPMQTVBGLYIA
 MTBUUTYGBFIDQYTOCVTIDIOIUCYGVSDSGNDBRVBNWUBYGARTCBNUGYBTUAOUSZAPESTVSTHKBFTVDUUAESIDQLFIAPEFPZACQYAHKQS
 SFGYAFIATBVSZABGYCTRMSNYTVQSWLOTSNDUBSTHKECTGSZBEGBTSGUTYBFGUNWNBPARUCKBEGBPYBUYGITFIIAUNNLITPAIAQDTMMA
 IEBUSVTGIYBEADKPBSCBFDITFNDIINUMQLNPYCUCBQLYIAOIVTDUITUMMUOBTXDUETVPRYSBFDIFGBNIEBIFIQIQDQIEOTYIACUFT
 VTUMUAEBICASAIHDQELSYAXCMZBFBCTYCSIZFTIFQLEBYGLXFLUBBFGUQAICBFUAFQLEFAQYSVSYAQNUTRNBFBFGUIFUNBHBUTYGUA
 INUTIPMQUCYGWFUDZTLDYACRBFIDBFYSPCUCRBFIDGLOLIPVTINARBTBIDBFYIOCIITYSRIONDBZLMTGUPATEGBTSIDUNTSEPDPAFS
 MABFGUIFIZEBTEPDARUCKBEGBPYBUYGIFQLCRBFTVBYIECDUNUCUPRYSYBMWTVDYIEBFTVCVGYCUBFYAHKAPDMEOPTPARTWSBTTFUGFS
 IDASAINUYPIYCZTBANSEIIVPDSPIFNTTILMGBAUFEYDIFPZSQIZGICPUPTIICTFLFUZBCCVCUENSTYAOIDMIEBFTIQSZIMWUNCUMZELES
 YASIFWUABGIOBRYQAQOISLSTHKBGRMTIBFIDEBBKAUDNNWUMVBYGQYCRBFTVBCEIUCYGWSUATEQLFRSZAPZMVMFSTUCMTCVCUBZFSEVKF
 LUEBLDTFVBNNWUIYIBIEYBOIMYLVNQAIOIQAQONSFUCZSMQNFAPRBFLYUCYIZLQPIFGBPABVQMTINUMYIUBIFGBMFSTUCSTHKDIACUG
 STIYCRBFDYUCVICAUIPEIKRQSIEUARSPADIGBOLIBQMEVFBQLFIAPDFLUEBQMRLBRSZFPTFPISDXVPAFEQLCRBFBYSZUSIOFWUABCUPA
 FYDSYIZLQPELFILVNQAQYDPIBFUZYNEFLICTVYTFUKIDGBSTUKITFGMYAYTTYSQAQYLGAPBKSADBELCRDSQBFTVQITFKGSZIYEMPMYT
 IHAPBKSADBYODUFNUKSQMWNTWLPFIANQSMYAIZTGZSPYLVDARIFLAIAPIZGUYAAPFTPLSDELFIIAUDIATOLVNQADVTUNURBTFTVTGSZIN
 PAFELSBEPZSDQMEVQLDELINWUMLSINYSRSPMFUAIASDQLECTSTUKMTNDFPSPIAMPBYSZUTYSFWDUNWUMMWUGYTVNTLOBESYLVIDGBNIGC
 ZLOIBMUOIRLNLAMRINEKBUBIAZLEBSTBRIOYIBKLDUIETRLNMNDBVGBLXIOBFTILSAPFIIVBYUOCBLDYADIBFYAOIACUGIZOIQYAPZLQD
 MYTFYPDQDUCKSPMNBEDUDMIEQIDBIATCBDSPDUIZBFTVDPABTNESINBFYAHKLSXBEESXYAQARTFICVDBNWUBICQIDESZUTOIACBFCYVT
 TNITPAOLBKUATVSEGBMPNQBEGBVTIZDUIMSYPSCUNTBGSDYOSTUKMTSEGBFSIUIPDIFTABFBDPAIFVIDEASAPRFBGPMYAVTICBYMNBVQM
 INBEPISDVGEDILHKEISNICINQPRISPBDAICVSITTBDSXLAPPABGTVCAGAPRLDYACICVEISNYMYAOSUAFTUKIDEMNQIVCUMWOMSNEPBE
 YQCUYSGPSALLOLIFIOLSMNBYPAFNICZIUUAHEIECBNDIYDYQISLAPZTTIUQVTBYCUFTZLMTBEZIUGTCMGLYIAOIACTYINBTLZUOQMVBYG
 SISPDYHMSABRSUTACMRQMSLPRTBOBDOBESPSVTYENDUACBFIDMNNDTUIYBFOYCUCTEMDGVCUYSGPPAMZZLTFQATYBFIIDYBIAMSIVBG
 TVBCPESACKATCUDBAPEDSACKATCUDBELFSIDGBSTUCRUDITLNUESVBVTBFGUAPFTUKCUYSGSPSGSTCUTOZMLNBQUZBCCVCULALDUBDSOL
 UADIUABFTUSIFSCUFEGMYAELUNESYASFBYMNBTTSBDBIOUDMYGMWTVTUZLPRMICVCYUAVHIOBFIDINYSUAGYTFLDUBMWIDSLHKEYACBDI
 NHUAQUDMIEMQUGLQSLQPYFZIGZLYIACTVBFBEYMNVBIOICTFMSLELUIBIDBYMNTGSZELCZBQTIOLVTNUFUPAULYTTQSZOSNUFTDNBLDU
 YIVTYSICSSQTFIATAGBIZTABRIOAPFTIYDIUEBPPOUBELDIMSNGIUEBPOUBGINWAMGSYAQNLSPAFFIDBYMNTGUNUBBZTFFMNEQBETTYDI
 HBCVGYUNOSVBWFMZVITBMWUNTUBSUOCBRTNWUBBZTFGMYAQNTFMQYMTNSNGIUIZBYSZTYUADBVTBFIAGLOLIPULIFIOGLOLIPDNLCBDSBPBG
 TVSLFIIATDBUTYFTINYMYAQNNTIACBDPYBQYTOIATCAEBQIOPMMZCNMTVTIINYSIOZUIFIOTFTIBFIAFPDSMSNGIUYSMSIVDUPSCXOSPR
 YTPRYTTCUGRUIBGBAIPSCXMMWMIQIQUINWUBALICFLCXLDUMQIZSSISABANFBESYUOELDIPRYSYBPABVQMIYBELCZBQTISSHZOSNUFTDNBUB
 ACITYINPOSYTTATPASDQLYGBFTVUFFBDSQAYSVBWNBWMBGPAATYBRBCLUIBBFBEBYMNBEGBVTCBDNIYDIYVSDUYIVTGLICUAHYAYTTVOS
 SFBTBFIDBFTVTYYSYBBFUAFUCKVTAHBBICVUTCZQAXLBFTVDUACYSVBYIVWUBTTFMWCUTFEIXIBTYACRDSBYSYBBFUAFEQLFICVBYBT
 PSVTQDUKTVYAIQFLLXQLBRBVTVCVCVYCDEUAHVBEQMEVFBVBMFYAABQAYAWTTYCRDSFLTBTYSLRIOBFIDQDUKTVYAPRYTBYUAFEFLUKBF
 CUMWUNTUQAICBFUAHPSUICLUFTPAZLMRZSUKTSGVVTCRDIHBMUNDGCTEGBOSVBSWSDUAUUKMSNEYAABICGCCYYBYTINBTLAQSSFLAIDBUSG
 SZYAXILOBUTYPRYSNWNWUBBOPAHXQMBFYAHKCRDSGCCYYBMFIESQKRQSVTYGSGFBVTUAMRFGBOIUAESYADNIKUFFBNIEBSIVBIDYSNI
 IFNYTISLQPPRIFSZYUNBSMYBURFSYUBCYRLBFIYOYCNWUQLCONICPUCTEMZNGEONIGBWSTFIATBAUSMBFBVSZABIOATTYSBTTELFTLZ
 PQDYCKTUZLOIZLVTBFDLRIOPAMUKTVUABFDYZTUAIZMVSZABTFZLIFGBICQLFTDNYBDSUBIZAUZLMFTVTUUKCUYSGPSASLIGBSXPDTY
 YSKTIZTGLNUMIFAESUUKBFTIZMWIQDACDSQDMVUNBHCYIZFTENNIWISUIKSZOLFTYGTIFTNWBWUZCUFRUFIVBTYACRBTVDUYSALWTVBE
 QMEVFBGBVTIOPAMSIVBGTVTPRITPOUBUABFTYLOBVUOBFIDMSNGIUSZQYTONUGCYBCUEAMKTVTDURFTVBFTVUVRNIFCUBDTIVTAPLADBT
 ICGYAFIATBVSZABMSEYDUIMPDDUYTTEYDRNIFURBDDUYTTEYDRNIFELDIVTICLFTVTUVFFBAPXBSPPSIPCUYTIAEACKSPSPSIPCUYTUASD
 GBVXSZTOVTINTFOUACQIIICIOSEGBYAFIATBVSZABIAEQCUYSGPADVTYTONZIEBQDAMUKBFIDIOYBNUUCGBCKBEFLUKIFQLBFBZBTINQYC
 IZBTCBCUQSVFSDEBAPWFUZBGDQBFVTUVRNIFCUBDTVAPLADBTACGBYAFIATBVSZABDNLCGYAACUAVGIPPQMSYBFDYBIAITZLPTSPMN
 BEBUSGSZQYCREBIPSAYGBKIAVGTQZSOLGLOLICADQYBANFCVCTIEBCKBEGBICQLFIDPVZVTDEVBNWNIIRIZIONFAPVMAVBVTINVTQSV
 GVIZTBQDOIQYTAIFLICPZMUGITQSVFSDSYBICASICFLMTIEATPACISZOLNWMNUYUAVGIPPQMSGLFLNUYGUAIOONDBNTVTIDBTMYT
 IHNURUUKYSYBMWTVBVBVTAPDMEODNLCSDBGSGBSLIFGBTYTCKDYHRQLDIGBWSTFTIAPRBQLEDQLYBICASAUDIAPBFTVBGWSTFNFAPZL
 IMMAURYUALQIZTESZDUBKMNNDBGMZEODQSYSPBEFLANFBELQDIUPZLSUFIIITPOUBTRMSIYEIEOCUYAHKSQBFBGTVYBFTDNSFMRYAABMS
 LNNFBEMDBDSPYOETITNMGEWBYYBBFUQNDMIETFMSNEYAABCBPBYLSPSSFINPZMQMQSXSXBYBUSEARUAKUMZIZVBTPACRBFUAFNNWIB
 NQBEQLBISUUKQAXLMRTFBKAPZLACUNIBFWMZUACVDBNWUBICNLKXPAINUMYASRSZEIUCIRSZBRSPABFIFGBAINWAMMLDUMRBUUMYAIZDU
 MSNGIUNUGCGLOLAIRFTVINUMYADBSFIUANOUVTOSYNNIWIILHKNZIAMVUOUFEGPRIFQLBRZIOYNYBFTYBURTICZSEADEASIPYSMWIDLRIA
 CRDSIOGPIINAMZTIMWDQEOLZIACRDSQAUAVGIPPQMSRSPAPZMAUBFUAFNBANFBEMAURODUNVTRSCVBUTREVTFMIECBUAQETIEIWTBTAR
 ODFISDZIIIPDBACUAHVSZABUMOLCREBSIDSNUYGIZTGUABTNUGTUALZSZABRLBFAZPMVZSTTYSZIASEFLUKIDINIBQLFIPMATDUUMYASLHK
 BFUZZBCCVCUBFUAFNUKMEYBYIAUQBTDS CBDPYBLSPSSFINIAQDVGUBPRMTMYEIEOUAHKTVSLCREBBKAUGATYELEIISZEAIEBUMLSPPSS

```
INITUKCUYSGPMQUZDUAIEIZIUBUCPMMSLYDUIPFLAUQUSLUGIFGBMVUNNMTVASTADNNWUMDPYBDSFNNWUMYALTBUUTYGBFSYBCIDBITYRL
NWBHDUQSWSSQCUESBSFIULPCUDIAUDBUBTIIVBUMYFSEVHUTUCKATCUDINWUMUIPLSDUAEBVTIFBTUALSPSSFINTFMVBUIMBDBGPATYFTIY
BTDSCBELDIRFTVQLRUUKCBCUDSNEGBTCMPOETITNMNTYGBIZFIAPRLPAUMFSIULXTFQLAII BVITIFTTPSNSTDIACLRTIFBSYUZBEQMYTMS
NGIUMWADVUAABFGEACTYTVYGCSYTFASAI CRBFBCSDDSPSGHCUYSGPSYYSSTNBEMYNEQLIFQMYTMAQPADRLMFDSTRHKDIACMWIDNWBWNO
YCAIRFTVOUOIYGBFBLRUNNWUBDQBFBYDSIAGYCUFLHKTIVIFUQTYAACYSLSLAPIYDSYBULIFLSUQUADUZMWICZQASLAIEBTCPVMSNGIUDM
BQTIIDERNZIAEOYAQNLRQMQLYGIDDIUABTCUNUHKEBLPCUDIIRSZYBLRIACRDSTXDUEBLZSZABMWTVBEGBVTDNLCDUSNNWNISRINYOSSF
QSWTSDKWGLOLPIAEQYAQNTFQLFIMWIDQITFFLIRIUBFTIUCDBPROIVSDSNUGYOTSTFINUMSTBKNUGYIRVTDYHRSZSTTYBFIDYLBKTIDB
SFUAPIVYBBFAQOIVTTAKRGVTITICELDSIVSLACOIVTINNFTYADVMSLGMASVMAUBSIMNIAAUAPLPNQUAUGSIBUYGISNTFLICGBSXPDTYYS
GHURUAIYYOTSTFLINWUMSTBKNUGYMYTELBRQLDSRTCRBFTIDPLAFSSUIRLNAMHSTILATBSIDIACUAFYLDYAFSBUUFASLXTFLIAMODYMN
BVQMSQOSYTVTNUFUPAPADYRFBGVTIOPAIRSZBFIDTICBDNIYTVBUUFASIPBFIDSLYIYTQITFQLVTVYOTDQUAQUMSAFIALAQSSFLAIDBY
IDEBZSSTFUAINWUMVTNUFUPAZLIFZIUUAHVTFBGBQYINSEQYOTSTFLIUHZSRBFLIYBINFTYOSSFYONIUGSIYBEGBNIEBIFBFDYSTTYDS
FSUIUCGBTSDMTATYDSYCSRBDFTPADSCBDPYBYOTSTFLINWUMSTBKIAATQCVBUQLNPSNUCPRYMAUCBTDSIYBEGBSZUTOIACPSZIIIMWNTFL
MRQLFSBUUFASIPVBVTYBIACRDSCBTFFYSIVEAESTIDSOLGCYIVTBFSYTUUKCUYSGPMDBDSPTYNEDBPRIRLNUGCISDYOTSTFLINWUMLSOLEI
VTNUGCBFSYTUICGBYAFIATBVSZABTFFLCXUFEBTFFELBFDUSTICQLFRZSPMYAIAMVUOITCKGULDZSEVFBTFNYUBFIKIAVGBFBESONTDSBF
DYIEITTBZMWITOGBOIMYOLAIVGELFIGBNDEIINUMTIUQAQVTTITTAUKUABFTYFWUABGTVYTIHIDBYMNBVTWLSZGUYAQNLSAPFPBFIAGLOL
APUCIFGBYSSSFBFIDPLNQAANUGRNIFUAVGIPPMQMSLDYAESDUSIQLYGONPRYTTOUADBMFBDLOBEYQCUYSGPSAICVIBFGYTVSEFLUKTIRTUC
IOAMUKIOBFCBIZSTTVYOMZTVVTOUAIFFLITOIYSMTKIPSRFCUIYBTEMMNGIUYSBFIDBFBGLYIAPTYAABAPECTVLNBVTZTVIAEAYTQDPT
PAYGAUCUBIMYSYIAUDDNAIIMMDDINUBQDQYTAITBCSEGBZLSEQMEVFLFIAPWSINDSDPLACRBFYIOBFTVGAQITPEBDQSQBFVTCYDYMN
BEQLFIPAUMQSVXGBLKLYIAPTYAICQYTOYIZLQPPSSTOUIQZSKVQYQDAMDNFTPAUABFDYIEITTBZMWITAGBIZTABFIOIODUAI RVUADUFIBF
TVNTTYGADUXNSYTFURBEFLCXUAQUYIYONTASIYUMOLAUNNIFTYGLMSRFFLCXUAHEMYBUNUUKBFUAFYFLUKUAQYICGBNUUCQAICMSNGAP
RLOBSAUKBFTIQSVSYA
```

Langkah Penyelesaian

Playfair Cipher sangat berkaitan dengan penggunaan bigram (karena aturan enkripsinya berpusat pada pemetaan bigram di sebuah matriks kunci 5×5). Kelemahan *cipher* ini:

- Rentan terhadap analisis frekuensi **bigram**, jika teks *cipher* cukup panjang.
- Bigram dan kebalikannya akan menghasilkan *cipher* yang saling berkaitan: jika bigram plaintext *XY* (baik hubungan searah baris, kolom, maupun *rectangle*) dienkripsi menjadi *cipher PQ*, maka bigram kebalikannya *YX* akan dienkripsi menjadi *QP*, kebalikan dari *cipher* sebelumnya.

Penyelesaian soal ini dibagi menjadi dua tahap. Tahap pertama adalah **analisis frekuensi bigram secara manual** untuk menambatkan sebagian huruf pada matriks kunci. Tahap kedua adalah **pencarian otomatis dengan *Simulated Annealing* (SA)** untuk melengkapi sel-sel yang tersisa, karena melanjutkan penebakan manual sel demi sel terlalu lambat.

Tahap 1: Analisis Frekuensi Bigram

1. Hitung frekuensi bigram **non-overlapping** pada *ciphertext* (`Solver.freq_bigram` pada `src/23/solver.py`). Perhitungan ini dilakukan pada *ciphertext* **sebelum** dilakukan substitusi apa pun, sehingga hasilnya tidak berubah walau hipotesis matriks kunci diubah-ubah. Terdapat **392** bigram unik, dengan sepuluh besar sebagai berikut.

Bigram (non-overlapping)				
Peringkat	Cipher	Freq (%)	Referensi	Freq (%)
1	BF	3.24	TH	3.56
2	UA	1.95	HE	3.07
3	GB	1.87	IN	2.43
4	VT	1.62	ER	2.05
5	TV	1.60	AN	1.99
6	YA	1.45	RE	1.85
7	IA	1.34	ON	1.76
8	DS	1.22	AT	1.49
9	ID	1.22	EN	1.45
10	QL	1.21	ND	1.35

2. Susun hipotesis awal dari empat pencocokan berikut:

Cipher	Freq (%)	Dugaan <i>Plain</i>	Dasar Dugaan
BF	3.24	TH	bigram tersering pada kedua sisi
GB	1.87	HE	peringkat 3 cipher vs bigram Inggris tersering kedua
VT	1.62	RE	pasangan resiprokal dengan TV
TV	1.60	ER	pasangan resiprokal dengan VT

UA belum dicocokkan karena cukup sulit memetakan huruf *plain* yang beririsan. Melanjutkan ke VT/TV, keduanya muncul dengan frekuensi yang mirip, sehingga hipotesis selanjutnya adalah memetakan VT/TV ke **RE/ER**, karena memang keduanya merupakan pasangan bolak-balik yang sama-sama sering muncul dalam Bahasa Inggris (2.05% dan 1.85%). Pencocokan GB → **HE** juga terverifikasi lewat kemunculan BG → **EH**.

3. Turunkan hubungan posisi antarkhuruf dari keempat hipotesis tersebut. Ketiganya ternyata merupakan hubungan *rectangle*, sehingga masing-masing mengunci baris dan kolom relatif antarkhuruf:

- BF → **TH**: B sebaris dengan T, dan F sebaris dengan H.
- GB → **HE**: G sebaris dengan H, dan B sebaris dengan E.
- TV → **ER**: T sebaris dengan E, dan V sebaris dengan R.

Gabungan ketiganya membentuk blok T-E-B pada satu baris, F-G-H pada baris lain, serta R-V pada baris ketiga, dengan kolom yang saling bersesuaian. Penambahan TF → **IT** kemudian menempatkan I. Total **9 dari 25** sel berhasil ditambahkan secara manual, dan inilah isi GRID pada `src/23/solver.py`.

Catatan: hipotesis sempat keliru ketika R dan V diletakkan pada baris yang sama dengan T, E, dan B. Susunan itu memaksa BE (0.88%) terbaca sebagai **VB**, sebuah bigram yang praktis tidak muncul dalam Bahasa Inggris. Kontradiksi inilah yang menjadi dasar penolakan hipotesis tersebut.

Hasil tahap ini adalah hipotesis matriks kunci berikut, dengan sel kosong menandakan posisi yang belum dapat disimpulkan dari analisis bigram:

I				
T	E	B		
F	G	H		
R	V			

Tahap 2: *Simulated Annealing*

Melanjutkan tahap 1 secara manual untuk 16 sel sisanya tidak efisien. Seringkali kesalahan posisi satu huruf baru ditemukan beberapa langkah kemudian. Karena itu, sisa pencarian diserahkan pada *Simulated Annealing*.

1. Hasil dekripsi dinilai dengan **fungsi objektif** berupa **log-probabilitas quadgram** (`Solver.score`), yaitu

$$S = \sum_i \log_{10} P(q_i),$$

dengan q_i adalah quadgram ke- i (jendela 4 huruf yang saling tumpang-tindih) pada hasil dekripsi. Nilai P diambil dari tabel **QUADGRAM** pada `src/frequencies.py` (*quadgram* yang tidak terdaftar diberi nilai dasar 0.001 sebagai penalti). *Quadgram* dipilih karena menilai kelayakan rangkaian huruf yang lebih panjang daripada bigram, sehingga lebih tegas membedakan teks Inggris dari teks acak.

2. Satu langkah dilakukan dengan menukar isi dua sel bebas secara acak (`Solver.swap`). Penukaran antarbaris atau antarkolom penuh hanya diaktifkan bila seluruh sel bebas, karena operasi tersebut akan menggeser sel-sel yang sudah dikunci.
3. Solusi tetangga dengan skor lebih tinggi selalu diterima. Kemudian, solusi yang lebih buruk tetap diterima dengan peluang $\exp(\Delta S/T)$, sehingga pencarian dapat keluar dari optimum lokal. Suhu awal $T = 20$ diturunkan bertahap sebesar 0.5 hingga mencapai 0, dengan 2500 percobaan pada tiap tingkat

suhu (total ± 100.000 evaluasi per *restart*). Proses ini diulang sebanyak 2 *restart*, dan skor terbaik yang akan dipakai.

4. Selama pencarian, penilaian hanya dilakukan pada **2400 karakter pertama** *ciphertext*. Ukuran ini sudah cukup untuk membedakan kunci benar dari kunci salah, namun memangkas waktu secara signifikan. Dekripsi penuh baru dijalankan sekali di akhir memakai kunci terbaik.

Hasil

Pencarian konvergen ke skor -6877.2 dengan waktu eksekusi sekitar **2 menit 30 detik**, dan menghasilkan pemetaan lengkap untuk seluruh **392** bigram unik. Pengujian berulang memberi kunci yang sama secara konsisten.

Perlu dicatat bahwa penguncian sembilan sel hasil tahap 1 sangat menentukan. Ketika seluruh 25 sel dibiarkan bebas, pencarian dengan jumlah *restart* yang sama kerap berhenti pada optimum lokal (misalnya pada skor -7098.3) dengan hasil dekripsi yang tidak terbaca. Dengan kata lain, analisis manual dan pencarian otomatis saling melengkapi: analisis manual memperkecil ruang pencarian dari 25 sel menjadi 16 sel, sementara SA menyelesaikan bagian sisanya.

Artefak

Skrip *solver* yang digunakan untuk menghitung frekuensi bigram *ciphertext*, menjalankan *Simulated Annealing* untuk melengkapi matriks kunci, menurunkan pemetaan bigram *cipher* $\rightarrow$ *plain* berdasarkan aturan Playfair, mendekripsi *ciphertext*, serta mencatat setiap hasil dekripsi ke `log/plain/` dan metadatanya (pemetaan, matriks kunci, frekuensi bigram, dan skor akhir) ke `log/metadata/`.

```

1 import json
2 import math
3 import os
4 import random
5 import sys
6 from datetime import datetime
7 from typing import Dict, List, Tuple
8
9 sys.path.insert(0, os.path.dirname(os.path.dirname(os.path.abspath(__file__))))
10 from frequencies import QUADGRAM
11
12 ALPHABET = "ABCDEFGHIJKLMNOPQRSTUVWXYZ"
13
14 _ = 0
15 GRID = [
16     [_, _, _, _, _],
17     ["I", _, _, _, _],
18     ["T", "E", "B", _, _],
19     ["F", "G", "H", _, _],
20     ["R", "V", _, _, _]
21 ]
22
23 class Solver:
24     def freq_bigram(self, string: str) -> Dict:
25         freq_data = {}
26         total_ngrams = len(string) // 2
27         for i in range(0, len(string), 2):
28             ngram = string[i:i+2]
29             freq_data[ngram] = freq_data.get(ngram, 0) + 1
30         for ngram in freq_data:
31             freq_data[ngram] = (freq_data[ngram] / total_ngrams * 100)
32         return dict(sorted(freq_data.items(), key=lambda item: item[1], reverse=
True))

```

```

33
34 def find_pos(self, grid: List, letter: str) -> Tuple[int, int]:
35     for r, row in enumerate(grid):
36         for c, val in enumerate(row):
37             if val == letter:
38                 return r, c
39     return None
40
41 def decrypt_pair(self, grid: List, a: str, b: str) -> str:
42     ra, ca = self.find_pos(grid, a)
43     rb, cb = self.find_pos(grid, b)
44     if ra == rb:
45         return grid[ra][(ca - 1) % 5] + grid[rb][(cb - 1) % 5]
46     if ca == cb:
47         return grid[(ra - 1) % 5][ca] + grid[(rb - 1) % 5][cb]
48     return grid[ra][cb] + grid[rb][ca]
49
50 def build_mapping(self, grid: List, bigrams: List) -> Dict:
51     return {bigram: self.decrypt_pair(grid, bigram[0], bigram[1]) for bigram in
    bigrams}
52
53 def decrypt(self, string: str, mapping: Dict) -> str:
54     return "".join(mapping[string[i:i+2]] for i in range(0, len(string), 2))
55
56 def score(self, plain: str) -> float:
57     total = 0.0
58     for i in range(len(plain) - 3):
59         total += math.log10(QUADGRAM.get(plain[i:i+4], 0.001))
60     return total
61
62 def random_grid(self, grid: List) -> List:
63     used = [val for row in grid for val in row if val != 0]
64     rest = [letter for letter in ALPHABET if letter not in used]
65     random.shuffle(rest)
66     return [[val if val != 0 else rest.pop() for val in row] for row in grid]
67
68 def free_cells(self, grid: List) -> List:
69     return [(r, c) for r in range(5) for c in range(5) if grid[r][c] == 0]
70
71 def swap(self, grid: List, free: List) -> List:
72     new_grid = [row[:] for row in grid]
73     if len(free) == 25 and random.random() < 0.1:
74         a, b = random.sample(range(5), 2)
75         if random.random() < 0.5:
76             new_grid[a], new_grid[b] = new_grid[b], new_grid[a]
77         else:
78             for row in new_grid:
79                 row[a], row[b] = row[b], row[a]
80     return new_grid
81     (ra, ca), (rb, cb) = random.sample(free, 2)
82     new_grid[ra][ca], new_grid[rb][cb] = new_grid[rb][cb], new_grid[ra][ca]
83     return new_grid
84
85 def anneal(self, string: str, bigrams: List, grid: List, temp: float = 20.0,
86             step: float = 0.5, rounds: int = 2500) -> Tuple[List, float]:
87     free = self.free_cells(grid)
88     current = self.random_grid(grid)
89     current_score = self.score(self.decrypt(string, self.build_mapping(current,
    bigrams)))

```

```

90     best, best_score = current, current_score
91     while temp > 0:
92         for _ in range(rounds):
93             trial = self.swap(current, free)
94             trial_score = self.score(self.decrypt(string, self.build_mapping(
trial, bigrams)))
95             delta = trial_score - current_score
96             if delta > 0 or random.random() < math.exp(delta / temp):
97                 current, current_score = trial, trial_score
98                 if current_score > best_score:
99                     best, best_score = current, current_score
100         temp -= step
101     return best, best_score
102
103     def solve(self, string: str, grid: List, restarts: int = 2) -> Tuple[List,
float]:
104         sample = string[:2400]
105         bigrams = list(self.freq_bigram(sample))
106         best, best_score = None, -math.inf
107         for _ in range(restarts):
108             result, result_score = self.anneal(sample, bigrams, grid)
109             if result_score > best_score:
110                 best, best_score = result, result_score
111         return best, best_score
112
113     def load_ciphertext(base_dir: str) -> str:
114         with open(os.path.join(base_dir, "f.txt")) as f:
115             return f.read().strip()
116
117     def write_plain(base_dir: str, timestamp: str, decrypted: str) -> str:
118         path = os.path.join(base_dir, "log", "plain", timestamp + ".txt")
119         os.makedirs(os.path.dirname(path), exist_ok=True)
120         with open(path, "w") as f:
121             f.write(decrypted)
122         return path
123
124     def write_metadata(base_dir: str, timestamp: str, mapping: Dict, grid_manual: List,
grid_filled: List, freq: Dict, score: float) -> str:
125         path = os.path.join(base_dir, "log", "metadata", timestamp + ".json")
126         os.makedirs(os.path.dirname(path), exist_ok=True)
127         payload = {
128             "timestamp": datetime.now().isoformat(timespec="seconds"),
129             "mapping": mapping,
130             "grid_manual": grid_manual,
131             "grid_filled": grid_filled,
132             "freq_ngram": freq,
133             "score": score,
134         }
135         with open(path, "w") as f:
136             json.dump(payload, f, indent=2)
137         return path
138
139     if __name__ == "__main__":
140         base_dir = os.path.dirname(os.path.abspath(__file__))
141         content = load_ciphertext(base_dir)
142
143         solver = Solver()
144         freq = solver.freq_bigram(content)
145         grid, score = solver.solve(content, GRID)

```

```

147 mapping = solver.build_mapping(grid, freq)
148 decrypted = solver.decrypt(content, mapping)
149
150
151 timestamp = datetime.now().strftime("%Y-%m-%d_%H%M%S")
152 plain_path = write_plain(base_dir, timestamp, decrypted)
153 metadata_path = write_metadata(base_dir, timestamp, mapping, GRID, grid, freq,
154                                score)
155 print(decrypted)

```

src/23/solver.py

Plaintext Hasil Dekripsi

PARTIWHATCHANGEWILLTHENNEWPOWERSBRINGTOGENSOKYOIDLIKETO THANKEVERYONEFORTAKINGTHETIMEOUTOFTHEIRBUSYLIVESTOPARTICIPATEIHIEDANOAKYUXUWISHTOCOLLECTYOURVALUEDVIEWSANDREPORTTHEMFORTHEBENEFITOFALXLGENSOKYOWITHTHATINMINDIDLIKETOTURNITTOVERTOXOURSYMPOSIUMSMODERATORHUMANMAGICIANANDONEOFGENSOKYOSLEADINGEXPERTSONYOUKAIMARISAKIRISAMEAHUMIAMTHEMODERATORMARISACAUSEUHIEMANBECAUSEIBELIEVEDTHESYMPOSIUMTOBEPRESENTEDHEREMAYBEINTERESTINGIHAVEACXCEPTEDTHEHONOROFTHISROLEANDUMXMIUSTSPEAKASYOUNORMALXLYDOHEARINGYOUTALKLIKETHISISDISGUSTINGOKAYTHENLETSSARTWITHTHEINTRODUCTIONSTAKEITAWAYIAMKANAKOYASAKAIMANAGETHESHINEONYOUKAIMOUNTAINIUSUALLYTHINKOFNOTHINGBUTSHINTOBUTIAMINTERESTEDTOHEARFROMTHOSEOFOTHERXRELIGIONSTODAYIAMBYAKURENHIXIXIRIXIAMTHECHIEFPRIESTOFTHEMYOURENTEMPLEONTHEOUTSKIRTSOFTHEVILXLAGEALTHOUGHIMTHECHIEFPRIESTIMSTILLANOVICEYETXTOATTAINENLIGHTENMENTBUTPLEASECONSIDERMYVIEWSTODAYIAMTOYOSATOMIMINOMIKOUMPRESENTLYIVERENOUNCEDMYHUMANITYANDAMAHHERMITIONLYAWOKERECENTLYSOIMSTILLHALFASLEXEPBUTIFELTTTHATIMIGHTDEEPENMYUNDERSTANDINGOFGENSOKYOBYATXTENDINGTODAYSOWASDETERMINEDTODOSOSPEAKINGOFWHICHOTHERRESIDENTSOFGENSOKYOSEXEMTOBEOGOODTERMSIFTHEYCANHAVEASYMPOSIUMLIKETHISWHOATHISISIUSTTHEFIRSTONEBUTWEREALWAYSUPPERTAKINONNEWCHALXLENGESTHUGHISTHATSOWHATCHANGEWILXLTHENNEWPOWERSBRINGTOGENSOKYOSOWEGOTHEREASHINTOGODABUDXDHISTPRIESTANDATAOISTHERMITSINCEWEVEPRETTYMUCHGATHEREDUPABUNCHOFBUSINESSXSRIVALSTHISUGHTABEGOXODATHREXEWAYTALKBETWEXENTHREERELIGIOUSFIGURESXSOWHATAREYALLGOINGTOTALKABOUTARENTYOUOURMODERATORSUPPOSEDTODECIDETHATOHYEAHOKAYTHENTHEFIRSTXTHEMEISWHATCHANGEWILLTHENNEWPOWERSBRINGTOGENSOKYOIUSTTHROWWHATEVERYAWANTOUTXTHEREASYOUALREADYARELAUGHTHENNEWPOWERSREFERSTOALXLOFUSRIGHTAHRIGHTIUSTXTOCLARIFYALITXTLEYALLTHREXEXARETHENNEWCOMERSTHEHIDDENMEANINGOFTHESYMPOSIUMISTHATYAMIGHTBEHOSTILEELEMENTSHERETODISRUPTOURPEACEFULGENSOKYOORSOMETHINANDHEREITHOUGHTYOUWERESTARTINGTOGETUSEDTOMEDONTMATTERHOWMANYYEARGOBYNEWCOMERSWILXLALWAYSNEWCOMERSFROMTHEOLDTIMERSPERSPECTIVEANYWAYWELLTHENASTHEREALNEWCOMERISHALXLSTARTGOFORITMYFIRSTIMPRESSIONOFGENSOKYOBESIDESTHEFACTTHATTHEAREYOUKAIWHOCAUSEHARMTOMANKINDISTHATXTHEMINDSOFHUMANSARECAPRICIOUSANDEASILYSWAYEDTHEHUMANSOFTHEVILXLAGEPENDTHEIRDAYSIINMENTALVAGRANCYFALLINGTOIMPROVETHEIRLOTINLIFEANDTHEYOUKAIAREALXLOWEDTOTAKEADVANTAGEOFTHEIRCAPRICIOUSNESSTOTHEIRHEARTSCONTENTFRANKLYITSAMIRACLETHATSOCIETYHASNTCOLLAPSEDYETANDIBELIEVETHECURRENTBALANCEISDANGEROUSLYUNSTABLEUHGUESXSWHYDOESGENSOKYOHAVENOADMINISTRATORSOCIETYCANNOTFUNCTIONWITHALXLHUMANPOSSESXINGTHESAMESOCIALSTANDINGANDRANKANADMINISTRATORHATSRIGHTHUMANSNEXEDLEADERSTOGUIDETHEMTOWARDSAHOPEFILXLEDFUTUREIFNOXONEXELSEISQUALIFIEDTHENIWOLDCERTAINLYIWONDERWHATKINDOFFXUTURELIESINTOREFORTHEHUMANSUNDERTHERULEOFSOMEONEWHOISNOLONGERHUMANRULEISSUCHANUGLYWORDITWOULDJUSTBEASTOPGAPMEASUREUNTILOMEONEFROMAMONGTHEHUMANISABLETOIMITATEANDOVERTURNMEMAYISAYAWORDYOU SAYTHAT THEREISNOADMINISTRATORINGENSOKYOBUTIDONTTHINKTHATSQUITETRUWHATDOYAMEANGENSOKYOSCURXRENTFORMCANXNOTBEMAINEDWITHOUTTHEYOUKAIPOWERGENSOKYOWASCREATEDBYYOUKAITOSAVETHEMSELVESAFTERALXLTHEHUMANSINGENSOKYOEXISTFORNOFURTHERPURPOSETHANTOPRESERVETHEYOUKAIITHEHUMANSAREBEINGRULEDBYTHEYOUKAIAGREXETHEYOUKAIKEEPWATCHONEACHOTHERTOPREVENTOTHERYOUKAIFROMRUNNINGRAMPANTANDRULINGOVERTHEHUMANSANDTOMAKESURETHEHUMANSDONTAWAKENTOTHEIRACTUALCONDITIONANDTURNDOWNTHEWRONGPATHTHEREFORETHEFACTTHATARULERHASNTEMERGEDFROMAMONGTHEHUMANISINGENSOKYOSBESTINTERESTSOTHATSSHOWITISBUTIFTHATSTHECASETHENTHEPOXORHUMANSHEREAREPITIALECREATURESTHINKOFITXTHISWAYARETHEANIMALSINAZOXOUNFORTUNATEANDAZOISAINSTITUTIONWHEREENDANGEREDANIMALSAREPROTECTEDANDHUMANS CANOBSERVE THEMANDLEARNABOUTXTHEMTAKINGANATURALLYFREXEBIRDANDLOCKINGITUPINSMALXLCAGEWHENYOULOOKATITFROMTHATPERSPECTIVEITSEXEMSQUTEUNFORTUNATEBUTNOWITDOESNTHAVETOWORRYABOUTPREDATORSORRUNNINGOUTOFFXFOODANDTHEREWILXLALWAYSBEPLentyOFHUMANSFORITTOLOOKATSOITWILXLNEVERGETBOREDIFYOUTHINKOFITLIKETHATARENTXTHEYFORTUNATESOGENSOKYOSHUMANSAREANIMALSINAZOXOWHATHAPPENEDTOHUMANIDIGNITYDIGNITYHUHIDONTTHINKTHATSSOMETHINGYOURECEIVEFROMOTHERSTHOUGHWHATDOYOUTHINKHMMXMEWHATSDIGNITYLIKEPRIDORSSELFRESPECTOFCOURSEIMPROUDOFEVERYTHINGIDOBUT EVENIFIWEREFORCEDINTASOMETHINIDBEPERFECTLYFINEWITHTHATSOLO

GASIWEREHAVINFUNISEXETHENPLEASEFORGETABOUTWHATISAIDWAITWHATWEREYATALKINABOUTLAUGHTHEREASONKANAKOCAMETOGENS OKYOANYWAYIWASHAVINAHARDTIMEFOLXLLOWINTHATHONESTLYIDONTGIVEAFLIPABOUTTHEHUMANSOFTHEVILXLGETALKABOUTYERSELV ESTHETHEMEWASWHATAREYOUBRINGINGTOGENSOKYOAFTERALLSOFIRSTOFALLWHYDYAMOVETOGENSOKYOANYWAYICAMETOGENSOKYOTOSE EKNEWFAITHINTHEOUTSIDEWORLDSDRINESHAVEBECOMEMERETOUISTXTRAPSORCALLEDNONSENSICALTHINGSLIKEPOWERSPOTSSOPLAC ESFORHUMANSTOXOFFERTHEIRFAITHTOTHEGODSHAVEDISAPXPEAREDGODSAREBEINGSTHATWILLCEASETOEXISTWITHOUTHUMANSBELIEV INGINTHEMISOYASAYBUTIDONTTHINKYAVEGOTTENMUCHFAITHINGENSOKYOEIFHERIUSTBELIEVINGTHATWEREREALISENOUGHINTHEOUTS IDEWORLDTHEYVESTOPPEDBELIEVINGTHATGODSEVENEXISTINDEXEDEVERYTHINGREALLYCHANGEDONCEWECAMETOGENSOKYOHHEREHUMAN SBELIEVETHATGODSEXISTASANEVERYDAYFACTWWELXLEVENIFIBELIEVEDINYAORNOTYERRIGH THEREAINTYATHATSNOTGOXODENOUGHFO RTHEOUTSIDEWORLDDEVENIFYOUPERFORMMIRACLERIGHTBEFORETHEIREYESTHEYLXLCONVINCE THEMSELVESITSANILLUSIONANDABSOL UTELYREFUSETOACXCEPTTHATITACTUALLYHAPXPENEDHUMANSAREEASILYSWAYEDCREATURESXSOTHATSTHELATESTFADAFADYESANENTI RELYINTENTIONALFADBUTYOUKNOWTHEBESTPARTISTHATEVENWHILETHEYDONTTHINKTHATGODSORYOUKAEXISTTHEYSTILLBELIEVEIN ALIENSOHNEVERHEARDABOUTTHATTHEMOMENTXTHATHUMANSREALIZEDTHATTHEVASTNESSOFSPACEWASBEYONDTHEIRABILITYTOCONTR OLTHEYSTARTEDTOCONVINCE THEMSELVESTHATX THERE MUST BEALIENSXSOMEWHEREOUTTHEREXEVEN THOUGHNOBODYS CONFIRMEDITYETR ATIONALISMSUDXDENLYTURNSTOMYSTICISMWHENITGOESTOXOFARFUNXNYISNTITIHAVEBEXENCURIQUSABOUTTHISFORAWHILENOWBUTY OURSHRINEHASTWOGODSAMIRIGHTWHYISTHATTECHNICALXLYWEVEGOTXTHREEBUTNEVERMINDTHATFORNOWOURSHRINESBEXENTHROUGHQ UITEABITWELXLIUSTLETMESAYSUWAKOISAGOODPARTNEROOHIKNOWTHISONESUWAKOSTHEONEWHOREALLYRULESTHESHRINERIGHTANDYE RIUSTXTHEONERUNXNINSTUFXFOUTINFRONTAHEMALTHOUGHTHATSNOTQUITEHOWITWORKSITSNOTWRONGTHATXTHEMORIYASHRINEBELON GSTOSUWAKOBUTSUWAKODOESNTHAVEMUCHINTERESTINGATHERINGFAITHYOUSEESOIHAVENCHOICEBUTXTOGATHERFAITHFORUSBYMYSE LFNCHOICESPEAKINGOFWHICHMSKANAKOYOUAREFROMSUWAOFSHINANOAREYOUNOTIMNOTBUTSANAEANDSUWAKOAREILIVEDTHEREFORAL ONGTIMETOXOTHOUGHACTUALXLYIMALSOFROMSHINANO THATBRINGSBACKSOMANYMEMORIESOHEREALXLYWHENEVERICAMETOSUWATHELAKE WASALWAYSFULLOFSUCHCLEANWATERITWASSUCHABEAUTIFULPLACEWHENWASTHISIMNOTSUREMAYBEABOUTATHOUSANDYEARSAGONOWADA YSITSMADEANCOMPLETEABOUTFACEANDBECOMEARATHERCOLDXCITYALTHOUGHIMPETTYPROUDOFITSEXCELXLENTFACTORIESANDTHEPRE CISIONMACHINERYTHEYMAKETHEREAREALSOSOMENICEHOTSPRINGSBYTHETRAINSTATIONTHECONSTRUCTIONOFTHEMYOURENTempleWHO ALETSNOTGETCAUGHTUPINTALKOFSOMECOUNTRYSIDETOWNNOONEHEREKNOWSABOUTNEXTUPISYOUPPOINTINGATBYAKURENWHYDOUCOMET OGENSOKYOIAMUSTASIMPLEMONKINTRAININGBUTBECAUSEOFAMINORDISAGREEMENTI WASBETRAYEDBYHUMANSANDSEALED AWAYHOWEVE RSOME OFMYOLDFRIENDSWHOMISXSEDMECAMETOMYRESCUEANDTHATPLACEIUSTHAPPENEDTOBEGENSOKYONOWWAITDOESTHATMEANYOUREH EREBYCOINCIDENCENOTCOINCIDENCEBUTRATHERINEVITABILITYWERETHEYNOTINGENSOKYOMYFRIENDSWOULDHAVEBEXENUNABLETOMA INTAINTHEIRSTRENGTHYOUBUILTYOURTEMPLEDIRECTLYABOVEWHEREI WASXSLEXEPINGWASTHATALSOINEVITABLEWHATMIGHTYOUBETA LKINGABOUTOHEYEAHIKOSGRAVEWASUNDERTHETEMPLEWASNTITYOUBUILTATEMPLEFORBUDXDHISMARELIGIONIUSEDFORPOLITICALMEA SURESONTOPOFITANDIHEARDITWASNOTIUSTANYTEMPLEBUTAYOUKAITEMPLEFULXLOFYOUKAIMONKSATEMPLE FILLEDWITHSUCHDETESTA BLEXEVILPOWERHOWRUDEWHILEITMAYBEFILXLEDWITHYOUKAITHOSEYOUKAIAREMORESINCERETHANHUMANSIUSTASHUMANS CANBECOMEH ERMITSANDCELESTIALSYOUKAICANBECOMEGODSANDBUDDHASMYOWNPOWERISALSOASVASTASITISTHANKSTOYOUKAISOYOUREAMONKPOX SESXSEDBYEVILTHENAMONKWHOHASFALXLENUNTHINKABLYFARORSOIDLIKETOSAYBUTASXSOMEONEWHOUSED BUDXDHISMFORPOLITICALE NDSISUPXPOSEIDONOTHAVETHERIGHTTODOSOWELXLTOTOBHONESTWEBUILTTHETEMPLETHEREBECAUSEOURTREASUREHUNTINGMOUSEREPO RTEDTHATTHERESSOMETHINGINCREDIBLEBURIEDXDOWNTHEREANDSOWESURVEYEDITWHENITTURNEDOUTXTOBESOMETHINGTHATSEXEMED MOREDANGEROUS THANWHATWEWEREXEXPECTINGWETHOUGHTWESHOULDDOOURBESTXTOHIDEITSOIWASTHATDANGEROUSXSEEMINGTHINGTH ENWHILEIDONTTHINKMYINTUITIONWASWRONGIREGRETXTABLYLACKEDTHEPOWERMEANYOUCOULDNTKEEPMESEALEDUNFORTUNATELYF OROURNAMELESSXSOCALXLEDMONKWHOTHOUGHTXTOKEEPMESHUTOFFXFOREVERTHEREVIVALOFTHESAINTYOU CANFIGHTLATERSWEATSOLA STUPISMIKOSXSTORYIPUTMYSELFOTOSLEEPWITHASPELXLIHADPLANXNEDTOWAKEUPASAHERMITANDENIOYTHEWORLDASIDESIREDBUTIAW OKEONCEI WASAWARETHATIWASNOLONGERINTHEORDINARYWORLDITWASTHENTHATICAMETO LEARNI WASHHEREINGENSOKYOIMINTHEPROCES SOFCAREFULLYSTUDYINGTHISWORLDANDTODAYWILLBEPARTICULARLYINFORMATIVEISTILXLAVENTFIGUREDOUTXTHEREASONIMHERET HOUGH THEREASONYOU'RE HEREISXSIMPLEINTHESAMEWAYTHATYOUKAIORGODSLIKE MEENDUPHEREAFTERTHEIREXISTENCEISRENOUNCEDI NTHEOUTSIDEWORLDYOUVEALSOCEASED TOEXISTIVECEASED TOEXISTBUTARENTERHERERIGHTNOWCOMMONSENSEANDUNCOMMONSENSEINOT HERWORDREALITYANDFANTASYTHEOUTSIDEWORLDANDGENSOKYOARESEPARATEDALONGTHATSAMELINEHENCETHINGSTHATHAVEDISAPPE AREDBEENFORGOTTENORHAVEHADTHEIREXISTENCERENOUNCEDINTHEOUTSIDEWORLD CANBEFOUNDINGENSOKYOSOTHATMEANSIWASCOMPL ETELYFORGOTTENSOIVECHANGEDFROMAHUMANTOAGODLIKEEXISTENCETHENTHATSCORXRECTWELXLASFARASIKNOWYOUWEREALREADYR EGARDEDASAGODINSHINTOXOHYESIFORGOTTOMENTIONTHISBUTIDONTNORMALXLYLIVEINGENSOKYOHUHLIVEINSPACEICREATEDMYSE LFLIVINGSOMEWHEREWHERE NOBODY CANBOTHER THEMISTHEESSENCEOFWHAT AHERMITISNOWTHATCHAMENTIONITANOTHERHERMITKNOWA LSO LIVESINAHARDTOREACHPLACEHOWDYADOSOMETHINLIKETHATANYWAYITSSIMPLEIFYOUSTUDYENOUGHOF THEHERMETICARTSWOULDYO ULIKETOBECOME MYDISCIPLEUHEMMETHINKABOUTITYOUCOULD BECOMEIMMORTALOXOHOPROSELYTIZINGNOWAREWEDONTXTHINKILXLT AKETHATLYINGDOWNSWITCHNOWANDILLTHROWINEXCELXLENTLUCKONALLYOURFORTUNESFORTWOMONTHSTHENALXLOWMEASWELLYOUAREA YOUKAILIKEHUMANAFTERALLYOUCERTAINLYAREPOPULARLAUGHAXANYWAYIGUESXSTHATWRAPSUPTHESELFINTRODUCTIONSSOWHATI WAN TXTOASKYALLNOWISWHADDYAWANTXTODOINGENSOKYOFROMNOWONITHINKIDLIKETOLIVEACAREFREFELIFEWITHOUTBEINGDISTURBEDIZ

ZATSOTRASCENDINGWORLDLYMATXTERSTOLIVEAREFINEDLIFEINAWORLDDWITHNOWORXRIESORANYTHINGATALLTHATSMYIDEALYOUIGH
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EDTOBEBUTAFTERHAVINGBEENTOUCHEDBYMYDISCIPLESHEARTSNOWIONLYTRAINWITHANALTRUISTICSPIRITANDIWISHTOSAVETHOSEYO
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WIMDOINGTHISFORYOU

Plaintext dengan Format Asli

Part I

What Change Will the New Powers Bring to Gensokyo?

I'd like to thank everyone for taking the time out of their busy lives to participate. I wish to collect your valued views and report them for the benefit of all Gensokyo. With that in mind, I'd like to turn it over to our symposium's moderator, human magician and one of leading experts on youkai.

Ah, um, I am the moderator. 'Cause... Uh, I mean, because I believed the symposium to be presented here may be interesting, I have accepted the honor of this role. And, umm...

Just speak as you normally do. Hearing you talk like this is disgusting.

Okay, then. Let's start with the introductions. Take it away.

I manage the shrine on Youkai Mountain. I usually think of nothing but Shinto, but I am interested to hear from those of other religions today.

I am the chief priest of the temple on the outskirts of the village. Although I'm the chief priest, I'm still a novice yet to attain enlightenment, but please consider my views today.

Um, presently, I've renounced my humanity and am a hermit. I only awoke recently, so I'm still half-asleep, but I felt that I might deepen my understanding of Gensokyo by attending today, so I was determined to do so. Speaking of which, the residents of Gensokyo seem to be on good terms, if they can have a symposium like this...

Whoa, this is just the first one. But we're always up fer takin' on new challenges, though.

Is that so...

[What Change Will the New Powers Bring to Gensokyo?]

So we got here a Shinto god, a Buddhist priest, and a Taoist hermit. Since we've pretty much gathered up a bunch of business rivals, this oughta be good. A three-way talk between three religious figures. So, what are y'all going to talk about?

Aren't you, our moderator, supposed to decide that?

Oh, yeah. Okay, then, the first theme is "What change will the new powers bring to Gensokyo?" Just throw whatever ya want out there.

As you already are. (laugh)

The "new powers" refers to all of us, right?

Ah, right. Just to clarify a little, y'all three are the newcomers. The hidden theme of the symposium is that ya might be hostile elements here to disrupt our peaceful Gensokyo or somethin'.

And here I thought you were starting to get used to me...

Don't matter how many years go by, newcomers will always be newcomers. From the old timers' perspective, anyway.

Well then, as the real newcomer, I shall start.

Go for it.

My first impression of Gensokyo, besides the fact that there are youkai who cause harm to mankind, is that the minds of humans are capricious and easily swayed. The humans of the village spend their days in mental vagrancy, failing to improve their lot in life, and the youkai are allowed to take advantage of their capriciousness to their heart's content. Frankly, it's a miracle that society hasn't collapsed yet, and I believe the current balance is dangerously unstable.

Uh, I guess...

Why does Gensokyo have no administrator? Society cannot function with all humans possessing the same social standing and rank.

An administrator?

That's right. Humans need leaders to guide them towards a hope-filled future. If no one else is qualified then I would certainly...

I wonder what kind of future lies in store for the humans under the rule of someone who is no longer human.

"Rule" is such an ugly word. It would just be a stopgap measure until someone from among the humans is able to imitate and overturn me.

May I say a word? You say that there is no administrator in Gensokyo, but I don't think that's quite true.

What do ya mean?

Gensokyo's current form cannot be maintained without the youkai's power. Gensokyo was created by youkai to save themselves, after all. The humans in Gensokyo exist for no further purpose than to preserve the youkai. The humans are being ruled by the youkai.

I agree. The youkai keep watch on each other to prevent other youkai from running rampant and ruling over the humans, and to make sure the humans don't awaken to their actual condition and turn down the wrong path. Therefore, the fact that a ruler hasn't emerged from among the humans is in Gensokyo's best interest.

So that's how it is. But if that's the case, then the poor humans here are pitiable creatures...

Think of it this way: are the animals in a zoo unfortunate?

And a zoo is?

An institution where endangered animals are protected and humans can observe them and learn about them. Taking a naturally free bird and locking it up in small cage... when you look at it from that perspective, it seems quite unfortunate. But now, it doesn't have to worry about

predators, or running out of food, and there will always be plenty of humans for it to look at, so it will never get bored. If you think of it like that, aren't they fortunate?

So Gensokyo's humans are animals in a zoo? What happened to human dignity?

Dignity, huh? I don't think that's something you receive from others, though... What do you think?

Hmm? Me? What's dignity?

Like pride, or self-respect.

Of course, I'm proud of everything I do, but even if I were forced into somethin', I'd be perfectly fine with that so long as I were havin' fun.

I see... Then please forget about what I said.

Wait, what were ya talkin' about?

(Laugh)

[The Reason for Coming to Gensokyo]

Anyway, I was havin' a hard time followin' that. Honestly, I don't give a flip about the humans of the village. Talk about yerselves. The theme was "what are you bringing to Gensokyo", after all. So first of all, why'd ya move to Gensokyo anyway?

I came to Gensokyo to seek new faith. In the outside world, shrines have become mere tourist traps or called nonsensical things like "power spots", so places for humans to offer their faith to the gods have disappeared. Gods are beings that will cease to exist without humans believing in them.

So ya say, but I don't think ya've gotten much faith in Gensokyo, either.

Just believing that we're real is enough. In the outside world, they've stopped believing that gods even exist. Indeed, everything really changed once we came to Gensokyo. Here, humans believe that gods exist as an everyday fact.

W-well, even if I believed in ya or not, yer right here, ain't ya?

That's not good enough for the outside world. Even if you perform a miracle right before their eyes, they'll convince themselves it's an illusion and absolutely refuse to accept that it actually happened. Humans are easily swayed creatures, so that's the latest fad.

A fad?

Yes, an entirely intentional fad. But, you know, the best part is that even while they don't think that gods or youkai exist, they still believe in aliens.

Oh? Never heard about that.

The moment that humans realized that the vastness of space was beyond their ability to control, they started to convince themselves that there must be aliens somewhere out there, even though nobody's confirmed it yet. Rationalism suddenly turns to mysticism when it goes too far. Funny, isn't it?

I have been curious about this for a while now, but your shrine has multiple divine presences, am I right? Why is that?

Technically, we've got multiple, but never mind that for now. Our shrine's been through quite a bit... Well, just let me say the companion deity is a good partner.

Ooh, I know this one. The companion deity's the one who really rules the shrine, right? And yer just the one runnin' stuff out in front.

Ahem. Although that's not quite how it works, it's not wrong that the shrine belongs to the companion deity. But the companion deity doesn't have much interest in gathering faith, you see. So I have no choice but to gather faith for us by myself. No choice.

Speaking of which, you are from that region of Shinano, are you not?

I'm not, but the others are. I lived there for a long time, too, though.

Actually, I'm also from Shinano. That brings back so many memories.

Oh, really?

Whenever I came there, the lake was always full of such clean water. It was such a beautiful place.

When was this?

I'm not sure. Maybe about a thousand years ago.

Nowadays, it's made a complete about-face and become a rather cold city... Although I'm pretty proud of its excellent factories and the precision machinery they make. There are also some nice hot springs by the train station....

[The Construction of the Temple]

Whoa, let's not get caught up in talk of some countryside town no one here knows about. Next up is you. Why'd you come to Gensokyo?

I am just a simple monk in training, but because of a minor disagreement, I was betrayed by humans and sealed away... However, some of my old friends who missed me came to my rescue. And that place just happened to be Gensokyo.

Now, wait. Does that mean you're here by coincidence?

Not coincidence, but rather inevitability. Were they not in Gensokyo, my friends would have been unable to maintain their strength.

You built your temple directly above where I was sleeping. Was that also inevitable?

What might you be talking about?

Oh, yeah. The grave was under the temple, wasn't it?

You built a temple for Buddhism, a religion I used for political measures, on top of it, and I heard it was not just any temple, but a youkai temple full of youkai monks. A temple filled with such detestable, evil power...

How rude. While it may be filled with youkai, those youkai are more sincere than humans. Just as humans can become hermits and celestials, youkai can become gods and Buddhas. My own power is also as vast as it is thanks to youkai.

So you're a monk possessed by evil then. A monk who has fallen unthinkable far... or so I'd like to say, but as someone who used Buddhism for political ends, I suppose I do not have the right to do so.

Well, to be honest, we built the temple there because our treasure-hunting companion reported that "there's something incredible buried down there!" and so we surveyed it. When it turned out to be something that seemed more dangerous than what we were expecting, we thought we should do our best to hide it.

So I was that dangerous-seeming thing, then?

While I don't think my intuition was wrong, I regrettably lacked the power.

Meaning you couldn't keep me sealed, unfortunately for our nameless so-called monk who thought to keep me shut off forever.

[The Revival of the Saint]

You can fight later. So, last up is the final story.

I put myself to sleep with a spell. I had planned to wake up as a hermit and enjoy the world as I desired, but I awoke once I was aware that I was no longer in the ordinary world. It was then that I came to learn I was here in Gensokyo. I'm in the process of carefully studying

this world, and today will be particularly informative. I still haven't figured out the reason I'm here, though...

The reason you're here is simple. In the same way that youkai or gods like me end up here after their existence is renounced in the outside world, you've also ceased to exist.

I've ceased to exist? But, aren't I here right now...?

Common sense and uncommon sense; in other words, reality and fantasy. The outside world and Gensokyo are separated along that same line. Hence, things that have disappeared, been forgotten, or have had their existence renounced in the outside world can be found in Gensokyo.

So that means I was...

Completely forgotten.

So I've changed from a human to a god-like existence, then?

That's correct. Well, as far as I know you were already regarded as a god in Shinto.

Oh, yes. I forgot to mention this but I don't normally live in Gensokyo.

Huh?

I live in a space I created myself. Living somewhere where nobody can bother them is the essence of what a hermit is.

Now thatcha mention it, another hermit I know also lives in a hard-to-reach place. Howd'ya do somethin' like that anyway?

It's simple, if you study enough of the hermetic arts. Would you like to become my disciple?

Uh, lemme think about it.

You could become immortal.

Ooh.

Oh, proselytizing now, are we? Don't think I'll take that lying down. Switch now and I'll throw in excellent luck on all your fortunes for two months.

Then allow me as well. You are a youkai-like human, after all.

You certainly are popular. (laugh)

A-Anyway, I guess that wraps up the self-introductions. So what I want to ask y'all now is whaddya want to do in Gensokyo from now on?

I think I'd like to live a carefree life without being disturbed.

Izzat so?

Transcending worldly matters, to live a refined life in a world with no worries or anything at all, that's my ideal.

You might have been thinking you've managed to become immortal and ageless by studying hermit techniques, but unfortunately, immortals in Gensokyo have a natural enemy.

You mean shinigami?

Yes, they will periodically come to kill you.

I believe those battles will be a form of entertainment. A nice, leisurely struggle.

Incidentally, I am not interested in immortality. Well, to be honest, I used to be. But after having been touched by my disciples' hearts, now I only train with an altruistic spirit.

And?

I wish to save those youkai who are in the undeserved position at the bottom of Gensokyo's society such as those who do not wish to fight.

And what'll happen after ya save them?

I will lead them to an even higher stage.

Sounds ominous to me. Just imagine a world dominated by youkai bodhisattva and youkai who have attained enlightenment. As for me, I want to create a world where there can be a pillar of support for both humans and youkai. That's what faith is. Faith is necessary to live spiritually rich lives free of worries.

That much is certain.

Our methods are different, but our minds are alike.

So are you holy rollers plannin' on makin' yerselves rich in the process of givin' us spiritually rich lives?

Yes, but what of it? This country is full of misunderstandings about religion, with many people who oppose religion. One of the reasons for that is because religions that misuse faith are spreading also. It's an unfortunate matter. Originally, religion was training with the goal of spiritual growth. Just like sports players use the latest theories to train their bodies to perfection, to become spiritually rich and feel happiness you need to use theories and morals, which is what religions were meant to be. Our duty is to make sure those in Gensokyo become successful yet not spiritually immature like those in the outside world.

I think I get it, but hearin' that from you... Anyway, since ya brought it up, let's change the topic to the outside world.

She says "manage", but she actually is a god residing in the shrine herself.

A temple with more youkai monks than human ones; the sort that could only exist in Gensokyo.

About a few years have passed since coming to Gensokyo.

In truth, everything is fine as long as the Barrier's power remains. This is a sophism often used by youkai.

So that the humans could not be ruled over and used as conveniently.

Counting divine pillars is normally used when counting sacred entities.

A frog-associated divine entity.

A location in the outside world. I heard it has a sea.

A treasure-hunting companion entity.

Positive thinking.

A suspiciously manipulative way of thinking by selling things, hinging on the line "don't you know I'm doing this for you?".

Sumber:

- https://en.touhouwiki.net/wiki/Symposium_of_Post-mysticism/Part_1

Kunci

Matriks kunci 5×5 hasil rekonstruksi:

S	Y	M	P	O
I	U	L	N	A
T	E	B	C	D
F	G	H	K	Q
R	V	W	X	Z

Susunan ini mengikuti pola pembentukan kunci Playfair pada umumnya: deretan kata kunci SYMPOIULNATE di awal, lalu dilanjutkan sisa alfabet secara berurutan (BCDFGHKQRVWXZ).

Referensi

Tabel referensi frekuensi bigram Bahasa Inggris yang digunakan pada Tahap 1 mengacu pada [5, 10]. Tabel QUADGRAM yang dipakai sebagai fungsi objektif pada Tahap 2 disusun dari 45 *quadgram* tersering pada berkas

statistik `english_quadgrams.txt` [9], dengan nilai persentase dihitung terhadap total kemunculan seluruh *quadgram* pada berkas tersebut. Konsep Playfair *Cipher* mengacu pada bahan kuliah [8].

2.4 Trinitarian Rhapsody

Deskripsi Soal

Di antara tumpukan lembar catatan risetnya, Marisa menemukan sebuah naskah analisis yang mengulas entitas penguasa tiga alam, sosok yang skalanya melampaui siapa pun di Gensokyo maupun *Lunar Capital*. Naskah yang menyimpan penjelasan mengenai eksistensi mutlak tersebut tidak dapat dibaca karena teksnya telah dienkrpsi menggunakan **Affine Cipher**.

Bantulah Marisa mengungkap pesan tersembunyi tersebut tanpa bermodalkan kunci awal maupun *known-plaintext*. Manfaatkan sifat matematis aritmatika modulo pada *Affine Cipher* beserta analisis frekuensi kemunculan karakter dalam bahasa Inggris untuk melakukan **Ciphertext-Only Attack**.

Berkas lampiran:

- Attachment: https://drive.google.com/file/d/16aQDPJHcHN2_EmEj2HHTeWeWChkZa_gZ/view?usp=sharing
- md5sum: 48e4f097521e85d43f5b432c67c6d6f9
- sha256sum: 3f10313362d08eb284c954b071345dc92f8c80169f1925d1be2e38971a6e037f

Berkas *Ciphertext*

```
MLLNUBUWDHOUAWNMFMAYNLUHYMVCZLNCMWLNCWKMFUZEMWDVMIKMLUMZFQKDMZCOZOJMVQKUZUFMAYNLUHOJUVUQLHIOZMWLNUBCH
LNLNHOUWJOHNHMGVUWEUHDHUWCZYQZMVQKUZULNUKOCMCMZFUCVUCLNICMMWJMPOAHLONUHMUGUDLMVQKUZUCZQOZLCZAOWBCHLND
MZYWLNUIHUKMCZUFUMLUFUMQNGUUDCZYLNUCHMHKQWQHOWUFYMCZLNCMWJUMHCZYLNMLLNUDMCZWOJNUHVMBOAEHQAFFHCPUVMVQKUZU
KMFHMZLOLNUKOCMCMZFUCVUCLNICMMZFMZQAZQFUFLNMBIFUWCHUOJXUAWMBOINMFBUBOZHLOMVQKUZUMZFLNMLLNUCHDHUHOYMLCP
UWNMFBUBOZMBOWCWNMFMLMVLCNWQOZWLHUMZLCOZOJQOAHWUOPUHQMKULNUKOCMCMZFNUICKKUFCLMUVIVULYOLNUCHMHKWMVQKUZUWD
MZYWQUMWUFMLQZUMZFNUHMGVUWEMWBOHZLNUKOCMCEUHMUYHCUPUFMLNCWMZFLOOGMEMILNUEOKMZVIDMHLWOJYVMVCZLNCMWWCZQUB
UCZYBALMKOHLMVWUNUMFFUQUCPUFLNUYOFWNLUILAHZUFNUHCZLOMFUQUCLJAVEUMWUVKMGZYNHUHVCPUZQHMZCZUWMZFYMPUNUHYHOL
UWSAUEMIOJKMLCZYWNUCWKOAZLUFNLHOAYNLNUMHWMZFYCPUBCHLNBIBHCZYCYZJOHLNNUHIOAZYLNUHIOAYNLNUHMLNUQMLUJUUVLWO
HHIJOHLNCLWLMZWJOHKMLCOZOJNUHMDUMHMZQUMZFMDDOCZLUFNUHMMWQHUFUWHPMZLOJNUHWUVJCNMPUNUMHFLNMLLNUDOVUQMLEMWOZ
QUMNAKMZBUCZYCLNMWVWOHUMQNUFKINUMHCZYLNMMLYVMEWNUHMKULNUZLNMWNUEMWFMUMVUHCZWDUVVWZFMWOHQHUWNLNMLWNU
MWUTLHUKUVIVMWQPCOAWMZFLNMLWNUEMWJJCVCQLUFECLNMBZOHKMWUTAMFVUWCHUWZOHNMWCLUWQMDUFKIZOLCQULNMLLNUMZYUHOJL
NUYOFFUWNUUGMLULHMZWJOHKUFCZLZOLNCWUPCVQHUMLAHUKMILNUYOFFUWBUYHMQCOAWLOKJMBVUWMZFLNUCHLUVVCZYCVMPULLOOL
NUHWNLUJCNWNLNMLEMKOWLQOKKOZVIBMZUFEMWLNHUFKAVVULENCNJCWLZUMLVICZLOLNUDMLLHYZCLFUVYCYNLUFCDZDOVVALUFLNCZY
WMZFEOAVFUMLLNUQOHDWUOJMCWNOHMKMZBVOOFQVOAHUFCFLUUVJCLEMWMWQHUFLOLNUBVOOFUMLCZYOFFUWNUQMLUCLWUUKWMWIKBO
VCQWAKMMLCOZOJMVVLNUZUMLCPUQNMHMLHUHCWLCQWJLNUQHUMLAHUWJLNUFUUDQCQUHONOHMQURAPUVMVMHLCMVDVCIWUZQMMZF
WAULOZCAWNMPUVUJLMBZFMZLMZFCZLUHUWLCZYLWUWLCZOZLOLNUHUFKAVVULJUPUHENCQNBUMZLOMJJUQLEUMVLNIHOKMZWFACZYLNU
UVMWLIUMHWJLNUHUBAVCQMFHUMVVIYHCDUFLNUKZLNUMHVUIKDCHULNUKMCZWIKDLOKWEUHMUDHUQQADMLCOZECLNCXULNUQOZ
WUSAUZHLCWULOMBWAHFNUCYNLWJLNUDHCUQWJVMHYUWUQUCKUZWMNMBCLJGUUDCZYHUFKAVVULCZQMDLCPCLIMZFLNUZROIKUZLOJL
NUNCYNVIWUQUCMVCXUFUWLNULCQUTDUHCUZQUCZFAQUBIEMLCNCZYLNUQOVHOJLNUFICZYJCNWQNMZYUMWYOFFUWUTDUQLUFLOMPU
HLNMHKJAVOHFULHAQLCPUWDCHCLWJHOKLNUNOAWUHQCLIOPUHENCQNWNUWLOOFYAMHFMZFLODHOLUQLLNUCZFCPCFAMVWNUOHNUDMW
WUFLNHOAYNFMZYUHOAWVKCKZMVDVMQUWNUQMLUEOAVFZMLAHMVBIBUQKUGZOEZMWYOFFUWENOQOAVFMVWOHUJAWULOMPULHNUFUOKOZ
WOHUPUZFHCPUMLNUKZMYMCZWLZJOHLAZMLUCZFCPCFAMVWCLEMWDHOBMBVINUHHOVUMWYAMHFCMZJUZLHMZQULNMLVUFLONUQMLUWCF
UZLJCQMLCOZBILNUKCFJCLNQUZLAHIECLNUZOFMMLNUWVMCMZYOFFUWUZOFCMWPUHIZMKUWAYYUWLNMLWNUEMLNUFOPUHUZLHM
ZQUWJOHCLUTDHUWUWUBOLNNUDOWWBCVCLILNMLWNUWLOFOZLNUKMCZHMFCZLOMQLIGUUDCZYMZUIUOZMVVENOUZLUHUFMZFCZLNUH
OMFCZJHOZLOJDHCPMLUNOAWUWDHOLUQLCZYLNUCHCZNMBCMLZLWYHUUGWNUVFLNUIELOBUWQHUFLONUQMLUNUMLLUFZFMZLWFMHDFEH
UMLNUWJJIUEMHOAZFLNUZUQWGOJBVMQGBAVVWENCQNLNUIWMAYNLUHFUCZNUHNOZOHMFZIEUBOAYNWEUHBABZUFOZJAZUHMVDIHUWLN
IUEEMWMMWQCMULFECLNLNUVDNMBULMZFLNUWQCZLJCQZMKUJOHIELOFMILMTAWEMWDHOBMBVIFUHCPUFJHOKLNUYHUUGEHOHFJOHIU
ELOTOWENCQNCWNMAZLCZYVIWCKCVMHLOLOTOZLNUCHEOHFJOHBOEMZFLUTCQOZLNUCHEOHFJOHDOCWOZCLCWDHUWAKUFLNMLLNUVMLLUHE
UHUZMKUFMJLUHLNUHUUBUQMAWUJCLWWADUHCCHCLIOHBOLEWZFDOWZMFMWLUHMCQOZQUCPUFZFBOHUNUQMLUENOKXUAWLNU
```


Pasangan kunci yang menghasilkan teks dengan kecocokan n -gram tertinggi dan membentuk *plaintext* yang bisa dibaca adalah $m = 15$ dan $b = 12$.

Artefak

Skrip *solver* yang digunakan untuk melakukan dekripsi dengan percobaan heuristik *known-plaintext* berdasarkan frekuensi karakter dan *brute-force fallback*.

```

1 import json
2 import os
3 import sys
4 from datetime import datetime
5 from typing import Dict, List, Tuple
6
7 sys.path.insert(0, os.path.dirname(os.path.dirname(os.path.abspath(__file__))))
8 from frequencies import BIGRAM, TRIGRAM, QUADGRAM
9
10 class AffineSolver:
11     def calculate_fitness(self, text: str) -> float:
12         score = 0.0
13         for i in range(len(text) - 1):
14             bg = text[i:i+2]
15             if bg in BIGRAM:
16                 score += BIGRAM[bg]
17         for i in range(len(text) - 2):
18             tg = text[i:i+3]
19             if tg in TRIGRAM:
20                 score += TRIGRAM[tg]
21         for i in range(len(text) - 3):
22             qg = text[i:i+4]
23             if qg in QUADGRAM:
24                 score += QUADGRAM[qg]
25         return score
26
27     def count(self, string: str) -> Dict:
28         data = {}
29         for c in string:
30             if c.isalpha():
31                 data[c] = data.get(c, 0) + 1
32         return dict(sorted(data.items(), key=lambda item: item[1], reverse=True))
33
34     def sort(self, string: str) -> List:
35         return list(self.count(string).keys())
36
37     def mod_inverse(self, a: int, m: int) -> int:
38         for x in range(1, m):
39             if (a * x) % m == 1:
40                 return x
41         return -1
42
43     def is_coprime(self, a: int, m: int) -> bool:
44         def gcd(a, b):
45             while b:
46                 a, b = b, a % b
47             return a
48         return gcd(a, m) == 1
49
50     def solve_keys(self, c1_char: str, c2_char: str, p1_char: str = 'E', p2_char:
51 str = 'T') -> Tuple[int, int]:

```

```

51     c1 = ord(c1_char) - ord('A')
52     c2 = ord(c2_char) - ord('A')
53     p1 = ord(p1_char) - ord('A')
54     p2 = ord(p2_char) - ord('A')
55
56
57     diff_c = (c1 - c2) % 26
58     diff_p = (p1 - p2) % 26
59
60     inv_diff_p = self.mod_inverse(diff_p, 26)
61     if inv_diff_p == -1:
62         return -1, -1
63
64     m = (diff_c * inv_diff_p) % 26
65
66     if not self.is_coprime(m, 26):
67         return -1, -1
68
69     b = (c1 - m * p1) % 26
70     return m, b
71
72 def decrypt(self, string: str, m: int, b: int) -> str:
73     m_inv = self.mod_inverse(m, 26)
74     if m_inv == -1:
75         return ""
76
77     result = []
78     for c in string:
79         if c.isalpha():
80             y = ord(c) - ord('A')
81             x = (m_inv * (y - b)) % 26
82             result.append(chr(x + ord('A')))
83         else:
84             result.append(c)
85     return "".join(result)
86
87 def brute_force_fallback(self, string: str) -> List[Tuple[int, int, int, str]]:
88     possible_m = [m for m in range(1, 26) if self.is_coprime(m, 26)]
89     results = []
90     for m in possible_m:
91         for b in range(26):
92             decrypted = self.decrypt(string, m, b)
93             score = self.calculate_fitness(decrypted)
94             results.append((score, m, b, decrypted))
95
96     return sorted(results, key=lambda x: x[0], reverse=True)
97
98
99 def load_ciphertext(base_dir: str) -> str:
100     with open(os.path.join(base_dir, "t.txt")) as f:
101         return f.read().strip()
102
103 def write_plain(base_dir: str, timestamp: str, decrypted: str) -> str:
104     path = os.path.join(base_dir, "log", "plain", timestamp + ".txt")
105     os.makedirs(os.path.dirname(path), exist_ok=True)
106     with open(path, "w") as f:
107         f.write(decrypted)
108     return path
109

```

```

110 def write_metadata(base_dir: str, timestamp: str, m: int, b: int) -> str:
111     path = os.path.join(base_dir, "log", "metadata", timestamp + ".json")
112     os.makedirs(os.path.dirname(path), exist_ok=True)
113     payload = {
114         "timestamp": datetime.now().isoformat(timespec="seconds"),
115         "key": {
116             "m": m,
117             "b": b
118         }
119     }
120     with open(path, "w") as f:
121         json.dump(payload, f, indent=2)
122     return path
123
124 if __name__ == "__main__":
125     base_dir = os.path.dirname(os.path.abspath(__file__))
126     content = load_ciphertext(base_dir)
127
128     solver = AffineSolver()
129
130     sorted_chars = solver.sort(content)
131     print("Karakter paling sering muncul di ciphertext:", sorted_chars[:5])
132     c1_char = sorted_chars[0]
133     c2_char = sorted_chars[1]
134     print(f"Mencoba memetakan {c1_char} -> E dan {c2_char} -> T")
135
136     m, b = solver.solve_keys(c1_char, c2_char, 'E', 'T')
137
138     decrypted = ""
139     if m != -1:
140         print(f"Ditemukan kunci: m = {m}, b = {b}")
141         decrypted = solver.decrypt(content, m, b)
142         print("Preview hasil dekripsi:")
143         print(decrypted[:100] + "...")
144     else:
145         print("Pemetaan E dan T tidak menghasilkan m yang valid. Mencoba Brute-
146         force Fallback...")
147         results = solver.brute_force_fallback(content)
148         top_result = results[0]
149         m, b = top_result[1], top_result[2]
150         print(f"Hasil terbaik dari brute-force: m = {m}, b = {b}")
151         decrypted = solver.decrypt(content, m, b)
152         print("Preview hasil dekripsi:")
153         print(decrypted[:100] + "...")
154
155     if decrypted:
156         timestamp = datetime.now().strftime("%Y-%m-%d_%H%M%S")
157         plain_path = write_plain(base_dir, timestamp, decrypted)
158         metadata_path = write_metadata(base_dir, timestamp, m, b)
159         print(f"\n[Teks lengkap tersimpan ke {os.path.relpath(plain_path)}]")
160         print(f"[Metadata tersimpan ke {os.path.relpath(metadata_path)}]")

```

src/24/solver.py

Plaintext Hasil Dekripsi

ATTHEBESPROETUSHADADAUGHTERGALINTHIASTHISMAIDENWASPLAYMATEANDCOMPANIONOFALCMENEDAUGHTEROFELECTRYONASTHEBIR
THTHROESFORHERAKLESWEREPRESSINGONALCMENETHEMOIRAIANDEILEITHYIAASAFAVOURTOHERAKEPTALCMENEINCONTINUOUSBIRTHP
ANGSTHEYREMAINEDSEATEDEACHKEEPINGTHEIRARMSCROSSEDGALINTHIASFEARINGTHATTHEPAINSOFFHERLABOURWOULDDRIVEALCMENE
MADRANTOTHEMOIRAIANDEILEITHYIAANDANNOUNCEDTHATBYDESIREOFZEUSABOYHADBEENBORNTOTALCMENEANDTHATTHEIRPREROGATIV
ESHADBEENABOLISHEDATALLTHISCONSTERNATIONOFCOURSEOVCAMETHEMOIRAIANDTHEYIMMEDIATELYLETGOTHEIRARMSALCMENESP
ANGSCEASEDATONCEANDHERAKLESWASBORNTHEMOIRAIWEREAGGRIEVEDATTHISANDTOOKAWAYTHEWOMANLYPARTSOFGALINTHIASSINCEB
EINGBUTAMORTALSHEHADDECEIVEDTHEGODSTHEYTURNEDHERINTOADECEITFULWEASELMAKINGHERLIVEINCRANNIESANDGAVEHERAGROT
ESQUEWAYOFMATINGSHEISMOUNTEDTHROUGHTHEEARSANDGIVESBIRTHBYBRINGINGFORTHHERYOUNGTHROUGHTHETHROATHECATEFELTSO
RRYFORTHISTRANSFORMATIONOFHERAPPEARANCEANDAPPOINTEDHERASACREDSERVANTOFHERSELF.IHAVEHEARDTHATTHEPOLECATWASON
CEAHUMANBEINGITHASALSOREACHEDMYHEARINGTHATGALEWASHERNAMETHENTHATSHEWASADEALERINSPELLSANDSORCERESSTHATSHW
ASEXTREMELYLASCIVIOUSANDTHATSHEWASAFFLICTEDWITHABNORMALSEXUALDESIRESNORHASITESCAPEDMYNOTICETHATTHEANGEROF
HEGODDESSHEKATETTRANSFORMEDITINTOTHISEVILCREATUREMAYTHEGODDESSBEGRACIOUSTOMEFABLESANDTHEIRTELLINGILEAVETOOT
HERSTHEFISHTHATWASMOSTCOMMONLYBANNEDWASTHEREDMULLETWHICHFITSNEATLYINTOTHEPATTERNITDELIGHTEDINPOLLUTEDTHING
SANDWOULDEATTHECORPSEOFAFISHORAMANBLOODCOLOUREDITSELFITWASSACREDTOTHEBLOOD-EATINGGODDESSHECATEITSEEMSASYMBO
LICSUMMATIONOFFALLTHENEGATIVECHARACTERISTICSOFTHECREATURESOFTHEDEEPCICEROHORACEJUVENALMARTIALPLINYSENECAAND
SUETONIUSHAVELEFTABUNDANTANDINTERESTINGTESTIMONYTOTHEREDMULLETFEVERWHICHBEGANTOAFFECTWEALTHYROMANSDURINGTH
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HEHIGHLYSPECIALIZEDAESTHETICEXPERIENCEINDUCEDBYWATCHINGTHECOLOROFTHE DYINGFISHCHANGEASAGODDESSEXPECTEDTOAVE
RTHARMFULORDESTRUCTIVESPIRITSFROMTHEHOUSEORCITYOVERWHICHSHESTOODGUARDANDTOPROTECTTHEINDIVIDUALASSHEORHEPAS
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SOREVENDRIVETHEMONAGAINSTUNFORTUNATEINDIVIDUALSITWASPROBABLYHERROLEASGUARDIANOFENTRANCESHATLEDTOHECATESID
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NCESFORITEXPRESSESBOTHTHEPOSSIBILITYTHATSHESTOODONTHEMAINROADINTOACITYKEEPINGANEYEONALLWHOENTEREDANDINTHER
OADINFRONTOFPRIVATEHOUSESPROTECTINGTHEIRINHABITANTSGREEKSHELDTHEYEWTOBESACREDTOHECATEHERATTENDANTS DRAPEDWR
EATHESOFYEWAROUNDTHENECKSOFBLOCKBULLSWHICHTHEYSLAUGHTEREDINHERHONORANDYEWBOUGHSWEREBURNEDONFUNERALPYRESTHE
YEWASASSOCIATEDWITHTHEALPHABETANDTHESCIENTIFICNAMEFORYEW TODAYTAXUSWASPROBABLYDERIVEDFROMTHEGREEKWORDFORYE
WTOXOSWHICHISHAUNTINGLYSIMILARTOTOXONTHEIRWORDFORBOWANDTOXICONTHEIRWORDFORPOISONITISPRESUMEDTHATTHELATTERW
ERENAMEDAFTERTHETREEBECAUSEOFTISSUPERIORITYFORBOTHBOWSANDPOISONANDASTERIA CONCEIVEDANDBOREHECATEWHOMZEUSTHE
SONOF CRONOSHONOREDABOVEALLHEGAVEHERSPLENDIDGIFTSTO HAVEASHAREOFTHEEARTHANDTHEUNFRUITFULSEASHERECEIVEDHONORA
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ANDINSEAWHOMSHEWILLSHEGREATLYAIDSANDADVANCESSHESITSBYWORSHIPFULKINGSINJUDGEMENTANDINTHEASSEMBLYWHOMSHEWILL
ISDISTINGUISHEDAMONGTHEPEOPLEANDWHENMENARMTHEMSELVESFORTHEBATTLETHATDESTROYSMENTHENTHEGODDESSISATHANDTOGIV
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HEMANDPROFITSTHEMANDHEWHOBYMIGHTANDSTRENGTHGETSTHEVICTORYWINSTHERICHPRIZEEASILYWITHJOYANDBRINGSGLORYTOHIS
ARENTSANDSHEISGOODTOSTANDBYHORSEMENWHOMSHEWILLANDTOHOSEWHOSEBUSINESSISINTHEGREYDISCOMFORTABLESEANANDWHOPRA
YTOHECATEANDTHELOUDCRASHINGEARTHSHAKEREASILYTHEGLORIOUSGODDESSGIVESGREATCATCHANDEASILYSHETAKESITAWAYASSOON
ASSEENIFSOSHEWILLSHEISGOODINTHEBYREWITHHERMESTOINCREASETHESTOCKTHEDROVESOFKINEANDWIDEHERDSOFGOATSANDFLOCKS
OFFLEECYSHEEPIFSHEWILLSHEINCREASESFROMAFEWMAKESMANYTOBELESSSTHENALBEITHERMOTHERSONLYCHILD SHEISHONOREDAM
ONGSTALLTHEDEATHLESSGODSANDTHESONOF CRONOSMADEHERANURSEOF THEYOUNGWHOAFTERTHATDAYSAWWITHTHEIREYESTHELIGHTOFA
LLSEEINGDAWNSOFROMTHEBEGINNINGSHEISANURSEOF THEYOUNGANDTHESEAREHERHONOURSHEIODSINCLUSIONANDPRAISEOFHECATEI
NTHETHEOGONYHASBEENTROUBLESOMEFORSCHOLARSINTHATTHESEEMSTOHOLDHERINHIGHRGARDWHILETHETESTIMONYOFOTHERWRITERS
ANDSURVIVINGEVIDENCESUGGESTSTHATTHISMAYHAVEBEENTHEEXCEPTIONONETHEORYISTHATTHEIODSORIGINALVILLAGEHADASUBSTA
NTIALHECATEFOLLOWINGANDTHATTHISINCLUSIONOFHERINTHETHEOGONYWASAWAYOFADDINGTOHERPRESTIGE BYSPREADINGWORDOFHERA
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VASIVEANDYETNOTMENTIONEDASMUCHASTEMPLEWORSHIPINATHENSHECATEALONGWITHZEUSHERMESATHENAHESTIAANDAPOLLOWEREVER
YIMPORTANTINDAILYLIFEASTHEYWERETHEMAINGODSOFTHEHOUSEHOLDHOWEVERITISCLEARTHATTHESPECIALPOSITIONGIVENTOHECAT

EBYZEUSISUPHELDTHROUGHOUTHERHISTORYBYDEPICTIONSFOUNDONCOINSOFHECATEONTHEHANDOFZEUSASHIGHLIGHTEDINMORERECENTRESEARCHPRESENTEDBYDESTEANDRANKINEINTHEHOMERICHYMNTODEMETERHECATEISCALLEDTENDERHEARTEDANEPITHETPERHAPSINTENDEDTOEMPHASIZEHERCONCERNWITHTHEDISAPPEARANCEOFPERSEPHONEWHENSHEASSISTEDDEMETERWITHHERSEARCHFORPERSEPHONE FOLLOWINGHERABDUCTIONBYHADESSUGGESTINGTHATDEMETERSHOULDSPEAKTOTHEGODOFTHE SUNHELIOSSUBSEQUENTLYHECATEBECAME PERSEPHONESCOMPANIONONHERYEARLYJOURNEYTOANDFROMTHEREALMSOFHADESSERVINGASAPSYCHOPOMPBECAUSEOFTHISASSOCIATIONHECATEWASONEOF THECHIEFGODDESSESOF THEELEUSINIANMYSTERIESALONGSIDEDEMETERANDPERSEPHONEANDTHEREWASATEMPLEDEDICATEDTOHERNEARTHEMAINSANCTUARYATELEUSIS

Plaintext dengan Format Asli

At Thebes Proetus had a daughter Galinthias. This maiden was playmate and companion of Alcmene, daughter of Electryon. As the birth throes for Herakles were pressing on Alcmene, the Moirai and Eileithyia, as a favour to Hera, kept Alcmene in continuous birth pangs. They remained seated, each keeping their arms crossed. Galinthias, fearing that the pains of her labour would drive Alcmene mad, ran to the Moirai and Eileithyia and announced that by desire of Zeus a boy had been born to Alcmene and that their prerogatives had been abolished. At all this, consternation of course overcame the Moirai and they immediately let go their arms.

Alcmene's pangs ceased at once and Herakles was born. The Moirai were aggrieved at this and took away the womanly parts of Galinthias since, being but a mortal, she had deceived the gods. They turned her into a deceitful weasel, making her live in crannies and gave her a grotesque way of mating. She is mounted through the ears and gives birth by bringing forth her young through the throat. Hecate felt sorry for this transformation of her appearance and appointed her a sacred servant of herself.

I have heard that the polecat was once a human being. It has also reached my hearing that Gale was her name then; that she was a dealer in spells and a sorceress; that she was extremely lascivious, and that she was afflicted with abnormal sexual desires. Nor has it escaped my notice that the anger of the goddess Hecate transformed it into this evil creature. May the goddess be gracious to me: Fables and their telling I leave to others.

The fish that was most commonly banned was the red mullet, which fits neatly into the pattern. It 'delighted in polluted things', and 'would eat the corpse of a fish or a man'. Blood-coloured itself, it was sacred to the blood-eating goddess Hecate. It seems a symbolic summation of all the negative characteristics of the creatures of the deep.

Cicero, Horace, Juvenal, Martial, Pliny, Seneca, and Suetonius have left abundant and interesting testimony to the red mullet fever which began to affect wealthy Romans during the last years of the Republic and really gripped them in the early Empire. The main symptoms were a preoccupation with size, the consequent rise to absurd heights of the prices of large specimens, a habit of keeping red mullet in captivity, and the enjoyment of the highly specialized aesthetic experience induced by watching the color of the dying fish change.

As a goddess expected to avert harmful or destructive spirits from the house or city over which she stood guard and to protect the individual as she or he passed through dangerous liminal places, Hecate would naturally become known as a goddess who could also refuse to avert the demons, or even drive them on against unfortunate individuals.

It was probably her role as guardian of entrances that led to Hecate's identification by the mid fifth century with Enodia, a Thessalian goddess. Enodia's very name suggests that she watched over entrances, for it expresses both the possibility that she stood on the main road into a city, keeping an eye on all who entered, and in the road in front of private houses, protecting their inhabitants.

Greeks held the yew to be sacred to Hecate ... Her attendants draped wreathes of yew around the necks of black bulls which they slaughtered in her honor and yew boughs were burned on funeral pyres. The yew was associated with the alphabet and the scientific name for yew today, *taxus*, was probably derived from the Greek word for yew, *toxos*, which is hauntingly similar to *toxon*, their word for bow and *toxicon*, their word for poison. It is presumed that the latter were named after the tree because of its superiority for both bows and poison.

And [Asteria] conceived and bore Hecate whom Zeus the son of Cronos honored above all. He gave her splendid gifts, to have a share of the earth and the unfruitful sea. She received honor also in starry heaven, and is honored exceedingly by the deathless gods. For to this day, whenever any one of men on earth offers rich sacrifices and prays for favor according to custom, he calls upon Hecate. Great honor comes full easily to him whose prayers the goddess receives favorably, and she bestows wealth upon him; for the power surely is with her. For as many as were born of Earth and Ocean amongst all these she has her due portion. The son of Cronos did her no wrong nor took anything away of all that was her portion among the former Titan gods: but she holds, as the division was at the first from the beginning, privilege both in earth, and in heaven, and in sea.

Whom she will she greatly aids and advances: she sits by worshipful kings in judgement, and in the assembly whom she will is distinguished among the people. And when men arm themselves for the battle that destroys men, then the goddess is at hand to give victory and grant glory readily to whom she will. Good is she also when men contend at the games, for there too the goddess is with them and profits them: and he who by might and strength gets the victory wins the rich prize easily with joy, and brings glory to his parents. And she is good to stand by horsemen, whom she will: and to those whose business is in the grey discomfortable sea, and who pray to Hecate and the loud-crashing Earth-Shaker, easily the glorious goddess gives great catch, and easily she takes it away as soon as seen, if so she will. She is good in the byre with Hermes to increase the stock. The droves of kine and wide herds of goats and flocks of fleecy sheep, if she will, she increases from a few, or makes many to be less. So, then, albeit her mother's only child, she is honored amongst all the deathless gods. And the son of Cronos made her a nurse of the young who after that day saw with their eyes the light of all-seeing Dawn. So from the beginning she is a nurse of the young, and these are her honours.

Hesiod's inclusion and praise of Hecate in the Theogony has been troublesome for scholars, in that he seems to hold her in high regard, while the testimony of other writers, and surviving evidence, suggests that this may have been the exception. One theory is that Hesiod's original village had a substantial Hecate following and that his inclusion of her in the Theogony was a way of adding to her prestige by spreading word of her among his readers. Another theory is that Hecate was mainly a household god and humble household worship could have been more pervasive and yet not mentioned as much as temple worship. In Athens, Hecate, along with Zeus, Hermes, Athena, Hestia, and Apollo, were very important in daily life as they were the main gods of the household. However, it is clear that the special position given to Hecate by Zeus is upheld throughout her history by depictions found on coins of Hecate on the hand of Zeus as highlighted in more recent research presented by d'Este and Rankine.

In the Homeric Hymn to Demeter, Hecate is called "tender-hearted", an epithet perhaps intended to emphasize her concern with the disappearance of Persephone, when she assisted Demeter with her search for Persephone following her abduction by Hades, suggesting that Demeter should speak to the god of the Sun, Helios. Subsequently, Hecate became Persephone's companion on her yearly journey to and from the realms of Hades, serving as a psychopomp. Because of this association, Hecate was one of the chief goddesses of the Eleusinian Mysteries, alongside Demeter and Persephone, and there was a temple dedicated to her near the main sanctuary at Eleusis.

Sumber:

- <https://en.wikipedia.org/wiki/Hecate>

Kunci

Kunci perkalian (*multiplier*) *m*: 15

Kunci pergeseran (*shift*) *b*: 12

Referensi

Tabel n -gram bahasa Inggris bersumber dari [10] dan [9]. Referensi mengenai konsep dasar dan implementasi dekripsi algoritma *Affine Cipher* mengacu pada [2]. Konsep *Affine Cipher* dan kriptanalisisnya mengacu pada bahan kuliah [8], sedangkan perhitungan invers modulo mengacu pada [6].

2.5 Silent Goddess

Deskripsi Soal

Di lembar terakhir catatan risetnya, Marisa menemukan salinan artikel investigasi mengenai sosok dewi misterius bersayap tunggal yang terpergok menyusup ke fasilitas di kedalaman Hutan Bambu. Potongan artikel berita tersebut kini tidak dapat dibaca karena telah dienkripsi menggunakan **Hill Cipher**.

Sang jurnalis yang menyusunnya enggan memberikan matriks kunci secara langsung. Ia hanya membocorkan satu petunjuk bahwa naskah investigasi tersebut diawali dengan kata “**EXPLOSIVE SCOOP**”.

Bantulah Marisa merekonstruksi matriks kunci menggunakan petunjuk awal *plaintext* tersebut melalui **Known Plaintext Attack** (KPA). Hitung invers modulo dari matriks kunci yang ditemukan, lalu gunakan hasilnya untuk mendekripsi sisa *ciphertext* hingga pesan lengkap artikel terungkap.

Berkas lampiran:

- Attachment: <https://drive.google.com/file/d/1aS17ZuSGAwfpcus4F6HEP3wPVXYIHE7n/view?usp=sharing>
- md5sum: 84e86bf1dafb89ec420c114abd60498c
- sha256sum: 6113d33c379b4727e120d3146cfc39ab3004842c0252848dc56c621ba7a5d25f

Berkas *Ciphertext*

```
VNUEXGMMXSASGLRPXOUAIOEKONMNANINLBTMTWTFACQMNZAUZIRRYXQOECANINLYACCDKPKICRKFAGAZXJKBJOPWJCAISEJCVEECKNINL
BTLPiHPRRFJATIIYQXGJULBVWHMBDPSMEEKGDLUVBAMUIWDOTYJKYZKITJKZJBNBYSQVSSXMWPGFQAWMQBNBDPSMEIMYXJMWOLWGBLOTMM
XTNJQYFNJVRLLJHZOLLRCOJHJZADHACNEVTMSPRFJUJZMSPCOGSARZWDDNAIUBMWUKEXLEPIQGGZTICEOWWROCRNOBJCDIIBDOHLBVLZMOJ
KSNKKYYQXPFLLLRCNDRMXSCXGKGUYQFZQFXRHTWPUJQWDOTYJKYZOKLPJTXPMXGMTKCRZSAINMRYUZEVLBVWHMGPVEFWVUBOOKARSI
VYHACWPJSYQGJHASHGQGOUJRCFBFLJJRJATANPQCBKGPVSUENOBNCNDRNUAEYSIXIXNFVQMNSDFKKYZWDGPVBIFMKTACPPGJYCJXJMSMHVM
QEGKOYPINBDNADHUVREKZQVCKRNIIZOLQSZTHUDJCDNTBODMGPVZDXPJXRHHKQDSFYXWKZIHDKTCVQJUIUVFCVASASKKVIYVJYDGBSLXC
YFYMWDUIZVBXCOBGCHHADGWHLBVWHMUEDACMKRNONEVBZSHYFFHMYGHTKTXPLMZFTUGSGPVBOTZWDBRLXRH
```

Langkah Penyelesaian

Hill cipher mudah dipecahkan dengan *known-plaintext attack*. Berdasarkan petunjuk yang diberikan, 14 karakter pertama dari *plaintext* adalah “**EXPLOSIVESCOOP**”. Misalkan untuk *Hill cipher* dengan $m = 3$ diketahui:

- $P = (4, 23, 15) \rightarrow C = (21, 13, 20)$ (dari blok “EXP” $\rightarrow$ “VNU”)
- $P = (11, 14, 18) \rightarrow C = (4, 23, 6)$ (dari blok “LOS” $\rightarrow$ “EXG”)
- $P = (8, 21, 4) \rightarrow C = (12, 12, 23)$ (dari blok “IVE” $\rightarrow$ “MMX”)
- Jadi, $K(4, 23, 15) = (21, 13, 20)$, $K(11, 14, 18) = (4, 23, 6)$, dan $K(8, 21, 4) = (12, 12, 23)$

$$\begin{pmatrix} 21 & 13 & 20 \\ 4 & 23 & 6 \\ 12 & 12 & 23 \end{pmatrix} = K \begin{pmatrix} 4 & 11 & 8 \\ 23 & 14 & 18 \\ 15 & 21 & 4 \end{pmatrix} \pmod{26} \rightarrow K = CP^{-1} \pmod{26}$$

- Matriks balikan dari P adalah $P^{-1} = \begin{pmatrix} 4 & 11 & 8 \\ 23 & 14 & 18 \\ 15 & 21 & 4 \end{pmatrix}^{-1} \pmod{26} = \begin{pmatrix} 8 & 24 & 1 \\ 1 & 0 & 24 \\ 24 & 1 & 25 \end{pmatrix}$

- Sehingga

$$K = CP^{-1} \pmod{26} = \begin{pmatrix} 21 & 4 & 12 \\ 13 & 23 & 12 \\ 20 & 6 & 23 \end{pmatrix} \begin{pmatrix} 8 & 24 & 1 \\ 1 & 0 & 24 \\ 24 & 1 & 25 \end{pmatrix} \pmod{26} = \begin{pmatrix} 18 & 22 & 1 \\ 25 & 12 & 7 \\ 16 & 9 & 11 \end{pmatrix}$$

Selanjutnya, sisa dari pesan *ciphertext* didekripsi secara langsung menggunakan matriks balikan (invers) dari K .

Langkah-langkah Dekripsi

Berikut adalah rincian langkah dekripsi untuk tiap blok karakter (Total 240 blok) menjadi vektor numerik, dikali dengan invers matriks kunci modulo 26, lalu dikembalikan ke bentuk teks:

Blok	Ciphertext	Vektor	Plaintext	Vektor
1	VNU	$(21, 13, 20)^T$	EXP	$(4, 23, 15)^T$
2	EXG	$(4, 23, 6)^T$	LOS	$(11, 14, 18)^T$
3	MMX	$(12, 12, 23)^T$	IVE	$(8, 21, 4)^T$
4	SAS	$(18, 0, 18)^T$	SCO	$(18, 2, 14)^T$
5	CGL	$(2, 6, 11)^T$	OPS	$(14, 15, 18)^T$
6	RPX	$(17, 15, 23)^T$	USP	$(20, 18, 15)^T$
7	OUA	$(14, 20, 0)^T$	ICI	$(8, 2, 8)^T$
8	IOE	$(8, 14, 4)^T$	OUS	$(14, 20, 18)^T$
9	KON	$(10, 14, 13)^T$	SHA	$(18, 7, 0)^T$
10	MNA	$(12, 13, 0)^T$	DYC	$(3, 24, 2)^T$
11	NIN	$(13, 8, 13)^T$	HAR	$(7, 0, 17)^T$
12	LBT	$(11, 1, 19)^T$	ACT	$(0, 2, 19)^T$
13	MWT	$(12, 22, 19)^T$	ERI	$(4, 17, 8)^T$
14	JFA	$(9, 5, 0)^T$	SFR	$(18, 5, 17)^T$
15	CQM	$(2, 16, 12)^T$	EQU	$(4, 16, 20)^T$
16	NZA	$(13, 25, 0)^T$	ENT	$(4, 13, 19)^T$
17	UIZ	$(20, 8, 25)^T$	ING	$(8, 13, 6)^T$
18	IRR	$(8, 17, 17)^T$	PHA	$(15, 7, 0)^T$
19	YXQ	$(24, 23, 16)^T$	RMA	$(17, 12, 0)^T$
20	OEC	$(14, 4, 2)^T$	CEU	$(2, 4, 20)^T$
21	ANI	$(0, 13, 8)^T$	TIC	$(19, 8, 2)^T$
22	NLY	$(13, 11, 24)^T$	ALF	$(0, 11, 5)^T$
23	ACC	$(0, 2, 2)^T$	ACI	$(0, 2, 8)^T$
24	DKP	$(3, 10, 15)^T$	LIT	$(11, 8, 19)^T$
25	KIC	$(10, 8, 2)^T$	YWE	$(24, 22, 4)^T$
26	RKF	$(17, 10, 5)^T$	HAV	$(7, 0, 21)^T$
27	GAZ	$(6, 0, 25)^T$	ELE	$(4, 11, 4)^T$
28	XJK	$(23, 9, 10)^T$	ARN	$(0, 17, 13)^T$
29	BJO	$(1, 9, 14)^T$	EDT	$(4, 3, 19)^T$
30	PWJ	$(15, 22, 9)^T$	HAT	$(7, 0, 19)^T$
31	CAI	$(2, 0, 8)^T$	AMY	$(0, 12, 24)^T$
32	SEJ	$(18, 4, 9)^T$	STE	$(18, 19, 4)^T$
33	CVE	$(2, 21, 4)^T$	RIO	$(17, 8, 14)^T$
34	ECK	$(4, 2, 10)^T$	USC	$(20, 18, 2)^T$
35	NIN	$(13, 8, 13)^T$	HAR	$(7, 0, 17)^T$
36	LBT	$(11, 1, 19)^T$	ACT	$(0, 2, 19)^T$
lanjut di halaman berikutnya,				

Tabel 2.1 – lanjutan dari halaman sebelumnya

Blok	Ciphertext	Vektor	Plaintext	Vektor
37	LPI	$(11, 15, 8)^T$	ERH	$(4, 17, 7)^T$
38	HPR	$(7, 15, 17)^T$	ASB	$(0, 18, 1)^T$
39	RFJ	$(17, 5, 9)^T$	EEN	$(4, 4, 13)^T$
40	ATI	$(0, 19, 8)^T$	VIS	$(21, 8, 18)^T$
41	YQX	$(24, 16, 23)^T$	ITI	$(8, 19, 8)^T$
42	GJU	$(6, 9, 20)^T$	NGE	$(13, 6, 4)^T$
43	LBV	$(11, 1, 21)^T$	IEN	$(8, 4, 13)^T$
44	WHM	$(22, 7, 12)^T$	TEI	$(19, 4, 8)^T$
45	BDP	$(1, 3, 15)^T$	GEN	$(6, 4, 13)^T$
46	SME	$(18, 12, 4)^T$	SOK	$(18, 14, 10)^T$
47	EKG	$(4, 10, 6)^T$	YOS	$(24, 14, 18)^T$
48	DLU	$(3, 11, 20)^T$	ONL	$(14, 13, 11)^T$
49	VBA	$(21, 1, 0)^T$	YDR	$(24, 3, 17)^T$
50	MUI	$(12, 20, 8)^T$	UGO	$(20, 6, 14)^T$
51	WDO	$(22, 3, 14)^T$	RGA	$(17, 6, 0)^T$
52	TYJ	$(19, 24, 9)^T$	NIZ	$(13, 8, 25)^T$
53	KYZ	$(10, 24, 25)^T$	ATI	$(0, 19, 8)^T$
54	KIT	$(10, 8, 19)^T$	ONS	$(14, 13, 18)^T$
55	JKZ	$(9, 10, 25)^T$	HED	$(7, 4, 3)^T$
56	JBN	$(9, 1, 13)^T$	ISP	$(8, 18, 15)^T$
57	OBY	$(14, 1, 24)^T$	LAY	$(11, 0, 24)^T$
58	SQV	$(18, 16, 21)^T$	SFA	$(18, 5, 0)^T$
59	SSX	$(18, 18, 23)^T$	SHI	$(18, 7, 8)^T$
60	MWP	$(12, 22, 15)^T$	ONU	$(14, 13, 20)^T$
61	GFQ	$(6, 5, 16)^T$	NCO	$(13, 2, 14)^T$
62	AWM	$(0, 22, 12)^T$	MMO	$(12, 12, 14)^T$
63	QBN	$(16, 1, 13)^T$	NTO	$(13, 19, 14)^T$
64	BDP	$(1, 3, 15)^T$	GEN	$(6, 4, 13)^T$
65	SME	$(18, 12, 4)^T$	SOK	$(18, 14, 10)^T$
66	IMY	$(8, 12, 24)^T$	YOW	$(24, 14, 22)^T$
67	XJM	$(23, 9, 12)^T$	ITH	$(8, 19, 7)^T$
68	WOL	$(22, 14, 11)^T$	ADI	$(0, 3, 8)^T$
69	WGB	$(22, 6, 1)^T$	STI	$(18, 19, 8)^T$
70	LOT	$(11, 14, 19)^T$	NCT	$(13, 2, 19)^T$
71	MMX	$(12, 12, 23)^T$	IVE	$(8, 21, 4)^T$
72	TNJ	$(19, 13, 9)^T$	SIN	$(18, 8, 13)^T$
73	QYF	$(16, 24, 5)^T$	GLE	$(6, 11, 4)^T$
74	NJV	$(13, 9, 21)^T$	WIN	$(22, 8, 13)^T$
75	RLJ	$(17, 11, 9)^T$	GED	$(6, 4, 3)^T$
76	HZO	$(7, 25, 14)^T$	APP	$(0, 15, 15)^T$
77	LLR	$(11, 11, 17)^T$	EAR	$(4, 0, 17)^T$
78	COJ	$(2, 14, 9)^T$	ANC	$(0, 13, 2)^T$
79	HJZ	$(7, 9, 25)^T$	EAN	$(4, 0, 13)^T$
80	ADH	$(0, 3, 7)^T$	DHA	$(3, 7, 0)^T$
81	ACN	$(0, 2, 13)^T$	SNO	$(18, 13, 14)^T$
82	EVT	$(4, 21, 19)^T$	TBE	$(19, 1, 4)^T$
83	MSP	$(12, 18, 15)^T$	ENS	$(4, 13, 18)^T$
84	RFJ	$(17, 5, 9)^T$	EEN	$(4, 4, 13)^T$
85	UIZ	$(20, 8, 25)^T$	ING	$(8, 13, 6)^T$
lanjut di halaman berikutnya,				

Tabel 2.1 – lanjutan dari halaman sebelumnya

Blok	Ciphertext	Vektor	Plaintext	Vektor
86	MSP	$(12, 18, 15)^T$	ENS	$(4, 13, 18)^T$
87	COG	$(2, 14, 6)^T$	OKY	$(14, 10, 24)^T$
88	SAR	$(18, 0, 17)^T$	OBE	$(14, 1, 4)^T$
89	ZWD	$(25, 22, 3)^T$	FOR	$(5, 14, 17)^T$
90	DNA	$(3, 13, 0)^T$	ETH	$(4, 19, 7)^T$
91	IUB	$(8, 20, 1)^T$	ERE	$(4, 17, 4)^T$
92	MWU	$(12, 22, 20)^T$	ISS	$(8, 18, 18)^T$
93	KEX	$(10, 4, 23)^T$	URE	$(20, 17, 4)^T$
94	LEP	$(11, 4, 15)^T$	LYN	$(11, 24, 13)^T$
95	IQG	$(8, 16, 6)^T$	OWA	$(14, 22, 0)^T$
96	ZTI	$(25, 19, 8)^T$	YTH	$(24, 19, 7)^T$
97	CEO	$(2, 4, 14)^T$	ISI	$(8, 18, 8)^T$
98	WWR	$(22, 22, 17)^T$	SJU	$(18, 9, 20)^T$
99	OCR	$(14, 2, 17)^T$	STA	$(18, 19, 0)^T$
100	NOB	$(13, 14, 1)^T$	NOR	$(13, 14, 17)^T$
101	JCD	$(9, 2, 3)^T$	DIN	$(3, 8, 13)^T$
102	IIB	$(8, 8, 1)^T$	ARY	$(0, 17, 24)^T$
103	DOH	$(3, 14, 7)^T$	PAT	$(15, 0, 19)^T$
104	LBV	$(11, 1, 21)^T$	IEN	$(8, 4, 13)^T$
105	LMZ	$(11, 12, 25)^T$	TIN	$(19, 8, 13)^T$
106	MOJ	$(12, 14, 9)^T$	WHI	$(22, 7, 8)^T$
107	KSN	$(10, 18, 13)^T$	CHC	$(2, 7, 2)^T$
108	KKY	$(10, 10, 24)^T$	ASE	$(0, 18, 4)^T$
109	YQX	$(24, 16, 23)^T$	ITI	$(8, 19, 8)^T$
110	PFL	$(15, 5, 11)^T$	SCL	$(18, 2, 11)^T$
111	LLR	$(11, 11, 17)^T$	EAR	$(4, 0, 17)^T$
112	CND	$(2, 13, 3)^T$	THA	$(19, 7, 0)^T$
113	RMX	$(17, 12, 23)^T$	TSH	$(19, 18, 7)^T$
114	SCX	$(18, 2, 23)^T$	EHA	$(4, 7, 0)^T$
115	GKG	$(6, 10, 6)^T$	SSO	$(18, 18, 14)^T$
116	UYQ	$(20, 24, 16)^T$	MEC	$(12, 4, 2)^T$
117	FZQ	$(5, 25, 16)^T$	ONN	$(14, 13, 13)^T$
118	FXF	$(5, 23, 5)^T$	ECT	$(4, 2, 19)^T$
119	XRH	$(23, 17, 7)^T$	ION	$(8, 14, 13)^T$
120	TWP	$(19, 22, 15)^T$	TOT	$(19, 14, 19)^T$
121	UJQ	$(20, 9, 16)^T$	HEO	$(7, 4, 14)^T$
122	WDO	$(22, 3, 14)^T$	RGA	$(17, 6, 0)^T$
123	TYJ	$(19, 24, 9)^T$	NIZ	$(13, 8, 25)^T$
124	KYZ	$(10, 24, 25)^T$	ATI	$(0, 19, 8)^T$
125	OKL	$(14, 10, 11)^T$	ONW	$(14, 13, 22)^T$
126	PWJ	$(15, 22, 9)^T$	HAT	$(7, 0, 19)^T$
127	TXP	$(19, 23, 15)^T$	CON	$(2, 14, 13)^T$
128	MXG	$(12, 23, 6)^T$	NEC	$(13, 4, 2)^T$
129	MTK	$(12, 19, 10)^T$	TIO	$(19, 8, 14)^T$
130	CRZ	$(2, 17, 25)^T$	NDO	$(13, 3, 14)^T$
131	SAI	$(18, 0, 8)^T$	ESS	$(4, 18, 18)^T$
132	NMR	$(13, 12, 17)^T$	HEH	$(7, 4, 7)^T$
133	YUZ	$(24, 20, 25)^T$	AVE	$(0, 21, 4)^T$
134	EVG	$(4, 21, 6)^T$	TOE	$(19, 14, 4)^T$
lanjut di halaman berikutnya,				

Tabel 2.1 – lanjutan dari halaman sebelumnya

Blok	Ciphertext	Vektor	Plaintext	Vektor
135	LBV	$(11, 1, 21)^T$	IEN	$(8, 4, 13)^T$
136	WHM	$(22, 7, 12)^T$	TEI	$(19, 4, 8)^T$
137	GPV	$(6, 15, 21)^T$	THE	$(19, 7, 4)^T$
138	EFW	$(4, 5, 22)^T$	REA	$(17, 4, 0)^T$
139	VUB	$(21, 20, 1)^T$	RER	$(17, 4, 17)^T$
140	OOK	$(14, 14, 10)^T$	UMO	$(20, 12, 14)^T$
141	ARS	$(0, 17, 18)^T$	RSA	$(17, 18, 0)^T$
142	IVY	$(8, 21, 24)^T$	BOU	$(1, 14, 20)^T$
143	HAC	$(7, 0, 2)^T$	NDT	$(13, 3, 19)^T$
144	PWJ	$(15, 22, 9)^T$	HAT	$(7, 0, 19)^T$
145	SQY	$(18, 16, 24)^T$	EIE	$(4, 8, 4)^T$
146	GJH	$(6, 9, 7)^T$	NTE	$(13, 19, 4)^T$
147	ASH	$(0, 18, 7)^T$	IHO	$(8, 7, 14)^T$
148	GQG	$(6, 16, 6)^T$	USE	$(20, 18, 4)^T$
149	OUJ	$(14, 20, 9)^T$	SLU	$(18, 11, 20)^T$
150	RCF	$(17, 2, 5)^T$	NAR	$(13, 0, 17)^T$
151	BFL	$(1, 5, 11)^T$	IAN	$(8, 0, 13)^T$
152	JJR	$(9, 9, 17)^T$	SWH	$(18, 22, 7)^T$
153	JTA	$(9, 19, 0)^T$	OFL	$(14, 5, 11)^T$
154	NPQ	$(13, 15, 16)^T$	EDF	$(4, 3, 5)^T$
155	CBK	$(2, 1, 10)^T$	ROM	$(17, 14, 12)^T$
156	GPV	$(6, 15, 21)^T$	THE	$(19, 7, 4)^T$
157	SUE	$(18, 20, 4)^T$	MOO	$(12, 14, 14)^T$
158	NOB	$(13, 14, 1)^T$	NOR	$(13, 14, 17)^T$
159	CND	$(2, 13, 3)^T$	THA	$(19, 7, 0)^T$
160	RCN	$(17, 2, 13)^T$	TIT	$(19, 8, 19)^T$
161	UAE	$(20, 0, 4)^T$	ISA	$(8, 18, 0)^T$
162	YSI	$(24, 18, 8)^T$	SEC	$(18, 4, 2)^T$
163	XIX	$(23, 8, 23)^T$	RET	$(17, 4, 19)^T$
164	NFV	$(13, 5, 21)^T$	MIL	$(12, 8, 11)^T$
165	QMN	$(16, 12, 13)^T$	ITA	$(8, 19, 0)^T$
166	DSF	$(3, 18, 5)^T$	RYB	$(17, 24, 1)^T$
167	KKY	$(10, 10, 24)^T$	ASE	$(0, 18, 4)^T$
168	ZWD	$(25, 22, 3)^T$	FOR	$(5, 14, 17)^T$
169	GPV	$(6, 15, 21)^T$	THE	$(19, 7, 4)^T$
170	BIF	$(1, 8, 5)^T$	LUN	$(11, 20, 13)^T$
171	MKT	$(12, 10, 19)^T$	ARC	$(0, 17, 2)^T$
172	ACP	$(0, 2, 15)^T$	API	$(0, 15, 8)^T$
173	PGJ	$(15, 6, 9)^T$	TAL	$(19, 0, 11)^T$
174	YCJ	$(24, 2, 9)^T$	IFE	$(8, 5, 4)^T$
175	XJM	$(23, 9, 12)^T$	ITH	$(8, 19, 7)^T$
176	SMH	$(18, 12, 7)^T$	ERO	$(4, 17, 14)^T$
177	VMQ	$(21, 12, 16)^T$	FTH	$(5, 19, 7)^T$
178	EGK	$(4, 6, 10)^T$	ESE	$(4, 18, 4)^T$
179	OYP	$(14, 24, 15)^T$	ARE	$(0, 17, 4)^T$
180	INB	$(8, 13, 1)^T$	TRU	$(19, 17, 20)^T$
181	DNA	$(3, 13, 0)^T$	ETH	$(4, 19, 7)^T$
182	DHU	$(3, 7, 20)^T$	ENJ	$(4, 13, 9)^T$
183	VRP	$(21, 17, 15)^T$	UST	$(20, 18, 19)^T$
lanjut di halaman berikutnya,				

Tabel 2.1 – lanjutan dari halaman sebelumnya

Blok	Ciphertext	Vektor	Plaintext	Vektor
184	EKZ	$(4, 10, 25)^T$	WHA	$(22, 7, 0)^T$
185	QVC	$(16, 21, 2)^T$	TIS	$(19, 8, 18)^T$
186	KRN	$(10, 17, 13)^T$	THI	$(19, 7, 8)^T$
187	IIZ	$(8, 8, 25)^T$	SPE	$(18, 15, 4)^T$
188	OLQ	$(14, 11, 16)^T$	RSO	$(17, 18, 14)^T$
189	SZT	$(18, 25, 19)^T$	NDE	$(13, 3, 4)^T$
190	HUD	$(7, 20, 3)^T$	PEN	$(15, 4, 13)^T$
191	JCD	$(9, 2, 3)^T$	DIN	$(3, 8, 13)^T$
192	NTB	$(13, 19, 1)^T$	GON	$(6, 14, 13)^T$
193	ODM	$(14, 3, 12)^T$	HOW	$(7, 14, 22)^T$
194	GPV	$(6, 15, 21)^T$	THE	$(19, 7, 4)^T$
195	ZDX	$(25, 3, 23)^T$	SIT	$(18, 8, 19)^T$
196	PJJ	$(15, 9, 9)^T$	UAT	$(20, 0, 19)^T$
197	XRH	$(23, 17, 7)^T$	ION	$(8, 14, 13)^T$
198	HKD	$(7, 10, 3)^T$	DEV	$(3, 4, 21)^T$
199	QSF	$(16, 18, 5)^T$	ELO	$(4, 11, 14)^T$
200	YXW	$(24, 23, 22)^T$	PSI	$(15, 18, 8)^T$
201	KZI	$(10, 25, 8)^T$	TCO	$(19, 2, 14)^T$
202	HDK	$(7, 3, 10)^T$	ULD	$(20, 11, 3)^T$
203	TCV	$(19, 2, 21)^T$	TUR	$(19, 20, 17)^T$
204	QJI	$(16, 9, 8)^T$	NOU	$(13, 14, 20)^T$
205	UVF	$(20, 21, 5)^T$	TTO	$(19, 19, 14)^T$
206	CVA	$(2, 21, 0)^T$	BEA	$(1, 4, 0)^T$
207	SAS	$(18, 0, 18)^T$	SCO	$(18, 2, 14)^T$
208	KKV	$(10, 10, 21)^T$	OPA	$(14, 15, 0)^T$
209	IVY	$(8, 21, 24)^T$	BOU	$(1, 14, 20)^T$
210	JYD	$(9, 24, 3)^T$	TIL	$(19, 8, 11)^T$
211	GBS	$(6, 1, 18)^T$	LEG	$(11, 4, 6)^T$
212	LXC	$(11, 23, 2)^T$	ALD	$(0, 11, 3)^T$
213	YFY	$(24, 5, 24)^T$	RUG	$(17, 20, 6)^T$
214	MWD	$(12, 22, 3)^T$	SBE	$(18, 1, 4)^T$
215	UIZ	$(20, 8, 25)^T$	ING	$(8, 13, 6)^T$
216	VBX	$(21, 1, 23)^T$	MAN	$(12, 0, 13)^T$
217	COB	$(2, 14, 1)^T$	UFA	$(20, 5, 0)^T$
218	GCH	$(6, 2, 7)^T$	CTU	$(2, 19, 20)^T$
219	HAD	$(7, 0, 3)^T$	RED	$(17, 4, 3)^T$
220	GWH	$(6, 22, 7)^T$	ATE	$(0, 19, 4)^T$
221	LBV	$(11, 1, 21)^T$	IEN	$(8, 4, 13)^T$
222	WHM	$(22, 7, 12)^T$	TEI	$(19, 4, 8)^T$
223	UED	$(20, 4, 3)^T$	ORS	$(14, 17, 18)^T$
224	ACM	$(0, 2, 12)^T$	OME	$(14, 12, 4)^T$
225	KRN	$(10, 17, 13)^T$	THI	$(19, 7, 8)^T$
226	ONE	$(14, 13, 4)^T$	NGM	$(13, 6, 12)^T$
227	VBZ	$(21, 1, 25)^T$	UCH	$(20, 2, 7)^T$
228	SHY	$(18, 7, 24)^T$	BIG	$(1, 8, 6)^T$
229	FFH	$(5, 5, 7)^T$	GER	$(6, 4, 17)^T$
230	MYG	$(12, 24, 6)^T$	WEW	$(22, 4, 22)^T$
231	HTK	$(7, 19, 10)^T$	ILL	$(8, 11, 11)^T$
232	TXP	$(19, 23, 15)^T$	CON	$(2, 14, 13)^T$
lanjut di halaman berikutnya,				

Tabel 2.1 – lanjutan dari halaman sebelumnya

Blok	Ciphertext	Vektor	Plaintext	Vektor
233	LMZ	$(11, 12, 25)^T$	TIN	$(19, 8, 13)^T$
234	ZFT	$(25, 5, 19)^T$	UET	$(20, 4, 19)^T$
235	UGS	$(20, 6, 18)^T$	OGA	$(14, 6, 0)^T$
236	GPV	$(6, 15, 21)^T$	THE	$(19, 7, 4)^T$
237	BOT	$(1, 14, 19)^T$	RIN	$(17, 8, 13)^T$
238	ZWD	$(25, 22, 3)^T$	FOR	$(5, 14, 17)^T$
239	BRL	$(1, 17, 11)^T$	MAT	$(12, 0, 19)^T$
240	XRH	$(23, 17, 7)^T$	ION	$(8, 14, 13)^T$

Artefak

Skrrip *solver* yang mengimplementasikan pencarian invers matriks secara modular dengan *Known Plaintext Attack* (KPA) pada bahasa pemrograman Python (`src/25/solver.py`). Skrip ini secara otomatis memecah teks menjadi blok $m = 3$ dan melakukan proses matriks balikan (*decryption*). Log penyelesaian (metadata kunci dan *plaintext*) disimpan di `src/25/log`.

Plaintext Hasil Dekripsi

EXPLOSIVESCOOPSUSPICIOUSSHADYCHARACTERISFREQUENTINGPHARMACEUTICALFACILITYWEHAVELEARNEDTHATAMYSTERIOUSCHARACTERHASBEENVISITINGEIENTEIGENSOKYOSONLYDRUGORGANIZATIONSHEDISPLAYSFASHIONUNCOMMONTOGENSOKYOWITHADISTINCTIVESINGLEWINGEDAPPEARANCEANDHASNOTBEENSEENINGENSOKYOBFORETHEREISSURELYNOWAYTHISISJUSTANORDINARYPATIENTINWHICHCASEITISCLEARTHATSHEHAS SOME CONNECTION TO THE ORGANIZATIONWHAT CONNECTION DOES SHE HAVE TO EIENTEI THERE ARE RUMORS ABOUT THAT EIENTEI HOUSES LUNARIANS WHO FLED FROM THE MOON OR THAT IT IS A SECRET MILITARY BASE FOR THE LUNAR CAPITAL IF EITHER OF THESE ARE TRUE, THEN JUST WHAT IS THIS PERSON? DEPENDING ON HOW THE SITUATION DEVELOPS IT COULD TURN OUT TO BE A SCOOP ABOUT ILLEGAL DRUGS BEING MANUFACTURED AT EIENTEI OR SOMETHING MUCH BIGGER WE WILL CONTINUE TO GATHER INFORMATION

Plaintext dengan Format Asli

Exclusive Scoop!
SUSPICIOUS SHADY CHARACTER
is frequenting pharmaceutical facility

We have learned that a mysterious character has been visiting Eientei, Gensokyo's only drug organization. She displays fashion uncommon to Gensokyo and a distinctive single-winged appearance, and has not been seen in Gensokyo before. There's surely no way this is just an ordinary patient, in which case it is clear that she has some connection to the organization.

What connection does she have to Eientei? There are rumors abound that Eientei houses Lunarians who fled from the moon or that it is a secret military base for the Lunar Capital. If either of these are true, then just what is this person? Depending on how the situation develops, it could turn out to be a scoop about illegal drugs being manufactured at Eientei, or something much bigger. We will continue to gather information.

Sumber:

- https://en.touhouwiki.net/wiki/Alternative_Facts_in_Eastern_Utopia/Article_and_Interview/Sagume_Kishin

Kunci

Matriks Kunci (K):

$$\begin{pmatrix} 18 & 22 & 1 \\ 25 & 12 & 7 \\ 16 & 9 & 11 \end{pmatrix}$$

Referensi

Konsep *Hill Cipher* dan *known-plaintext attack* mengacu pada bahan kuliah [8] dan [1]. Perhitungan invers matriks modulo 26 mengacu pada materi invers modulo [6].

Bab 3

Problem Set (Bonus)

3.1 Ex Machina

Deskripsi Soal

Di tengah *hollow night*, sebuah transmisi aneh muncul dari salah satu wilayah yang telah lama ditinggalkan. Sinyalnya berulang secara teratur, tetapi tidak seorang pun mengetahui siapa yang mengirimkannya ataupun sudah berapa lama mesin tersebut aktif.

Setelah diteliti, pesan tersebut diketahui telah dienkripsi menggunakan **Enigma Machine M3**. Namun, setiap percobaan untuk mendekripsinya menggunakan konfigurasi Enigma M3 selalu menghasilkan teks yang tidak masuk akal. Entah karena pengaruh *hollow night*, kerusakan pada mekanismenya, atau memang karena mesin tersebut sejak awal dibangun dengan aturan yang berbeda, Enigma ini telah dimodifikasi.

Analisislah perilaku mesin tersebut, **temukan perubahan yang membedakannya dari Enigma M3 standar**, kemudian lakukan **kriptanalisis** untuk merekonstruksi konfigurasi yang digunakan dan memulihkan pesan aslinya sebelum *hollow night* berakhir.

Berkas lampiran:

- Attachment: https://drive.google.com/file/d/1TcBU6s876b5ah9EVsuX_xHDTc_0LaWsI/view?usp=sharing
- md5sum: 1074d1cfcaf2482f7229c8f7d40ba097
- sha256sum: fd31f4b0b6475fc15926caea6b28179ea2a1228fe9f063cad8b2d5c215d9031b
- Kode mesin: src/31/enigma-machine/

Berkas *Ciphertext*

```
UTJX CENI RYOW WSVU YEHC YRPR BJWZ NXDJ UNVX WXFE
WOUO DQDZ XJTI QKBI USMI AXEI JFZX XQWJ IBXM PCBW
BAFH OAXG BPBD KASZ LQOY ASNA ZPPC UTVF ISQZ HILW
GYCJ AUQV UGXH VQWW OXMX WEBW XSBQ EGZM OSKQ OYWK
MLKZ YPRC MMHW EHEQ DYJU XLVA CJPB SSVH WFFM WKGS
SAVE XVMF ZGCI SPEJ EFHD RXBS LZDX WYYU UWHB WTJG
ZFJM MTUV HHEZ GVKU IRON NNUU VVFD XZGW BUHB KSSG
VBHP KCQN CJCP GILT KZZK FMLW YVNS OXYJ AXQF RNML
NRYI RLIP CYPC WHQI YLUS JDDH HIDZ OWMH HHML CXXG
BJPY HWFI CMGP XGBP LBHZ ESGG XATC ERAY OSFN MKBP
WWKA PXHU UVIG ICOZ PDLW IMYC XZZH YQUV GUPM WYUE
RYPG TMWO GMTC PMXK ZLPS ACWW UFKO SJST OHGW AXOL
JNBX SQUP UIAP OMLU THXE LCGR HVDP HYLK HQXU OJMO
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NCWD WIYK BZXM RPBO SYGI ZZYF AKCQ HVCM OHBT XEGI
GHXE YBZL OHSA DDZF AZPB AQVL AMCI TBFN TBCS ZNQO
FRCD WZRB KJEL MWDC RDAB NQIH DJKQ PYYB GNLN GWWL
CNCZ NEDW FXXR XPTH LVWN ZCXB TNYH PQUG TNFV AXFE
KSGO HNZP PKHK HYQL SWQL RWXV LVAC WYWN IFBU JEPX
YSNJ IYXB JVNN QMEI FHMY KJTR YBAM HQBF FMWJ ZZVF
WUMS UPVJ BGUQ OJNP QWMM NABE YCCF RJTQ UHGL MMYM
KMIH VDKC EAOJ YCPB CDQV OEIL FLLO MGHG PPPA QOUS
UYIN MKBE HBGW VGVX ALOJ DGRD FBUQ XYJQ AYHV EKZY
SUJG LADI LTQT DXFZ AHLY LZWZ TBRJ BGKZ NRAB ZKXH
YYTB POXJ MXCP HDPR FCUG CSRB TQQY ZTBH ZZKP XEBO
TLZN XLOY ZSSV MQIU NVMP PSOG HBJZ CTPD EGXV AGWQ
MJBZ AMSI DYKS TZGB LPIW RMKO ZRYY OLTO YKKB YZWJ
EDDR RRLY WZRA PKNF IJGF SBJD IOQW BUCF PUEQ MYOZ
DBVQ GYUB CQJX VCSA TQCY XSUL RFNL ODFC KTKD CCIS
AZFE MHPR SAPV CGFQ HCAB JVBK CSWW CERU UOJX FERO
FGYS PIRG MDTP YWWV BOIK LOBZ QQON QCSN YZPM EEZP
NAJC TDLK VMAL VMRT OIHJ YKGL AYGM YCJE RKZD BBKE
QFNC IIHG JXMQ BMAS KJVG GITP HUTX YTMR YGIA YTHP
DRAF OZEG IHJV SARO HTPJ IVRN XGTJ EXJI MWBK CYCH
NOPN SVMM FRZH RAET IYKA GAKC HFVX BUTT JGGK HKWW
VKBY POVG IYIC KOZA ZDPK VXXE TZZR AJMQ RAVC FJIC
QNNR KJLG ZFMW QOQP KGFT DULW DWWP LHKB UIUP QDCE
CGOG IAIC UCWW GESB YSDW FAFX DYAF QGVZ JBVV VRAF
NAYL JQTN XDYU KANE WKZJ FVAI MSGZ PMKV SRAS PRPD
UVXV NXJI VNAY PVQZ WBPQ FXRG AOGP JXII CPFM ZKWP
IHQY KHOK IHGQ KTQH QTMD MWLG YUUG BOVY QBHG VKXF
LDUG LSWX JAQY SAVN PYFR CCLW OPPB OOMV IHQT SCKN
JTSL ZQLC ISKC LPUI NKJF CQZZ NEDW WXYI SBCO KVPY
RHLV CHMA BOCX GZXT TCHA FJGC GFGU LEGB GESE LXUQ
LZZO BIIP NLYI HDEC OSUH JJZX EPZP PCWP XAVT VNWG
AJQU HXCC ATXZ FMKD NFXC CCSX TNON RNRQ KJQB SODI
HOZQ HCUQ WQJQ YCYD BXHK NSQK RUOM NOCE IXRL PQHZ
NKAJ TAGS ZFSI VQHG ULJV PZPJ KEBX ZXXT UCWX MXBR
MEPQ APOG COGY PJVO MSVW CWOK JYKV MNBR DZTO GATI
XXPN ZHLD OFLX LHWY SFJF IFCZ PDYG BHFQ YCRB GCHA
JAGX PNTK RFOL RWRD ZHPD NOGG JJTO PXPQ RWPU JTQY
YUIE GQJD YCYC MLHM FZLS HNFV QCUY FBFS XXGO FJPT
CQPS YHOD AVPR ALGP JPKB FFDH KLGS ZGXS KYUX BWSX
YPSY MYFY NWYO BNJY FDLR UPNV QRMU YWET RWEF PMCS
SKTK KNVO FPWB QBXA ENXM KMOZ SCPJ SDYX JFCV XPRJ
LDFQ DNCV VMXH WFLH CYIJ XHFC CYAD QRQO AGKT OXGL
VGJC ZFQE MTNZ OBZT AMMC RZMP SFEP YNZO LOIA QSPE
QRVQ LTEM DPZS RJBP YJEV WROV HRFO WJPS BENC AUJH
EDWS GFGJ RNYL BBKG KJXW VLXI EKQL LTUB PSNG XJRR
JZXP PLYP QPVS AXEM QSDI MFOF RMKC PMIU BGCI KMHU
VZRL QRAJ AEUP JQMY SUTY SYCQ FWYQ VDXA PXJV KQBS
HYDU QEUR SZUA FWQF OQDT DUFQ VOKW RWAC BFIF XZNM
VPWU ADXS LNCV LBAK OJNZ JMFO XQWL TZDT SGRB OTKG
DVMA XSQB BDNJ PPGB GIUN JGSV YFER AVPS WDYL XVVA
ZLQB GIBL TGJH DZNH OFKM DXMP VYGQ GHIC IZXJ FDDB
XJUL WOZZ XUGC TTHW YGKI NSXI ZREE YYWR IKCS ONQE
NJTZ FZWC ATSM MQLA VLNN BCCS ZBKK FEHZ GTYE ZYVC
XNYB RWXG RHQT RDWK FHJP THQA YPEG ZVQW ZXNX WMRO
YZMU HDDC BAWN MATS XAWM GHDJ TSEJ BOAE EVWU YYMD
YJQU NFEC QLWD WTKG CBTN VPLA DQDY HAYB UPLJ ZHTB
XMLB MTUP AMWM IONR MJRB XOYL PYZU JTRY KATU UAOZ
MLBV JZLL BDYV MXYR OQGX GRUK STVT MLSX OLZJ IEJT
NLRV HNUV PIKX VOBQ EXMU BHZK FHNC UNCN WVEY BHRU
XXSW IXFC DAUC ZYLR IECI ZVDW CTUN FVDE RJZP TRQY
EHMF MDDS DJTE BVIO ZFVO YPXM KKXA SYNS BFFW HOLQ
FZVA IFHS PZJB YAYH CMMD OTLD GEVL LLXJ SEKD WFAD
HYYG ZIIV ZOOM WANE XHND CDTU CGAV JOKK MTLE YDDW
QEFP NIVT CCUL DENX SCSU BPUH GIIT TTCN NCQP UBKC
YCXX PWVO TGVS KNSP GDMS DHIG OMQY IVQH IVCG CARZ
HRXX WSRU WWKO EOQC QKQF GRCF KOQD NRUH MNCT BZGP

```

```

JPGH BYXW RMVV QOWW PNCX BFQM BNGV HLXX WFZE WSAY
SMXY VWOZ LNPV HVHL MSAF EZIJ EVHY EWSI SHUJ VKIG
SNZI IHGV BIGX DELG NXTU KEWJ LJRG YLES JURS TGMN
JUBX QSYU BKAQ YMGM BPBI PCMK UYCY YPIS IVDH NNLO
KNKT MJIE QTKQ IXQF INFF ZOFV NBZS UQLP ZNLF DXCI
FPRW MIHU RRNL VNFQ DCEB MNZH AEAU FRND JIIP KIEJ
WPPI ZPON TRZN BNNT DANG VYEJ OGDY SZVG ZEPZ ZOME
JJJT QHZZ POGO NMYJ DIIJ PQEM YTHF QCFW WTDQ EQML
CENZ ZOOR FUXJ FLZT YEAQ YOEK QTBQ JBDC DLFV PTDA
FFFU DYUS MXPV HBGJ EFJZ POMU HMSJ CYVY QOBV XDEW
LWKV GZXU LHPR WTPV BKMV VESX PFNS PUEN VZCU OEKV
CHJR YODV METT PSYW WFMS EEZV YGJK DBNA KHER SSXE
BZVY NLPL FVRV KUBC YOTP TMOT BCXV GGJU YGYH GVBZ
CLAR HADF VAOX HPGH PVEB XVVY BGTL AMPC UCIL UMCX
FPCP GHGM MXYV IXVH HVTI MUPU XDCN GISL LLMI JUEH
RTMV CKFE ULKD NLYP ZZQC VFUI IWZG TGMH LPSI TEJD
UZTP GBVI XEDU QNCL RIAR UNQA LHAF RFLJ JSTJ BXHL
QSQV XQZA KLLK YGKX BTHY CAPR PORQ DAKA NMYW KHJE
EIGX VGKG NXGL FTEA HEFX BNQU GIKS TJJH CNAJ XQQK
XRUV ZKAA YMTX XUHN RPBQ RGOI QTUJ XEIU NLBZ OFEV
PIHB UUMW NHHH RACR LRIR UQWQ NVYS QAQI LWQL RTLR
XWHD WNSF AURV FPMI DAKA LPJL SZDH RJRV QNEY OTGQ
AZOF SYPB QMSK GSUK HINN TNIJ HRWI KYPM VRBV CLQL
VINE VGYN CEAQ UWIE QJUI SYGU WEJJ OGTA MOAQ XDMU
XQML BINO TTYV JYEE LXOW NXXV ZSLO GMLO ZQJH OTLZ
KVXC GNDJ MAQE IXWS SBPI NZST VOUJ EFJC LWVP TGVR
IBOV DRAP VLDQ AKQB DTJM YNGM GBIY IQZD BBRN KHBB
ZOEY LFCM FULG FTSU DFIW BSDN RTDF PJWM FKJH FCLR
PYCN FPYR AJIB SEGV DSGY TZPD CYMH BCLV FXFG DFSU
HUXM SNCN PGCS QOQG BEET WFMV BLUK ZSVA PWVX RHDE
CPUS CFOV ZGKV QSNX ZORB SIGH PXQD WYPY IPQY QTBE
FPLK LHUK BGAG CIVW RQCB MLCB TJKQ XXGL GOTL PXQN
UKWH UASG CXBO NEWQ PMPB DUKA RQCB LJZW PXUK YLWC
GHCH EZPR GFFL DGQH TRUQ FFGG YOQX ZQWY UVMX ZIZH
PVTI HKPK RXVO ZHBP ZFIM NJZZ BLRY YBJE DLNX RVBG
ONOK YHLR QENX TSXX PMQP YYAD PYIH PRHA ZRSX WGTW
ADKA GLBT VKPH WNVS XJNE SNNT SRCK FAVZ HROC MSHS
WVRS GJXY ZELC OYWT JXOU VVPM AWSV ZXJU WNSE KQCJ
EUSP UWTG DLYY QWTM UWAC OFIY BFCY JKXS TSYW EGSG
DKME FPDQ DLGR UPFN VWUD AYPC UFCP AVIV QMWU XONL
CGSC EFWR DRUG YQHM SLCT IQOS WRBF WSAM RXWA AIVO
GCPY SCLZ DWTX RLDZ MLST HHJW XRYD NYMK SIIE DXCD
YHLC JNLN HNVH YBQW FFFF KSOM VPOR OQUA ODVC NZEM
QYPS VTAK ZNXX PWOL QYKB JGDP WGHK IRAS HCID LLGC
VFUW TMUF LOMK NOWC ISNB OUQR ZRVW KUUU BYEO EYDF
JVDV NGXR BYEE RNJA XWVG IMXR BQFR ZZAK FPNQ GNOI
OALF VWOK PTRF MFBM PHRL MJAT CHJG CEYD NJVE WJBR
CGPS LONH XDHN EANP SMYL VJNJ FWVI NWRH LGBB YBHW
PDGI FWEX BCCC ZPOQ RSAY OUGR QWPX JOHS NAHO EFVG
KIKV XCAD DTQC VUUQ RCJC LNSU XKUM PRKM HPQF GAQI
CFGD TOKS LAMZ KAJG HSER HUNH PMXM SGFE VBAU SOYB
GXNI APKW FNEW HGHP DXEP GPIU CNIB DDIG YCCE TOWB
OSJZ CHQU ALIO XGUI APYG UDZD QUZZ FBAS EAKX CXSU
EQDT EUIP SWUT BESM BDFG FQZB SITO ZHEB AIPO QISJ
BBZS BWHQ CTHR PBKK DRPQ UTTG RXJJ SKVC OJUW MLRR
IEUK GFTL NUUZ BMIF GYWI BFUK MMXJ ZSTL VFLR FBFR
OTZA SJBH FABD RXPV UCTN KFMM CPWH ZVYB PDCR TZOM
XTQU PKJT MGVE UFKC XZJY KOVN TUDL YUML USPF GZVZ
KCJX YQRY YEHF SEKL JPRS SOTJ NPUK MHGX CNCD VXDW
NMTV BYMA RGWM UGBH POWX XYEW YPXV FKHS ETBV YBUJ
PXGE GCOL UOHY BIPX NIKP VDBR OPGQ XSSX PEWQ PCXZ
NITG WNRK QVSW SBWY FWNC CYZV SNFL SQUP UQIA QDXM
DYFV YDPK YXWP YQFQ SKQA RWGZ JJJJ YJBF VTSU HOVW
EIRO RIKC MPPF MMQY VDOU QAEJ DHUO PINZ BPQD ENDH
CDJG THOB DERH RMZJ BKDK PDFF RPQN UQRG KNFC NMPE
NCKM NPZJ RKGM RKPI WNXS PEGG BRPL HQUC KVJJ UMFM

```

```

KVZR YOGC YJXM UZVN DRUK WBHQ XZPH DPYI LQUV KBPP
TYNF UVZZ YQWO JTSN IFNQ LPLR JZEJ ZZQJ LNKQ OKQT
UMLF QVOZ WSLP THJF DJUC KQEX ISCX QCEP VCQC NMRR
QLXZ XFOP FZVP ZXGJ RQSS BLPS JDZZ PLSW RCPC MCNG
FCJD PQIO GNCA MYHA GOCF BVZE REZY WIDC DRYX GQJW
VLKC KMXZ KTRT SQDH EDKW GDBT KBRX CZRK HDYU HVIF
WNFX QGFT OYQU NQLB SBRQ QFGR FTXF HCCV XQWW CZQN
SZCG VFMZ EZGD XSWI WFZU SPBQ CCOJ GLZI GRUY DREP
LRVR OFSN KKVJ BANW OINR ONFF OUKA BKPM EPWJ FJZX
AJKZ CNCR TAMK FXHY ZCAH GHTU ZTKF PCBR HPEK VXCD
HXKB QTVH CHWU ZJCX HRPT U MVC UBIO XSGO AINB APYK
CZJS DRBV JCLT QFNR YQRS FKXT TBWY LDVC DJNJ CVED
TXQB AVIW VVDB WZHY WGQQ ZZYF NVXW OPBQ CRDD BDLW
SCQG CGXD UTMB RBGA VMWJ MFEE QHBM PYDL YMGC IJKF
OUES HWWO ELRO XHHF CLSV QBTT BPTO NGKM DDWB FGAP
DEYU JNHR FQAY DPTA TRND NBCI SBHR OASR CMJU HPXQ
KLDD IVHA SHNT HGPZ PXWU MGYL JWLF DSWE SMQN AQOM
MIQX MIPV PWPA YBMN VWSQ WGJX EPHY ICAB MEMB NMVI
XLUO XOUB SOOY PXIS CBQP ZDNU PDYU JQUA GXHX KSJK
FGUJ YQDW EIXP VBSE IANY MWIR ITTP MUVF SUYV XUAR
YKSO JEVZ AMFS QZPY TKIN JZVO IGFK LNGL MNUK RIEU
CKVQ BQVL VIWT MZKP VKZZ FRBX CJXC ORRD SHMO LWBQ
OSVO VOVM XUIZ CNLJ FWVF QSNB NVXN CHFG XYYO GVLU
EDZK OWBC AWOX XTSI XRQO XSHS NBJP CQKW XPLB BBJT
QLYS NBLB QXSN UXPL KSYF BFUH INXV ZPTN YQTF RQCO
AYDG ZFVB MCAG ETBM XZYJ EVEP BCXU ZVQJ PEBD DZUX
JFNL VMZY CKXS QRIN MZWL KUVY VPEJ UHLR REWR BUGH
LCGI TBPV VEAG ETCM VBDR CFVM AZTS JLPY OJPM JTKB
NKKP MJXW FFFA RAYR YSYX HCBH QWUY LKBG ZIKI ICYH
LXOD KHYL HEWJ BNXQ RGXN GWUD LKPJ XVXD MHPE ORRL
JTZV MONB YISX SVEE CHXG JNLY QBYA WPQZ STDO BEJF
JEMV WPFO BJZP LIDL GGOG CPVR JQEP OLNH HVCR QNWQ
LQHJ BXFD F

```

Analisis Modifikasi Mesin

Ketika zip diekstrak dan Makefile dijalankan, terdapat dua *executables* yang dihasilkan, yakni *machine* dan *encode*.

```

src/31/enigma-machine  ? main  !? >
ls
[ encode
[ encode.c
[ enigma.c
[ enigma.h
[ machine
[ machine.c
[ Makefile
[ plugboard.c
[ plugboard.h
[ reflector.c
[ reflector.h
[ rotor.c
[ rotor.h
[ secret.txt

```

Untuk setiap *header* (*.h) dan *kode* (*.c) berkoresponden pada komponen asli dari Enigma Machine M3, yakni sebagai berikut:

1. *Plugboard* (`plugboard.h` dan `plugboard.c`): Mensimulasikan *plugboard* (*Steckerbrett*), yakni papan kabel yang menukar pasangan huruf secara dua arah (jika A dihubungkan dengan B, maka $A \rightarrow B$ dan $B \rightarrow A$). Huruf yang tidak dihubungkan akan dipetakan ke dirinya sendiri. Pemetaan disimpan pada `wiring[26]`. Fungsi `connect()` membentuk pemetaan dari tepat 10 pasang huruf, sedangkan `identity()` menghasilkan *plugboard* tanpa kabel (setiap huruf dipetakan ke dirinya sendiri). *Plugboard* dilewati **dua kali** dalam satu kali enkripsi karakter, yaitu sebelum sinyal masuk ke rotor dan setelah sinyal kembali dari reflektor.
2. *Reflector* (`reflector.h` dan `reflector.c`): Mensimulasikan reflektor (*Umkehrwalze*), yaitu komponen statis yang memantulkan sinyal kembali melewati rotor dengan pemetaan berpasangan. Di sini, hanya ada **Reflector B** (YRUHQSLDPXNGOKMIEBFZCWVJAT) melalui fungsi `refl_b()`. String tersebut berarti $A \rightarrow Y$, $B \rightarrow R$, $C \rightarrow U$, dan seterusnya. Jika dibalik, $Y \rightarrow A$ dan $R \rightarrow B$. Jadi, ke-26 huruf dibagi menjadi 13 pasangan ($A \leftrightarrow Y$, $B \leftrightarrow R$, $C \leftrightarrow U$, ...), dan tidak ada huruf yang dipetakan ke dirinya sendiri (tidak ada $A \rightarrow A$).

Pemetaan berpasangan ini membuat Enigma ini memiliki dua sifat penting:

- Enkripsi sama dengan dekripsi. Jika dengan posisi rotor tertentu A dienkripsi menjadi X, maka dengan posisi rotor yang sama X akan dienkripsi menjadi A. Oleh karena itu, untuk mendekripsi pesan cukup memasukkan *ciphertext* ke mesin dengan pengaturan awal yang sama.
 - Huruf tidak pernah dienkripsi menjadi dirinya sendiri. Karena reflektor tidak pernah memantulkan huruf ke dirinya sendiri, sinyal selalu kembali melalui jalur yang berbeda sehingga huruf keluaran selalu berbeda dengan huruf masukan.
3. *Rotor* (`rotor.h` dan `rotor.c`): Mensimulasikan rotor (*Walzen*). `RotorSpec` menyimpan spesifikasi rotor I–VIII, yaitu *wiring* dan posisi *notch* (I: Q, II: E, III: V, IV: J, V: Z, VI–VIII: M dan Z). `Rotor` menyimpan *wiring* maju (`forward_wiring`), *wiring* balik (`backward_wiring`), posisi *notch*, *ring setting* (*Ringstellung*), dan posisi rotor (*Grundstellung*). Fungsi-fungsi pentingnya adalah sebagai berikut.
 - `build()`: membangun rotor dari spesifikasi, *ring setting*, dan posisi awal.
 - `fwd()` dan `rev()`: memetakan sinyal melewati rotor pada arah maju (menuju reflektor) dan arah balik (dari reflektor). Keduanya memanggil `map()` yang menggeser masukan sebesar $shift = position - ring$, melewatkannya ke *wiring*, lalu menggeser balik keluarannya.
 - `notch()`: memeriksa apakah rotor sedang berada pada posisi *notch*.
 - `step()`: memutar rotor satu posisi.
 - `get()`: mengambil spesifikasi rotor berdasarkan namanya.

Secara fisik, rotor terdiri atas dua bagian, yaitu **inti kabel** (*wiring*) yang mengacak huruf dan **cincin huruf** (*ring*) A–Z di sekelilingnya yang juga membawa *notch*. *Ring setting* adalah besar pergeseran *ring* terhadap *wiring*-nya, sedangkan *position* adalah posisi putar rotor saat ini (dinyatakan dengan huruf A–Z). *Ring setting* tetap selama satu pesan, sedangkan *position* berubah setiap kali rotor berputar. Karena $shift = position - ring$, pengacakan sebenarnya bergantung pada **selisih** keduanya. Contohnya, dengan rotor dan posisi yang sama tetapi *ring* berbeda, hasil enkripsinya berbeda:

```
echo AAAAA | ./machine I II III AAA AAA # BDZGO
echo AAAAA | ./machine I II III BBB AAA # EWTYX
```

Sebaliknya, jika *ring* dan *position* digeser dengan jumlah yang sama, selisihnya tetap sehingga pengacakannya juga sama:

```
echo AAAAA | ./machine I II III AAA AAA # BDZGO
echo AAAAA | ./machine I II III AAB AAB # BDZGO
```

Yang berubah hanyalah **waktu perputaran** rotor di sebelahnya. *Notch* menempel pada cincin, sehingga `notch()` memeriksa `rotor_position`, bukan posisi *wiring*. Hal ini berdampak pada kriptanalisis:

- *Ring* rotor kiri tidak perlu dicari. Tidak ada rotor lain di sebelah kiri rotor kiri, sehingga *notch*-nya tidak berpengaruh. Yang penting hanya selisih *position* dan *ring*-nya. Oleh karena itu, *ring*

rotor kiri dapat ditetapkan menjadi A dan cukup *position*-nya yang dicari. Pada *plaintext* 2000 huruf, konfigurasi AAA AAA dan BAA BAA menghasilkan *ciphertext* yang identik. Ruang kunci *ring* pun turun dari 26^3 menjadi 26^2 .

- *Ring* yang salah tetap menghasilkan teks yang sebagian besar benar. Selama selisih *position* dan *ring* benar, hanya waktu perputaran rotor tengah yang meleset. Pada 2000 huruf, konfigurasi AAA AAA dan AAB AAB baru berbeda mulai huruf ke-21, dan secara keseluruhan 1923 dari 2000 huruf (96%) tetap sama. Oleh karena itu, pencarian dapat dipecah menjadi beberapa tahap: urutan rotor dan *position* dicari terlebih dahulu dengan *ring* AAA, kemudian *ring* rotor kanan dan tengah, lalu *plugboard*. Setiap tahap dinilai menggunakan statistik (menggunakan CI jika mengikuti PPT). Dengan demikian, seluruh kombinasi tidak perlu dicoba sekaligus.

4. *Enigma* (`enigma.h` dan `enigma.c`): Merangkai seluruh komponen menjadi satu mesin dengan susunan tiga rotor (kiri, tengah, kanan), *plugboard*, dan reflektor.

- `rotate()`: mengatur pergerakan rotor sebelum setiap karakter dienkripsi. Rotor kanan selalu berputar, rotor tengah berputar ketika rotor kanan berada pada *notch*, dan ketika rotor tengah berada pada *notch*, rotor tengah dan kiri berputar bersamaan (*double stepping*).
- `run()`: mengenkripsi satu karakter dengan alur *plugboard* → rotor kanan → rotor tengah → rotor kiri → reflektor → rotor kiri → rotor tengah → rotor kanan → *plugboard*.
- `new()` dan `plain()`: membuat mesin dengan *plugboard* 10 pasang dan tanpa *plugboard*.
- `cli()`: mem-*parse* argumen, memvalidasi konfigurasi (rotor tidak boleh dipakai ulang, huruf *plugboard* tidak boleh duplikat), lalu mengenkripsi masukan dari `stdin` karakter demi karakter. Hanya huruf kapital yang dienkripsi, spasi dan *newline* diteruskan apa adanya tanpa memutar rotor, sedangkan karakter lain menyebabkan program berhenti dengan galat.

5. `machine.c`: Titik masuk program `machine` yang hanya memanggil `cli()`. Program ini dapat dijalankan dalam dua bentuk:

```
./machine <rotor_kiri> <rotor_tengah> <rotor_kanan> <rings> <positions>
./machine <rotor_kiri> <rotor_tengah> <rotor_kanan> <rings> <p1> ... <p10> <positions>
```

6. `encode.c`: Program pra-pemrosesan *plaintext*, bukan bagian dari mesin Enigma. Program ini mengubah huruf menjadi kapital, membuang karakter selain huruf, lalu memformat keluaran menjadi kelompok 4 huruf dengan 40 huruf per baris. Format ini sama dengan format `secret.txt`.

7. `Makefile`: Mengompilasi `machine` (dari seluruh berkas komponen Enigma) dan `encode`.

Langkah Penyelesaian

Karena tidak ada petunjuk *plaintext*, serangan yang digunakan adalah *ciphertext-only attack* berbasis *Index of Coincidence* (CI). CI mengukur peluang dua huruf yang diambil secara acak dari suatu teks bernilai sama:

$$CI = \frac{\sum_{i=A}^Z f_i(f_i - 1)}{N(N - 1)}$$

dengan f_i adalah frekuensi huruf ke- i dan N adalah panjang teks. Teks acak memiliki CI sekitar $1/26 \approx 0,0385$, sedangkan teks bahasa Inggris memiliki CI sekitar 0,065. *Ciphertext* soal ini (7089 huruf) memiliki CI 0,03864, yang berarti terlihat acak. Jika *ciphertext* didekripsi dengan konfigurasi yang benar, CI hasilnya akan naik mendekati 0,065, sedangkan konfigurasi yang salah tetap menghasilkan CI sekitar 0,0385.

Langkah-langkahnya adalah sebagai berikut.

1. Pilih 3 dari 8 rotor dengan memperhatikan urutan, sehingga terdapat $8 \times 7 \times 6 = 336$ urutan rotor. Setiap urutan memiliki $26^3 = 17.576$ posisi awal, sehingga total terdapat $336 \times 17.576 = 5.905.536$ konfigurasi.

Pada tahap ini, *ring* ditetapkan AAA dan *plugboard* diasumsikan tidak ada. Alasannya adalah sebagai berikut.

- Ruang untuk segala kemungkinan kunci terlalu besar untuk dicoba seluruhnya. Mesin ini hanya menerima *plugboard* kosong atau tepat 10 pasang. Banyaknya cara memilih 10 pasang dari 26 huruf adalah

$$\frac{26!}{6! \cdot 10! \cdot 2^{10}} = 150.738.274.937.250 \approx 1,5 \times 10^{14}.$$

Jika digabung dengan urutan rotor, posisi, dan *ring* rotor tengah dan kanan (26^2), totalnya sekitar $5,9 \times 10^6 \times 676 \times 1,5 \times 10^{14} \approx 6 \times 10^{23}$ konfigurasi. Jumlah ini mustahil dicoba satu per satu. Pencarian yang hanya mencakup rotor dan posisi sudah memakan sekitar 2,5 menit.

- *Ring* yang salah tidak menghalangi pencarian solusi. Seperti dijelaskan pada bagian rotor, *ring* rotor kiri tidak berpengaruh, sedangkan *ring* rotor tengah dan kanan yang salah hanya menggeser waktu perputaran rotor. Sebagian besar teks tetap terdekripsi dengan benar, sehingga CI hasilnya tetap jauh di atas CI teks acak.
- Karena konfigurasi yang mendekati benar sudah menaikkan CI, pencarian dapat dipecah, yakni dengan urutan rotor dan posisi dicari terlebih dahulu, kemudian *ring*, lalu *plugboard*.
- Jika hasil tahap ini sudah menghasilkan teks yang terbaca utuh, tahap *ring* dan *plugboard* tidak perlu dilakukan (lihat langkah 4).

2. *Solver* ditulis dalam bahasa C dan langsung menggunakan kode mesin yang diberikan (`plain()` dan `run()`), sehingga perilaku mesin yang disimulasikan sama persis dengan mesin pada soal. Seluruh 7089 huruf *ciphertext* didekripsi untuk setiap konfigurasi, lalu CI hasilnya dihitung. Pencarian diparalelkan per urutan rotor menggunakan OpenMP dan selesai dalam sekitar 2,5 menit pada 16 *thread*.

3. Keluaran *solver* adalah sebagai berikut.

```
Terbaik sejauh ini: rotor I II III posisi TCW CI 0.03868
Terbaik sejauh ini: rotor I II V posisi VVS CI 0.03873
Terbaik sejauh ini: rotor I VII V posisi PPE CI 0.03876
Terbaik sejauh ini: rotor VI II IV posisi KQF CI 0.06838

10 kandidat terbaik:
1. rotor VI   II   IV   posisi KQF CI 0.06838
2. rotor I    VII  V    posisi PPE CI 0.03876
3. rotor II   VII  VIII  posisi A0Z CI 0.03876
4. rotor VII  III  I     posisi NBI CI 0.03875
5. rotor III  I    VII   posisi MVX CI 0.03874
...
```

Hanya satu konfigurasi yang menonjol, yaitu rotor **VI II IV** dengan posisi **KQF** (CI 0,06838). Seluruh konfigurasi lain memiliki CI di bawah 0,0388, atau setara dengan teks acak. CI 0,06838 sudah mendekati CI bahasa Inggris, sehingga *plaintext* dapat dipastikan berbahasa Inggris.

4. Awalnya, *ring* dan *plugboard* direncanakan untuk dicari pada tahap berikutnya. Namun, dekripsi dengan rotor VI II IV, *ring* AAA, posisi KQF, dan tanpa *plugboard* sudah menghasilkan teks bahasa Inggris yang terbaca utuh dari awal hingga akhir, dengan CI 0,06838 yang sudah mendekati CI bahasa Inggris. Hal ini menunjukkan bahwa kedua asumsi pada langkah 1 benar.

- Jika *ring* rotor tengah atau kanan salah, akan ada bagian teks yang rusak secara berkala, yaitu setiap kali rotor tengah atau kiri berputar pada waktu yang salah. Pada hasil dekripsi, tidak ada bagian yang rusak.
- Jika *ciphertext* dibuat dengan *plugboard*, dekripsi tanpa *plugboard* akan memasukkan huruf yang salah ke rotor, sehingga sebagian besar huruf hasil dekripsi akan salah dan CI-nya jauh di bawah CI bahasa Inggris. Kenyataannya, seluruh teks terbaca dengan benar.

Oleh karena itu, pencarian *ring* dan *plugboard* tidak diperlukan. Dekripsi akhir dilakukan dengan mesin yang diberikan:

```
./machine VI II IV AAA KQF < secret.txt
```

Artefak

Solver ditulis dalam bahasa C (`src/31/solver.c`) dan di-link langsung dengan kode mesin pada `src/31/enigma-machine/`. Keluaran *solver* disimpan di `src/31/log/solver_output.txt`, dan *plaintext* hasil dekripsi disimpan di `src/31/log/plain.txt`.

```
cc -O3 -fopenmp -Ienigma-machine solver.c \
    enigma-machine/{enigma,rotor,plugboard,reflector}.c -o solver
./solver enigma-machine/secret.txt
```

Plaintext Hasil Dekripsi

SCENARIO NIGHTTIME VOID HUNDREDS OF YEARS AGO THEN NIGHT BLADE WAS DIVIDED INTO TWO GROUPS THERE FORMISTS AND MODERATES WITH LINNE AND HER BROTHER KUON ON THE SIDE OF THE MODERATES WITH KUON AS THEIR LEADER A GREAT BATTLE BROKE OUT BETWEEN THE TWO SIDES RESULTING IN MANY LIVES LOST AS SURVIVING VILLAGE ELDER DECIDED TO SPLIT THE KNOWLEDGE OF EXS INTO THE BODIES OF KUON AND LINNE FROM THE ISLINNESS SOUL WAS CURSED TO BECOME A PARASITE THAT UPON THE DEATH OF HER CURRENT BODY WOULD TAKE OVER THE BODY AND SOUL OF ANOTHER CURRENTLY LIVING PERSON CAUSING HER SOUL TO PERSIST ETERNALLY KUON BURDENED BY HIS OWN CURSE LEFT AND DECIDED TO LEAVE HIS CLOSELY WALDSTEIN TO FOLLOW AND CARE FOR LINNE THE TWO SIBLINGS WENT SEPARATE PATHS AND SINCE THEN LINNE HAS BEEN SEARCHING FOR A WAY TO FINALLY PUT AN END TO HER CURSE OF REINCARNATIONS SCENARIO CHAINED SOUL TEN YEARS PRIOR TO ARRIVING IN KANZAKA IONCE MORE LINNE ENCOUNTERED DAYOUNG SETH IN HER PREVIOUS BODY SHE TOLD HIM ABOUT THE STORY OF THE PRINCESS OF THENIGHT BLADE WHILE CONCEALING HER TRUE IDENTITY UNDER THE NAME CHITOSE AFTER THEY YOUNG SETH EXPRESSED A DESIRE TO TAKE ON THE PRINCESS BURDEN LINNE TOLD HIM THAT MATURE AND TRAIN BEFORE HE CAN TAKE ON THIS ROLE ABIT LATER SHE WAS CALLED OVER BY WALDSTEIN AND PARTED WAYS WITH SETH WHO LATER REALIZED HER TRUE IDENTITY SCENARIO POWER BEAT IN HER PRESENT DAY BODY LINNE AND WALDSTEIN ARRIVE JUST OUTSIDE KANZAKA AFTER HEARING OF THE AMOUNT OF NEW BIRTHS SPROUTING IN THE CITY THAT USED TO BE THE LAND OF THENIGHT BLADE CENTURIES AGO THEY REMINISCE ON THEIR PAST IN THE NIGHT BLADE AND HOW SOCIETY HAS DRASTICALLY CHANGED SINCE THEIR AGETHETWO EXPRESS SOME CONCERN OVER ENTERING THE CITY KNOWING THAT IT IS POPULATED WITH NIGHT BLADE MEMBERS WHO BELIEVE THE TWO BETRAYED THE CLAN THEY CONTINUE TO DISCUSS ASPECTS OF THENIGHT BLADE IN THE PRESENT DAYS SUCH AS LINNE'S ENCOUNTER WITH SETH AS HER PREVIOUS INCARNATION EVENTUALLY LINNE FALLS ASLEEP BEFORE SETTING OFF TO KANZAKA THEN NEXT MORNING THE NEXT DAY THE TWO WALK THROUGH THE CITY WALDSTEIN EXPRESSES SOME IRRITATION AT HOW MUCH THEY STAND OUT AND ATTRACT ATTENTION WITH LINNE OFTEN HAVING TO MAKE UP AN EXPLANATION ON WALDSTEIN'S GEAR AND LARGE BUILD GOING SO FAR AS TO PRETEND HE IS A FOREIGN WRESTLER KNOWN AS LIONMAN WHEN THE TWO SETTLED DOWN AT A RESTAURANT THEY DISCUSS WHAT THEY FOUND REGARDING INBIRTH ACTIVITY IN THE CITY WALDSTEIN MENTIONS SHE HEARD ABOUT A GATHERING OF INBIRTH SHAPPENING THAT WEEKEND AND LINNE BRINGS UP THAT IT MAY BE FOR A POWERFUL ORGANIZATION SHE HEARD BROUGHT TUPE EARLIER THAT DAY MNE'S AS SHE HAD EVEN HEARD OF ITS LEADER PARADOX AND THE PREVALENT OFFICERS INCLUDING THE ONE KNOWN AS PERSISTENT SURVIVOR AND HIS DE MISE LINNE AND WALDSTEIN AGREE TO INVESTIGATE THIS GATHERING WHEN THE DAY COMES AND THEY SPEND THE REST OF THE WEEK AT A NEARBY INN UNTIL THE WEEKEND OF THE GATHERING THE DAY OF THE GATHERING LINNE AND WALDSTEIN MAKE A PLAN TO SNEAK INTO THE MANSION WHERE THE EVENT IS BEING HELD BEFORE ENTERING LINNE DEVICES A PLAN FOR HER TO SCOPE OUT THE AREA TO MAKE SURE THINGS THERE WILL BE SAFE HOWEVER SHE WAS DISTRACTED BY THE FOOD SERVED AT THE EVENT THUS SHE WAS CAUGHT HELD BY TWO GUARDS AND BROUGHT TO HILD A FOR QUESTIONING HILD A QUICKLY DISCERNS THAT SHE IS NO ORDINARY CHILD AND AN INBIRTH IN CONVERSATION WHEN HILD A MENTIONSSHE WILL BE BECOMING A REBIRTH LINNE ACCIDENTALLY IMPLIES SHE IS THE PRINCESS OF THENIGHT BLADE WALDSTEIN WHO WAS WATCHING FROM A DISTANCE IMMEDIATELY INTERVENES PICKING HER UP AND TAKING HER AWAY FROM THE SCENE WHEN BROUGHT TO SAFETY LINNE EXPRESSES GUILT AT LETTING HER HUNGER GET THE BETTER OF HER AND REVEALING HER IDENTITY TO PARADOX HOWEVER LINNE IS AT LEAST GLAD TO HAVE SEEN HER IN PERSON AND TO KNOW OF HER PLAN TO BECOME A REBIRTH LINNE AND WALDSTEIN HAVE ONE LAST CONVERSATION OF THE POWERFUL EXSEMANATING FROM PARADOX AND THE LIKELIHOOD THAT SHE MAY BE CONNECTED TO KUON'S SCENARIO CHILD OF DARKNESS A FEW MONTHS AFTER ARRIVING AT KANZAKA LINNE FINDS HERSELF PONDERING OUT LOUD ABOUT RECENT NIGHT ACTIVITY IN A NIGHTER CLASS MEETING AT THE SCHOOL SHE MUST ATTEND DUE TO THE AGE OF HER VESSEL HER SCHOOL MATE CHI HARU OVERHEARS HER SOMEWHAT CONFUSED LINNE ATTEMPTS TO KEEP THE SITUATION UNDER CONTROL AS SHE NEEDS TO KEEP UP THE FRONT THAT SHE IS A CHILD OF SCHOOL AGES SHE CONTINUES TO REFLECT ON THE CURRENT STATE OF THENIGHT WHISPERS TO HERSELF AND TRYING TO HIDE WHAT SHE'S SAYING WHENEVER CHI HARU OVERHEARS HER SHE OFTENSURPRISES LINNE BY COMING TOTALK TO HER WHILE SHE'S LOST IN THOUGHT BUT LINNE RECOGNIZES THAT SHE'S TRYING TO BE FRIENDLY EVENTUALLY CHI HARU LEAVES TO GO HOME AND LINNE DECIDES TO WANDER OFF INSTEAD OF RETURNING HOME TO WALDSTEIN LINNE TAKES A STROLL THROUGH THE CITY AND ENDS UP DECIDING TO ENTER A TOY STORE OUT OF CURIOSITY AS SHE WAS LOOKING THROUGH THE STORE A CLERK NOTICED HER TALKING TO HERSELF ABOUT HOW TOYS WERE BACK IN HER TIME SHE ATTEMPTS TO LEAVE AFTER REALIZING THIS BUT ACCIDENTALLY RUNS STRAIGHT INTO A BOY IN A SCHOOL UNIFORM HE EXPRESSES CONCERN FOR HER HELPING HER UP AND LEADING HER TO THE EXIT LINNE PROMPTLY LEAVES THE AREA AS QUICKLY AS SHE CAN TO GET AWAY FROM HIM SHE WORRIES SHE ACTED TOO COLDLY TO HIM DURING THEIR BRIEF INTERACTION AND SPENT TIME WALKING AROUND THE CITY MORE FOR THE REST OF THE DAY LATER IN THE AFTERNOON WHEN TRYING TO RETURN HOME LINNE WANDERS NEAR A SCHOOL WITH STUDENTS THAT HAVE THE SAME UNIFORM AS THE BOYS

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Kunci / Konfigurasi

- Reflektor: B
- Urutan rotor (kiri, tengah, kanan): VI, II, IV
- *Ring setting*: AAA
- Posisi awal: KQF
- *Plugboard*: tidak ada

Hasil Decrypt Sesuai Format Sumber Asli

Scenario: Nighttime Void

Hundreds of years ago, the Night Blade was divided into two groups: The Reformists, and Moderates, with Linne and her brother Kuon on the side of the Moderates with Kuon as their leader. A great battle broke out between the two sides, resulting in many lives lost. A surviving village elder decided to split the knowledge of EXS into the bodies of Kuon and Linne. From this, Linne's soul was cursed to become a parasite that, upon the death of her current body, would take over the body and soul of another currently living person, causing her soul to persist eternally. Kuon, burdened by his own curse, left and decided to leave his close ally Waldstein to follow and care for Linne. The two siblings went separate paths, and since then, Linne has been searching for a way to finally put an end to her curse of reincarnation.

Scenario: Chained Soul

Ten years prior to arriving in Kanzakai once more, Linne encountered a young Seth in her previous body. She told him about the story of the Princess of the Night Blade while concealing her true identity under the name Chitose. After the young Seth expressed a desire to take on the Princess' burden, Linne told him to mature and train before he can take on this role. A bit later, she was called over by Waldstein and parted ways with Seth, who later

realized her true identity.

Scenario: Power Beat

In her present day body, Linne and Waldstein arrive just outside Kanzakai after hearing of the amount of new In-Births sprouting in the city that used to be the land of the Night Blade centuries ago. They reminisce on their past in the Night Blade, and how society has drastically changed since their age. The two express some concern over entering the city, knowing that it is populated with Night Blade members who believe the two betrayed the clan. They continue to discuss aspects of the Night Blade in the present day, such as Linne's encounter with Seth as her previous incarnation. Eventually, Linne falls asleep before setting off to Kanzakai the next morning.

The next day, the two walk through the city. Waldstein expresses some irritation at how much they stand out and attract attention, with Linne often having to make up an explanation on Waldstein's gear and large build, going so far as to pretend he is a foreign wrestler known as "Lion Man". When the two settle down at a restaurant, they discuss what they found regarding In-Birth activity in the city. Waldstein mentions he heard about a gathering of In-Births happening that weekend, and Linne brings up that it may be for a powerful organization she heard brought up earlier that day, Amnesia. She had even heard of its leader, Paradox, and the prevalent officers, including the one known as Persistent Survivor and his demise. Linne and Waldstein agree to investigate this gathering when the day comes, and they spend the rest of the week at a nearby inn until the weekend of the gathering.

The day of the gathering, Linne and Waldstein make a plan to sneak into the mansion where the event is being held. Before entering, Linne devises a plan for her to scope out the area to make sure things there will be safe. However, she was distracted by the food served at the event, thus she was caught, held by two guards, and brought to Hilda for questioning. Hilda quickly discerns that she is no ordinary child and an In-Birth. In conversation, when Hilda mentions she will be becoming a Re-Birth, Linne accidentally implies she is the Princess of the Night Blade. Waldstein, who was watching from a distance, immediately intervenes, picking her up and taking her away from the scene.

When brought to safety, Linne expresses guilt at letting her hunger get the better of her and revealing her identity to Paradox. However, Linne is at least glad to have seen her in person, and to know of her plan to become a Re-Birth. Linne and Waldstein have one last conversation of the powerful EXS emanating from Paradox and the likelihood that she may be connected to Kuon.

Scenario: Child of Darkness

A few months after arriving at Kanzakai, Linne finds herself pondering out loud about recent Night activity in an after class meeting, at the school she must attend due to the age of her vessel. Her schoolmate Chiharu overhears her, somewhat confused. Linne attempts to keep the situation under control, as she needs to keep up the front that she is a child of school age. She continues to reflect on the current state of the Night, whispering to herself and trying to hide what she's saying whenever Chiharu overhears her. She often surprises Linne by coming to talk to her while she's lost in thought, but Linne recognizes that she's trying to be friendly. Eventually, Chiharu leaves to go home, and Linne decides to wander off instead of returning home to Waldstein.

Linne takes a stroll through the city, and ends up deciding to enter a toy store out of curiosity. As she was looking through the store, a clerk noticed her talking to herself about how toys were back in her time. She attempts to leave after realizing this, but accidentally runs straight into a boy in a school uniform. He expresses concern for her, helping her up and leading her to the exit. Linne promptly leaves the area as quickly as she can to get away from him. She worried she acted too coldly to him during their brief interaction, and spent time walking around the city more for the rest of the day.

Later in the afternoon, when trying to return home, Linne wanders near a school with students that have the same uniform as the boy she ran into earlier. While there, an angry-looking student bumps into her, knocking her down. The angry student attempts to fight her over this, and Linne considers a way to escape. But before any fight can occur, the boy from earlier

came in to intervene. To save Linne from the situation, he told the angry student that he was here to pick up his brother "Yoshio", being Linne. This failed, so he instead picked her up and left the scene.

When brought to safety, Linne is somewhat grateful, but mostly embarrassed and irritated. The boy accidentally finds out Linne is a girl when attempting to put her down, resulting in Linne strangling him to the point of unconsciousness. When he wakes up, Linne thanks him for earlier and tries to go her own way, but the boy insists they walk around Kanzakai together. The two then properly introduce themselves. The boy's name is Hyde Kido.

The two spend the rest of the day walking around the city and eating at a crepe restaurant. When the sun sets, Hyde insists she stay at his place, but Linne declines and the two go their separate ways for the night.

The next day, Linne decides to investigate the school Hyde goes to, and to check on him if she can. She sees Hyde walking home with a female classmate, and immediately notices her refined and efficient way of movement, as if she may have been through some kind of training. It bothers Linne that someone like her would be close to Hyde.

When Linne sees Hyde go off on his own, she notices a frightening disturbance in the EXS surrounding the area, and attempts to run to Hyde to save him from a potential Void attack. She comes in in time and manages to save him from one Void, but is bitten by another. A sword immediately emerges from Hyde as he groans in pain, as Linne realized that he has become an In-Birth. She brings him to safety, ready to teach him about the Hollow Night that he had accidentally stumbled into.

Scenario: A Reluctant Hero

Linne spent a few weeks training Hyde and teaching him how to control the EXS within him. A day before the upcoming Night, she appeared in front of him as he was returning home with groceries he bought. She warns Hyde about the upcoming Night, and that he should not go out if she's not present. Hyde says he'll go anyway if she's in danger, bothering Linne as she doesn't want him risking his life. Just then, Waldstein reveals himself, who was hiding and watching them. Linne forgot to tell Hyde that he was there. He's disappointed by the lack of food Hyde brought back, and Hyde decides to go back and buy more groceries. Before he leaves, Linne tells him to be careful.

Hyde returns after a long time, bringing bad news on why he was late. He had ran into Paradox because a mysterious child told him to go to her location, and accidentally disclosed his identity, putting a target on the back of him and Linne. This greatly disappointed her, but she also immediately recognized this mysterious child as her brother Kuon, based on the description Hyde gave. She told him everything about him, about the Secret Arts and why the two separated. She once again warns Hyde to not get himself into danger under any circumstances, with him responding the same way as he did previously. Irritated, but somewhat understanding, she said she would follow and accept his choice.

Sumber:

- <https://inbirth.wiki.gg/wiki/Linne>

Referensi

Konsep mesin Enigma (*plugboard*, rotor, dan reflektor) mengacu pada bahan kuliah [8]. Spesifikasi *wiring* rotor I–VIII, posisi *notch*, dan Reflector B yang digunakan untuk memverifikasi kode mesin mengacu pada [12]. Rumus dan nilai acuan *Index of Coincidence* mengacu pada [13]. Strategi *ciphertext-only attack* bertahap (urutan rotor dan posisi terlebih dahulu dengan CI, kemudian *ring*, lalu *plugboard*) mengacu pada [4], sedangkan perhitungan ukuran ruang kunci *plugboard* mengacu pada [11].

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Lampiran

Repository GitHub

Seluruh kode sumber *solver*, berkas *ciphertext*, log hasil dekripsi, teks sumber *plaintext*, serta sumber L^AT_EX laporan ini dapat diakses pada repository berikut.

<https://github.com/reletz/IF4020-3-Tugas1>